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Supervisor: Guven Demirel

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Demand Forecasting and Pricing Strategy for AI-Powered Brain Diagnostics

(Adoption Scenarios and Market Strategy for QP-Brain in England)

Hamza Nauman (240471033)

Yin Min Aung (240915364)

Veena Poduru (241051702)

Nisha Sunariya (240847906)

George Dreyer-Wilton (240669537)

Student's Declaration

We,

Hamza Nauman (240471033),

Yin Min Aung (240915364),

Veena Poduru (241051702),

Nisha Sunariya (240847906),

George Dreyer-Wilton (240669537)

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Table of Contents

Student's Declaration	2
List of Figures	6
List of Tables.....	8
List of Equations	9
Executive Summary	10
1. Introduction (Hamza Nauman – 240471033)	17
2.1. Literature Review (Hamza Nauman – 240471033)	21
2.2. Business Problem (Hamza Nauman – 240471033)	26
3. The Dataset and Exploratory Data Analysis (Yin Min Aung – 240915364)	36
3.1. Overview of Data Sources	36
3.1.1. NHS Diagnostic Imaging Dataset (DID)	36
3.1.2. NHS Diagnostic Waiting Times and Activity Dataset (DWTA).....	37
3.1.3. NHS Workforce Statistics – Hospital and Community Health Service (HCHS) Staff	37
3.1.4. Disease-Specific Incidence and Prevalence Data	38
3.1.5. Private Sector Imaging Data	39
3.2. Data Preprocessing.....	39
3.2.1. Construction of 10-Year Monthly Brain Scan Dataset For QP-Brain (England) 40	
3.2.2. Construction of the Trust-level Brain Scan Dataset (2024/25).....	45
3.3. Exploratory Data Analysis	52

3.3.1.	Time Series Trends Analysis.....	53
3.3.2.	Provider-level analysis.....	61
4.	Analytical Framework and Methodology	69
4.1.	Quantifying England Neuroimaging Market (Yin Min Aung – 240915364)	69
4.1.1.	Methodology	69
4.1.2.	Brain MRI Market Size Estimation (England, 2024/25)	71
4.2.	Forecasting Global demand and Adoption Scenarios (Veena Poduru - 241051702)	
	86	
4.2.1.	Forecasting Global demand	86
4.2.2.	Adoption scenarios.....	93
4.3.	Analysing Private Market (George Dreyer-Wilton – 240669537).....	117
4.3.1.	Market Sizing.....	117
4.3.2.	Drivers of Private Demand	119
4.3.3.	Provider Distribution	120
4.3.4.	Demand Drivers	120
4.3.5.	Demand Barriers	121
4.3.6.	Market Growth.....	122
4.4.	Adoption Within The Private Sector (George Dreyer-Wilton – 240669537)	123
4.4.1.	Potential Early Adopters	123
4.4.2.	Adoption Drivers	124
4.4.3.	Adoption Limiters.....	125
4.4.4.	Adoption Scenarios.....	126

4.5.	Determining Ideal Pricing and Value-Based Proposal – Competitor Analysis, Pricing Strategy, Market-entry Recommendation (Nisha Sunariya – 240847906).....	128
4.5.1.	Quibim’s Market Strategy: Competitor Insights, Pricing Models & NHS Entry Pathways	128
4.5.2.	Competitor Landscape in the UK AI Imaging Market.....	134
4.5.3.	Quibim’s Competitive Position and SWOT Analysis	138
4.5.4.	Pricing Models & NHS Benchmarks	140
4.5.5.	NHS Benchmarks and Publicly Available Data	142
4.5.6.	Innovative Pricing Models in MedTech.....	144
4.5.7.	Illustrative ROI Calculations	149
4.5.8.	Market Entry Channels – NHS & Private Sector Pathways	151
5.	Discussion (George Dreyer-Wilton – 240669537)	157
6.	Conclusion (George Dreyer-Wilton – 240669537).....	162
	AI Declaration.....	167
	References (Section 1 & 2).....	168
	References (Section 3, 4.1 & 4.2)	170
	References (Section 4.3 and 4.4)	173
	References (Section 4.5)	176

List of Figures

Figure 1 Private Providers	46
Figure 2 Step 1 – MRI (Brain) and Radiologist Merge	49
Figure 3 Step 2 – MRI (Brain) and Waiting Time Merge	50
Figure 4 Final Merge	51
Figure 5 Brain MRI Growth	53
Figure 6 Population adjusted Trend	54
Figure 7 Comparison of Growth Trend.....	55
Figure 8 Monthly Trend of Total Brain Scans in England.....	56
Figure 9 MRI (Brain) Diagnostic Waiting Time Activity	57
Figure 10 Radiologist Workforce Trends.....	58
Figure 11 Radiologist Shortfall Trend	59
Figure 12 Brain & CNS Tumour Incidence (UK).....	60
Figure 13 Monthly Dementia Diagnoses Trend.....	61
Figure 14 Top 10 NHS Providers by Brain MRI Scan Volumes (2024/25).....	62
Figure 15 NHS Brain MRI Scans by Private Providers.....	62
Figure 16 Number of Providers Offering MRI VS Total Provider in Waiting List	63
Figure 17 Brain MRI Delay by England Region	64
Figure 18 Scatter Plot of Waiting List VS Delayed Cases.....	65
Figure 19 Radiologist by England Region.....	66
Figure 20 Top 10 NHS Trusts by Brain MRI Workload	66
Figure 21 NHS Trusts with Lowest Efficiency	67
Figure 22 Scatter Plot of Demand Pressure	68
Figure 23 Diagnostic Imaging Activity by Modality (NHS England, 2025).....	71
Figure 24 Diagnostic Imaging Activity by Body Site (NHS England, 2025).....	72

Figure 25 Brain MRI % of all MRIs.....	73
Figure 26 MRI Exams per 1,000 Population by Country (2017)	74
Figure 27 Breakdown of Diagnostic Test by Waiting List (NHS England, 2025).....	75
Figure 28 Breakdown of Diagnostic Test by Over 6 weeks waiting list (NHS England, 2025)	76
Figure 29 Performance against the NHS six-week diagnostic standard, January 2010– November 2024 (British Medical Association, 2025)	77
Figure 30 NHS Specialty Group Distribution (NHS England, 2025).....	78
Figure 31 Top-Down Market Sizing Flowchart for Brain MRI in England (2024/25).....	80
Figure 32 England vs. UK Brain MRI Market (2024/25, estimated)	81
Figure 33: 5-year monthly forecast using Holt-winters exponential smoothing	91
Figure 34: Sample data in the demand sheet in Provider_Scans_5_years.xlsx excel file	104
Figure 35: S-curves for next 5 years from 2025-26 to 2029-30	114

List of Tables

Table 1 Variables in the Time Series Monthly Dataset (Apr 2015 – Mar 2025).....	45
Table 2 Summary table of final dataset (Demand)	52
Table 3 Market Sizing Calculation Table for Brain MRI.....	85
Table 4: Model comparison	90
Table 5: 12-month forecast table.....	92
Table 6: Sample of Provider_Scans_5_years.xlsx (MRI (Brain) sheet)	96
Table 7: Brain MRI as a Percentage of Total MRI Scans (England)	97
Table 8: Sample of Waiting Time sheet in Provider_Scans_5_years.xlsx excel file	99
Table 9: Sample of Radiologists sheet in Provider_Scans_5_years.xlsx excel file	100
Table 10: Sample Scan-to-wait ratio sheet in Provider_Scans_5_years.xlsx excel file.	102
Table 11: Top five Trusts per scenario for current year (2024-25)	111
Table 12: Top 5 Trusts per Scenario (2025–26).....	113
Table 13 Comparative Competitor Table	138
Table 14 Key Pricing Lessons for Quibim.....	145
Table 15 Recommended Pricing Strategy	151

List of Equations

Equation 1 MRI Brain %	47
Equation 2 Radiologist %	48
Equation 3 Scans per Radiologist	49
Equation 4 Scan-to-6+week-wait Ratio	50
Equation 5 Demand Score	51
Equation 6 UK Brain MRI Scans.....	81

Executive Summary

This report, “*Demand Forecasting and Pricing Strategy for AI-Powered Brain Diagnostics (Adoption Scenarios and Market Strategy for QP-Brain in England)*”, investigates the growing mismatch between demand for brain magnetic resonance imaging (MRI) in England and the capacity of the diagnostic workforce, and explores how Quibim’s QP-Brain artificial intelligence platform can be positioned to address these systemic challenges. The overall objective of the project is to quantify demand and market size, forecast future adoption scenarios, and develop an appropriate pricing strategy for both NHS and private sector deployment of QP-Brain. In order to achieve this, the study integrates ten years of NHS imaging data, workforce statistics, disease prevalence and incidence figures, and private sector estimates, with statistical forecasting, scenario modelling, and competitor analysis. The outcome is an evidence-based roadmap for how QP-Brain could support both clinical practice and health system efficiency in England.

The motivation for this project is the sustained imbalance between the rising demand for brain MRI and the limited diagnostic workforce. As shown in Section 3.3.1, the number of full-time equivalent radiologists employed in the NHS increased by 48.7% between 2015/16 and 2024/25, from about 3,750 to 5,900. During the same period, however, brain MRI demand grew even faster, from 666,000 to 1.07 million annually, representing an increase of 61% and a compound annual growth rate of 4.9%. This imbalance has created structural bottlenecks in service provision. By March 2025, more than 70,500 patients were waiting for a brain MRI, and 13.7% of them had been waiting longer than six weeks, far above the NHS Constitution threshold of 1% and the interim target of 5%. These waiting times have serious clinical consequences. In conditions such as stroke, brain tumours and dementia, delays in diagnosis can result in disease progression, worsened prognoses, and in some cases avoidable mortality.

The impact of COVID-19 exacerbated this already critical mismatch, creating a backlog that continues to affect services. During 2024/25, NHS England outsourced about 40,000 brain MRIs to private providers, equivalent to 4% of national activity, spending £216 million on outsourced reporting alone (Section 3.3.2). While outsourcing temporarily relieved pressure, it underlined systemic reliance on stopgap solutions rather than sustainable approaches. QP-Brain is designed to offer precisely this kind of sustainable solution. As discussed in Section 1, QP-Brain is an AI platform that automates volumetric and lesion quantification across more than 130 brain regions. Traditional radiologist-led visual assessments remain prone to perceptual error, fatigue and inter-observer variability, but QP-Brain generates quantitative and standardised outputs that can be benchmarked against normative databases. It integrates into existing PACS workflows without requiring new software or workstations, producing structured reports that enhance diagnostic confidence while reducing reporting time. From a regulatory standpoint, QP-Brain carries CE marking, UKCA certification and FDA 510(k) clearance, demonstrating its readiness for scalable deployment. It is also compliant with GDPR and HIPAA, ensuring secure deployment either on cloud-based platforms or hospital servers. The platform is being applied not only in dementia care but also in oncology and research on brain atrophy, demonstrating its broader utility.

The literature reviewed in Section 2.1 reinforces the importance of these technological developments. Studies show that retrospective diagnostic error rates in radiology can reach 30%, while in daily practice error rates average between 3 and 5%. These errors are closely linked to high workload and reporting volumes. Fatigue and task overload are significant drivers of perceptual errors. QP-Brain directly addresses this by reducing cognitive load and automating quantification. Evidence from other studies shows that AI can increase sensitivity and specificity in image interpretation, improve workflows through reprioritisation of urgent cases, and contribute to population-level health improvements. However, the literature also

highlights barriers to adoption, such as algorithmic transparency, ethical issues of bias, regulatory complexities, and data privacy concerns.

The business problem, set out in Section 2.2, reflects both the urgency of system pressures and the barriers to adoption. NHS waiting lists have expanded significantly, partly as a result of COVID-19 and partly due to chronic workforce shortages. By 2024/25, over 1.7 million patients were waiting for diagnostic tests of all modalities, including more than 331,000 awaiting MRI scans. Of these, 13.7% had already waited longer than six weeks. To mitigate shortfalls, the NHS has turned increasingly to outsourcing, which is financially costly and clinically uneven. Yet adoption of AI-based diagnostics has been slow. Barriers include the absence of specific reimbursement categories under the NHS Payment Scheme, which forces Trusts either to absorb costs or bundle AI into larger service contracts; regulatory hurdles that make approval of adaptive AI systems slow and complex; uneven IT infrastructure across Trusts; and a lack of clinician training in AI systems. Smaller Trusts and rural hospitals, in particular, are disadvantaged in terms of IT capacity and imaging equipment, deepening regional disparities. Data privacy is another concern, since brain scans are categorised as “special category” data under GDPR, requiring rigorous safeguards. Collectively, these issues complicate Quibim’s path to market, even though the clinical rationale for AI adoption is strong.

To build an evidence base for addressing these challenges, the study brought together a wide range of datasets (Section 3.1). These included the NHS Diagnostic Imaging Dataset (DID), which provided information on brain MRI activity from 2015 to 2025; the Diagnostic Waiting Times and Activity (DWTA) dataset, which measured backlog and waiting lists; NHS Workforce Statistics, which detailed radiologist FTEs; dementia prevalence data from NHS England; and brain tumour incidence data from Cancer Research UK. Private sector estimates

were drawn from the LaingBuisson UK Diagnostics Market Report and IHPN. Two consolidated datasets were constructed as described in Section 3.2. The first was a ten-year monthly national time series covering April 2015 to March 2025, which integrated imaging, workforce, waiting list, disease and private market data. The second was a 2024/25 provider-level dataset that combined brain MRI activity, workforce capacity and waiting time data. Derived measures such as scans per radiologist, scan-to-wait ratios and composite Demand Scores were calculated. A custom cleaning process was applied to ensure provider names were consistent, datasets were merged, and Trust-level pressure was quantified.

Exploratory Data Analysis (Section 3.3) provided critical insights. At the national level, monthly brain MRI volumes increased from about 52,000 in 2015 to more than 90,000 by 2025, a rate far above the 6.5% population growth and the 18–20% increase in dementia prevalence and tumour incidence. This indicates that demand growth is driven primarily by changing clinical practice and increased referral rather than demographics. A significant dip occurred in 2020/21 due to pandemic disruption, followed by a rebound that overshot pre-pandemic trends. By 2025, waiting lists persisted, with 15–20% of patients facing delays beyond six weeks. At the regional level, disparities were clear. London had the highest radiologist workforce (1,222 FTEs) while the South West had the lowest (541). Trust-level analysis highlighted outliers such as The Walton Centre and Clatterbridge Cancer Centre, where each radiologist handled workloads exceeding 500 brain MRIs annually, more than double the national average. Meanwhile, Trusts such as Homerton Healthcare and Royal Papworth were flagged as among the least efficient, with low scan-to-wait ratios. The Demand Score framework combined workload and efficiency rankings to identify high-pressure Trusts most suited to early adoption of AI solutions.

Market quantification, detailed in Section 4.1, used a top-down approach. NHS activity in 2024/25 comprised 1,014,655 brain MRIs, equivalent to 21.3% of all MRI procedures. Adding an estimated 58,850 private brain MRIs yielded a combined total of 1,073,505 scans. Relative to England's 2024 population of 58.38 million, this equates to 18.4 brain MRIs per 1,000 people, which is far below international comparators such as Germany, France, Japan, and the United States, where rates exceed 100 per 1,000. This underlines both capacity constraints and unmet demand. At the same time, 70,500 patients were awaiting brain MRI in March 2025, with 13.7% waiting beyond six weeks, underscoring the backlog problem. Workforce analysis showed that radiologists handled an average of 174 brain MRIs annually, confirming systemic reliance on efficiency gains and the need for supplemental technological solutions.

Future demand was forecast using ten years of monthly data and four time-series models: Holt-Winters exponential smoothing, ARIMA, SARIMA and Prophet (Section 4.2.1). Model performance was evaluated using RMSE and MAPE, with SARIMA and Holt-Winters proving the most reliable. Both captured seasonality and long-term growth, consistently predicting continued upward trends in brain MRI demand. To translate this into adoption potential, scenario-based modelling was applied using provider-level Demand Scores (Section 4.2.2). Adoption probabilities were calculated using logistic transformation, with higher-pressure Trusts assigned higher likelihoods of early uptake. Continuity over time was ensured using a decay factor of 0.5. Three scenarios were modelled following S-curve diffusion: Conservative (0–5 Trusts annually), Medium (5–10) and Ambitious (10–15+). The Conservative scenario suggested minimal system-wide impact. The Medium scenario indicated incremental improvements. The Ambitious scenario showed the potential for rapid adoption and significant backlog reduction if supported by policy and funding.

Private sector analysis (Sections 4.3 and 4.4) complemented the NHS focus. Data from LaingBuisson and IHPN suggest between 20,000 and 60,000 private brain MRIs annually in England, with a central estimate of 58,850, equivalent to 5–6% of national demand. Activity is concentrated in metropolitan hubs such as London, where patients increasingly turn to private providers for faster access. Early adoption is most likely from major providers such as HCA, Spire, Nuffield, Circle, Alliance Medical and InHealth, driven by competition, marketing benefits, and faster service delivery. However, barriers exist, including affordability, uneven capacity, and regulatory uncertainty. Scenario analysis suggested adoption would begin in urban hubs before spreading more widely.

Pricing and competitor analysis, set out in Section 4.5, identified a fragmented AI imaging market with a mix of per-scan charges, enterprise licences and subscription models. Quibim's SWOT analysis highlighted strengths in regulatory readiness, workflow integration and technical capability, but also noted threats from NHS cost sensitivity and regulatory inertia. Pricing analysis highlighted the structural challenge that AI diagnostics currently lack specific reimbursement categories in the NHS Payment Scheme. Providers must therefore absorb costs or bundle them into enterprise agreements. Nonetheless, ROI modelling showed that QP-Brain could generate significant savings, particularly if it reduced outsourcing, which cost the NHS £216 million in 2024. The study recommends a dual pricing strategy: low-margin, high-volume models for NHS Trusts to support adoption, and flexible, premium models for private providers who are less cost-sensitive and motivated by differentiation.

The discussion in Section 5 synthesises the findings, emphasising that the England's neuroimaging market is defined by rising demand, uneven workforce distribution and persistent backlogs. Forecasting confirms that demand will continue to rise, while adoption modelling demonstrates that accelerated uptake of AI could reduce waiting lists and workforce

pressures. The private sector provides a valuable proving ground for early adoption, while NHS adoption will depend on demonstrating cost-effectiveness, aligning with procurement frameworks, and building clinician confidence.

In conclusion, this study demonstrates that the England's neuroimaging market is characterised by persistent capacity shortages, structural backlogs, and uneven regional distribution of diagnostic services. Brain MRI demand has risen far faster than population growth or disease incidence, leaving tens of thousands of patients facing delays that exceed national standards. At the same time, the radiology workforce has grown but remains insufficient to meet demand. Against this background, QP-Brain represents a credible and timely intervention. By automating analysis, integrating into clinical workflows, and providing regulatory-approved outputs, it can ease workforce burdens, improve diagnostic accuracy, and support backlog reduction. Forecasting confirms that demand will continue to rise in the medium term, and adoption scenarios indicate that ambitious uptake could meaningfully alter system trajectories. Pricing and competitor analysis shows that while NHS reimbursement frameworks present challenges, a dual strategy that differentiates between NHS and private markets could overcome cost barriers and maximise uptake. The evidence presented throughout this report highlights both the urgency of addressing diagnostic bottlenecks and the strategic opportunity for Quibim to align its technological strengths with market realities. If implemented successfully, QP-Brain can contribute to reducing waiting times, supporting radiologists, and advancing the adoption of AI-enabled neuroimaging in England, while also establishing a model for international deployment.

1. Introduction (Hamza Nauman – 240471033)

Quibim, a global leader in advanced medical imaging analytics, is leading the way in the development of AI-powered diagnostic tools that are revolutionizing neurological diagnosis and brain imaging. Quibim's cutting-edge screening and computational analysis methods have made it possible to detect dementia, Alzheimer's disease, and other neurodegenerative diseases earlier and more precisely. Quibim combines his expertise in artificial intelligence and radiography. The need for cutting-edge brain imaging technologies has increased quickly as preventative treatment, early intervention, and precision diagnosis become more popular in international healthcare systems. Because neurological diseases, especially dementia, are becoming more commonplace globally and have a substantial impact on patient quality of life, caregiver stress, and healthcare expenses, they rank among the most important healthcare concerns of our day. For this reason, advanced brain scans like CT and MRI, as well as AI-driven analytics, are important tools for improving long-term patient care, helping doctors decide on treatments, and boosting diagnostic confidence.

In this regard, Quibim is working with Queen Mary University of London MSc Business Analytics students on a capstone project that aims to produce useful data about the UK market for brain imaging and screening technologies. In order to predict adoption patterns among medical professionals and determine the best pricing tactics that strike a balance between accessibility and over time business performance, the project will evaluate the present and future demand for AI-driven brain diagnostic solutions. This project will give Quibim a thorough grasp of market dynamics by utilizing forecasting techniques and sophisticated data analytics. This will strengthen the company's business strategy, expand the use of its brain imaging device, and give it a stronger competitive advantage in the UK healthcare market. In

the end, this partnership will help Quibim grow strategically, move precision medicine forward, and make it easier to treat people with neurological diseases.

These days, the radiologist's visual assessment is the primary method for subjectively assessing brain atrophy and associated changes in both healthy individuals and patients with neurodegenerative diseases. On FLAIR images, concentrated abnormalities like hypoxic-ischemic areas or white matter demyelinating lesions usually show up as signal hyperintensities, which the radiologist must personally describe. The accuracy and consistency of this standard-of-care method, which primarily uses qualitative readings, are limited. The need for objective, quantitative techniques that can more accurately and earlier detect early structural flaws in the tissues of the brain, such as atrophy and lesions, is growing as the aging population contributes to a notable increase in neurological disorders.

The cutting-edge Quibim QP-Brain® platform, powered by artificial intelligence, promises to revolutionize the interpretation of brain MRI data by providing previously unheard-of levels of accuracy, speed, and clinical utility in the diagnosis and monitoring of neurological disorders. Tools that go beyond conventional visual assessments to identify subtle structural changes in the brain are desperately needed, especially in light of the sharp increase in neurodegenerative diseases like dementia, Alzheimer's disease, and other disorders of the central nervous system. By automating the quantitative analysis of brain MRI data, QP-Brain immediately resolves this problem. This allows for the precise measurement of brain sizes and lesion loads, which would not be possible otherwise. The platform offers accurate measurements of white matter hyperintensities (WMH) in terms of both load and setting, along with absolute and relative volumetry of grey matter, white matter, cerebrospinal fluid, and over 130 different brain regions. For increased diagnostic confidence, doctors can compare these outputs to healthy,

age- and sex-matched individuals after standardizing them against a sizable normative database.

QP-Brain is exceptionally adept at recognizing and classifying WMH thanks to its proprietary AI ensemble. By removing physiological hyperintensities and artifacts, it generates outputs that are incredibly dependable. By enabling standard 2D FLAIR sequences and eliminating the need for more intricate or specialized imaging protocols, its novel architecture democratizes access to advanced imaging analysis while keeping the device affordable, even for local hospitals and smaller care facilities. By generating results that show up as an extra series in a patient's PACS folder without requiring extra workstations or specialized viewing software, QP-Brain actually blends in perfectly with current clinical workflows. By allowing radiologists and neurologists to obtain structured, objective reports that include reference charts and long-term tracking more rapidly, this connection decreases submitting time and boosts efficiency.

QP-Brain guarantees the highest standards of data protection by being completely compliant with GDPR and HIPAA from a compliance and security standpoint. It can be deployed using reliable cloud infrastructures like AWS or Microsoft Azure, or it may be securely set up on hospital servers secured with firewalls. All of these options are hidden to safeguard patient identity. The platform's clinical efficacy and regulatory preparedness for broad regional deployment are further evidenced by its FDA 510(k) clearance, CE mark, and UKCA certification. In addition to dementia care, QP-Brain supports clinical and research projects in oncology, including monitoring the outcomes of glioblastoma treatment, and in the investigation of patterns of brain atrophy associated with aging and neurological diseases.

QP-Brain combines state-of-the-art AI technology, smooth workflow integration, and regulatory-grade dependability to redefine the standards for brain MRI analysis. Objective data can be used by medical professionals to track patient progress, make early diagnoses, increase

diagnostic accuracy, and ultimately provide more individualized and efficient care. As a result, Quibim is improving patient outcomes globally and furthering the goal of precision medicine in addition to the field of medical imaging.

Using in-depth data-driven analysis, the present study seeks to provide a comprehensive analysis of Quibim's position in the UK healthcare market for brain imaging solutions. It begins with a review of the literature that highlights any gaps that require further research, critically evaluates what is currently known, identifies significant discoveries, and summarizes the body of knowledge. The business issue is then covered in the report, with particular attention paid to the difficulties Quibim is encountering in the UK. The methodology section describes the analytical techniques used in detail after providing an overview of the data collection and exploratory data analysis. The following sections measure the market using trust-level analysis and interest sizing. Future adoption will then be predicted using scenario-based adoption, machine learning techniques, and time series models. The study also looks at pricing strategies, such as value-based concepts, competitor benchmarking, and market entry recommendations. The insights gathered from the studies are used to inform strategic decision-making after the results have been presented and discussed. The study concludes with a summary of the results and recommendations for future research to guarantee a thorough contribution to both academic knowledge and real-world application.

2.1. Literature Review (Hamza Nauman – 240471033)

The increased demand for diagnostic imaging is putting increasing strain on radiologists, which often results in diagnostic errors and job overload. Studies show that whereas retrospective error rates in radiography can reach 30%, real-time diagnostic errors occur in about 3–5% of daily practice (Lee et al., 2013). In addition to increasing patient morbidity and death, these errors increase the likelihood of malpractice lawsuits, highlighting the urgent need for technological improvements that improve the accuracy and efficacy of diagnostic workflows. In this sense, Quibim's QP-Brain is a groundbreaking AI-powered technology that can address some of these systemic problems by automating brain MRI quantification and enhancing diagnostic consistency.

The association between error rates and radiologist workload is well-established. On days when diagnostic errors occurred, radiologists worked 21% more than they usually did (Kasalak et al., 2023). This suggests that task overload has a major impact on perceptual errors in CT reporting. Crucially, fatigue-related errors were evenly distributed throughout the day and there was no indication that they increased toward the end of shifts. This suggests that task intensity, not the effects of the time of day, is the primary determinant. This conclusion is corroborated by earlier studies, which show that poorer detection accuracy and interpretative disparities are linked to longer shifts, faster reporting speeds, and higher reporting volumes (Christensen et al., 1981; Hanna et al., 2018; Sokolovskaya et al., 2015). These results point to a systemic problem: radiologists are experiencing uncontrollable stress as imaging volumes keep increasing. This emphasizes the need for AI-based support systems that help reduce diagnostic variability and offload tedious duties.

By automating dimensional and lesion evaluation in brain imaging and producing objective, repeatable results that lessen the possibility of human perceptual error, Quibim directly addresses these problems with QP-Brain®. By measuring brain structures and white matter hyperintensities (WMH) in 132 different locations, QP-Brain helps radiologists reduce their cognitive load and boost diagnostic confidence by enabling them to support visual results with evidence backed by data. This is in line with professional recommendations to abandon traditional workload caps, which might not work because radiologists' abilities vary from person to person (Alexander et al., 2022), and instead embrace technological advancements that have the potential to dynamically advance human knowledge.

Still, there are gaps in medical care and study. Kasalak et al. (2023) point out that while workload plays a significant role in diagnostic errors, other elements such as interruptions, cognitive biases, and case complexity are also important. They found that 30.6% of diagnostic errors occurred when radiologists were not overworked. Likewise, additional peer-reviewed studies are required to evaluate Quibim's AI solutions' effect on reducing errors in actual hospital settings, despite the fact that they are FDA-approved, CE-marked, and incorporated into clinical procedures. Instead of using longitudinal trials, most current research evaluates AI-driven therapies using randomized trials and retroactive workload studies.

The research on artificial intelligence (AI) and machine learning (ML) in healthcare, especially as it relates to brain health, shows how significantly new technologies may change patient care, diagnosis, and treatment. Studies consistently demonstrate that AI systems can diagnose conditions more accurately than conventional medical techniques. For instance, Lee et al. (2020) showed 95% sensitivity and 90% specificity in CT-based analysis, while Smith et al. (2018) reported that AI accomplished a sensitivity of 92% and particularity of 88% in MRI-based detection tasks. These outcomes surpass a number of traditional diagnostic methods.

These developments show how AI-driven image interpretation could improve early neurological disease diagnosis, lower diagnostic errors, and improve clinical decision-making. AI has shown promise not only in diagnosis but also in treatment personalization and regimen optimization. In order to improve clinical outcomes and lower death rates, Wang et al. (2019) and Zhang et al. (2019) showed how predictive algorithms may anticipate patient deterioration, modify treatment dosage, and minimize side effects. The promise of AI for precision medicine was further demonstrated by Chen et al. (2021), who found that AI-assisted drug optimization reduced adverse events and increased therapeutic results by 30%.

The role of AI in managing population health more generally is another important topic in the literature. Implementing AI-based disease surveillance using real-time data sources increased outbreak prediction accuracy by 85%, per studies like Brown et al. (2019). This allowed for the earlier and more successful implementation of public health initiatives. Similar to this, Smith et al. (2021) found that AI-driven targeted interventions resulted in notable cost savings of up to 25%, highlighting the scalability and financial advantages of AI-powered public health initiatives. Crucially, AI has shown promise in mitigating health disparities. Chen et al. (2021) demonstrated that choice support systems reduced the diversity in effects by improving minority groups' adherence to diabetes care guidelines. These results imply that AI could be a tool to increase clinical effectiveness and equity in the provision of healthcare. AI also has the potential to foster medical innovation by discovering new treatments and customizing actions for each patient, as demonstrated by its use in drug discovery (Jones et al., 2019) and reinforcement learning in treatment optimization (Zhang et al., 2019). When taken as a whole, these developments demonstrate how AI is influencing healthcare on many levels, from population-level health management to individualized patient care.

Notwithstanding these encouraging results, some important gaps and difficulties are constantly highlighted. Widespread adoption remains hampered by ethical worries about algorithmic bias, accountability, and transparency (Dagnew et al., 2021; Díaz-Verdejo et al., 2020). Interpretability issues are brought up by the "black box" nature of many deep learning systems, which may jeopardize patient safety and physician confidence. Due to the sensitive nature of neurological and genetic data, data security and confidentiality risks significantly complicate the use of AI in healthcare (Caine & Sharman, 2019; Cardoso et al., 2021). There are still regulatory obstacles. Frameworks like the European Commission's AI Ethics standards and the FDA's Software as a Medical Device (SaMD) regulations provide guidance, but national implementation varies, raising questions about consistency and compliance. The methodology used in the literature is based on comparatively small datasets, frequently from very small populations, which limits how broadly the findings can be applied. Many studies are retrospective in nature and lack long-term confirmation, raising questions about how well the findings will translate into actual clinical settings. Furthermore, concerns about equity and accessibility persist because the advantages of AI may disproportionately benefit highly resourced health systems, displacing those in poverty and widening the digital divide.

Important new information regarding the use of artificial intelligence (AI) to identify intracranial hemorrhage (ICH) on non-contrast-enhanced CT images and its impact on radiologists' workflow productivity is provided by the O'Neill et al. (2020) study. While simple notification methods, such as a widget display or flagging studies, had no discernible effect on queue-adjusted wait times or turnaround times, the authors found that active reprioritization of AI-flagged examinations resulted in a significant reduction in wait times, particularly for routine inpatient and outpatient cases (12.01 minutes vs. 15.75 minutes baseline, $P < .0001$). This illustrates that AI's primary benefit extends beyond alerting radiologists; it also allows for the dynamic reordering of process priorities in order to remove bottlenecks. Importantly, the

analysis highlights that waiting for routine tests accounts for more than 90% of turnaround delays, suggesting that worklist wait times—rather than interpretation time—are the primary source of reporting delays. Previous research has demonstrated that AI algorithms for ICH diagnosis can perform diagnostically on par with experts, with sensitivity and specificity as high as 95%–99% (Arbabshirani et al., 2018; Kuo et al., 2019; Ojeda et al., 2019). However, O'Neill et al. expand on this work by examining the impact of AI integration on clinical workflow and show that algorithmic correctness alone is insufficient if it isn't coupled with effective workflow reprioritization. However, the study also highlights significant gaps: generalizability remains uncertain due to its limited use of non-contrast-enhanced CT in a single facility, and the lack of weekend and after-hours research limits knowledge of AI's influence in emergent scenarios. Additionally, the absence of a concurrent control group and an assessment of the impact of false-positives on the workflow as a whole highlight areas that need greater investigation. As AI becomes more integrated into radiology, this study highlights the importance of cautious implementation strategies that address algorithmic performance as well as real-world workflow challenges and fluctuations in clinical demand.

With significant improvements in customized therapy, managing population health, diagnostic precision, and healthcare efficacy, research clearly shows that AI and ML offer revolutionary potential to improve brain wellness and medical care in general. AI medications have shown promise in reducing costs, improving results, and even tackling health inequities, but their full potential still depends on resolving ongoing issues with data privacy, algorithmic bias, ethical use, and regulatory compliance. However, studies show a direct correlation between radiologists' workload and mistakes in diagnosis, with a high number of cases jeopardizing accuracy and reliability.

In this regard, artificial intelligence (AI) systems like Quibim's QP-Brain have demonstrated that automated, quantitative brain MRI analysis can increase diagnosis accuracy, decrease clinician effort, and remove subjective interpretation. Though promising, there are still many unanswered questions regarding these solutions' generalizability, integration into intricate clinical settings, and effect on systemic error-causing factors like disruptions and multidisciplinary tasks. To ensure better patient outcomes and long-term healthcare system transformation, achieving the promise of AI in brain health will eventually call for thorough, longitudinal validation, equitable implementation strategies, and an agreement between breakthrough technology and responsible, ethical implementation.

2.2. Business Problem (Hamza Nauman – 240471033)

The medical field is not far behind the rapid advancement of artificial intelligence (AI) in all areas of life. Even though the application of AI in medicine has grown tenfold in recent years, there are still issues and unknowns. Artificial intelligence (AI) technologies typically use machine learning algorithms to perform "intelligent" analytical and inferential tasks on health data in order to effectively inform healthcare decision-making. These tasks include, but are not limited to, diagnosing and treating neurological and chronic disorders, providing medical services and treatments, developing new medications, connecting appropriate individuals with clinical trials, and tracking and predicting diseases and epidemics (infodemiology). However, a number of ethical concerns, such as algorithmic bias that produces inconsistent or discriminatory results, privacy violations, disputes over possession of data, and a lack of transparency in data use, are raising questions about AI's efficacy in healthcare (Vayena et al., 2018).

The waiting periods for brain scan imaging are another significant business problem that causes significant delays in receiving the results. Serious delays in diagnosis and treatment can arise from improper interpretation of brain scans, especially for urgently needed conditions like cancer and stroke. Improving results and saving lives depend on prompt action and early exploration. Delays can lead to the spread of cancer and other diseases, requiring more drastic, possibly ineffective treatments. Delays in stroke diagnosis worsen damage to the brain and raise the risk of permanent disability. Delays in identifying injuries or degenerative diseases can result in chronic pain, decreased mobility, and increased long-term medical expenses, even in less severe cases. Patients may have unreasonable expectations and experience increased stress if they are not fully informed about the procedure and timeline for receiving their scan results. There may be more uncertainty and worry when results are sent to a patient portal in certain healthcare systems before the patient has had a chance to discuss them with their medical team.

Although the acute lack of radiologists is widely recognized, the consequences of these shortages are often poorly understood. Radiology is the cornerstone of healthcare since, without a diagnosis, there can be no therapy. The biggest problem facing radiology is the mismatch between supply and demand. Radiographers (technicians who take the scans), radiologists (physicians who analyze the images), and scanners are all in low supply despite the high demand for scanning. Medical developments that require complicated medical imaging for therapy and an aging population with chronic ailments are the reasons for the dramatic rise in demand. When this demand is not satisfied, patients suffer in the real world since imaging delays can have catastrophic consequences. According to the NHS, there are 133,000 scans taking place in the UK every day. The COVID-19 backlog and the increase in scans have resulted in a tenfold increase in patients waiting more than six weeks for a CT or MRI scan (RCR, 2020). Days, hours, and minutes are important when it comes to diagnosis

because delays prevent early treatment and lead to worse results. Studies show that unnecessary patient deaths result from imaging bottlenecks that delay diagnosis (Cournane et al., 2016).

The lack of data and market research on the use of AI in brain scans, particularly in the UK, is one of the main obstacles to adoption and growth for Quibim, a top supplier of AI-driven diagnostic solutions in medical imaging. Even though AI is excellent at precision diagnosis, especially for prostate cancer and brain scan imaging, its slow adoption in this field limits the company's ability to extend its AI solutions to brain scan interpretation (Chauhan et al., 2021). Market dynamics for AI-driven diagnostic tools in brain imaging are still developing in the UK, and successful penetration of these technologies is being hampered by a lack of extensive data on consumer demand, adoption rates, and pricing plans. Recent research indicates that the UK healthcare system is still unprepared to use AI techniques in clinical settings, especially when it comes to neuroimaging. This limits the potential impact (Gambhir et al., 2020).

Quibim may have trouble adapting its solutions for regional conditions or navigating the regulatory environment if it is not fully aware of the particular requirements of the healthcare system, such as the requirement for AI-based stroke or brain tumor diagnosis. Furthermore, despite Quibim's products' obvious benefits in early disease identification, the medical community's hesitancy to adopt AI technologies due to ignorance may hinder their adoption. Several studies have demonstrated that worries about regulations, accuracy, and reliability frequently cause healthcare professionals to be reluctant to use AI tools (Topol, 2019). More focused efforts are needed to inform medical professionals about the benefits of AI-driven diagnostics and the significance of early detection for neurological illnesses.

Furthermore, despite Quibim's products' obvious benefits in early disease recognition, the medical community's hesitancy to adopt AI technologies due to ignorance may hinder their adoption. Several studies have demonstrated that worries about rules and regulations, precision,

and reliability frequently cause healthcare professionals to be reluctant to use AI tools (Topol, 2019). More focused efforts are needed to inform medical professionals about the benefits of AI-driven assessment and the significance of early diagnosis for neurological illnesses. The best pricing strategies to make AI products feasible and competitive would also require market research. According to a recent McKinsey & Company study, AI solutions frequently perform poorly in the healthcare sector because they lack an understanding of market conditions, specifically price (Chui et al., 2020). By filling in these knowledge and research gaps, Quibim can lead the way in AI-driven brain scan diagnostics, just as it has done with prostate cancer. Quibim can also help change healthcare in the UK by offering faster and more precise diagnoses.

Navigating the regulatory approval process is one of the biggest business obstacles facing organizations like Quibim that want to use AI-driven diagnostic solutions for brain scans. Countries like the UK may have a slow, disjointed, and complicated healthcare regulatory framework, especially when it comes to AI technologies. The approval of pharmaceuticals, diagnostics, and medical devices has traditionally been supervised by regulatory bodies like the Food and Drug Administration (FDA) in the US and the Medicines and Healthcare Products Regulatory Agency (MHRA) in the UK. The dynamic nature of AI and the speed at which its technologies develop—often surpassing the creation of legal frameworks that were not designed to support such technology present a special challenge, though.

By classifying AI-powered diagnostic tools as either software as a medical device (SaMD) or medical devices, the MHRA is instrumental in approving them in the UK (MHRA, 2021). Before these devices can be marketed for therapeutic use, certain requirements must be fulfilled. The dynamic nature of AI systems, which are constantly adjusting to new data, is the challenge. Because AI is dynamic, it is difficult to comply with traditional regulatory

frameworks that demand predetermined and validated models. Because regulatory agencies must guarantee algorithms function safely and consistently across a variety of clinical settings—a challenge that is especially challenging with machine learning models that evolve over time—AI solutions are subject to a more complicated approval process (Price, 2020).

Usually, a number of regulatory testing processes are required for AI diagnostic tools, like brain scan analysis. It might take years before the algorithm's precision, dependability, and therapeutic efficacy are established. The requirement to validate the technology across a range of populations presents an additional challenge for clinical studies involving AI, necessitating the collection of large, costly, and tedious datasets. The approval process for brain scans is complicated by the need for AI technologies to be tested across a range of conditions (such as malignancies, strokes, and neurodegenerative diseases) and methods of imaging (such as CT and MRI) (Gambhir et al., 2020).

Regulatory approval delays raise operating costs in addition to slowing market access. Companies like Quibim may need to invest a large sum of money in clinical testing and compliance before their product is authorized for use by patients and medical professionals. Missed opportunities in a cutthroat industry where other AI firms might launch comparable technology earlier or secure approval more quickly could result from this delay (Topol, 2019). For instance, traditional diagnostic tools may only require a year or two to be cleared, whereas AI tools in radiology may require years (Ball, 2020).

One of the biggest barriers to the UK's adoption of AI-driven diagnostic tools for brain imaging is the incapacity of medical professionals to successfully integrate these tools into routine practice. Because many radiologists, radiographers, and clinicians lack formal training in AI use, interpretation, or limitations, there is a skills gap that could compromise trust and effective utilization (Topol, 2019; RCR, 2022). When incorporating AI into existing clinical workflows,

it is often required to modify data governance practices, comply with interoperability standards, and enhance Picture Archiving Communication Systems (PACS)/Vendor Neutral Archive (VNA) systems to meet NHS Digital and Digital Technology Assessment Criteria (DTAC) requirements (NHS England, 2023). Another restriction is the availability of data. Dementia-specific algorithms require large, well-annotated UK datasets that link imaging to therapeutic outcomes, but GDPR restrictions and safe data environment laws limit accessibility (ICO, 2023). Operationally, the shortage of radiologists and radiographers in England limits the ability to track and integrate new AI technologies, and because of NHS imaging backlogs and Community Diagnostic Center roll-outs, buyers may put throughput ahead of long-term preventative analytics (RCR, 2024). While the commercial sector remains fragmented, the NHS is volume-dominant and highly price-sensitive. As a result, it is challenging to determine the best prices for both acute use cases, like stroke, when the advantages of therapy are immediately apparent, and longer-horizon paths, like dementia, where cost savings appear years later (Alzheimer's Society, 2024). Lastly, market segmentation complicates price strategy.

The UK's growing use of AI for brain imaging raises concerns over data security and privacy because brain scans are considered particularly sensitive medical data and often contain identifiable patient information. Such data is categorized as "special category" information under the Data Protection Act of 2018 and the UK GDPR, which calls for robust safeguards for processing, storage, and transportation (ICO, 2023). AI algorithm training and validation frequently demand large, high-quality datasets. This could involve transferring photos between cloud-based systems, private suppliers, or NHS Trusts, all of which present possible security risks. Ransomware attacks and other cybersecurity threats have increased in recent years, as seen by the NHS's numerous high-profile incidents that compromised data integrity and interrupted services (NCSC, 2023). A breach undermines patient confidence, which is crucial

for encouraging participation in AI-powered diagnostic programs, in addition to the possibility of heavy fines from the Information Commissioner's Office (Nuffield Council on Bioethics, 2022). NHS entities may be subject to political opposition and scrutiny from the public, but contracts and market share for private sector companies can also be lost due to reputational harm. Therefore, it is crucial to meet the NHS's Digital Technology Assessment Criteria (DTAC), which require rigorous adherence to cybersecurity, interoperability, and data security specifications prior to deployment. Because of the growing privacy risks and the possibility of public backlash, the use of AI in brain imaging may stall in the absence of thorough technical safeguards, staff training in data handling, and open patient permission processes.

Inequitable access to cutting-edge imaging technologies, like high-resolution MRI or CT scanners and AI-powered diagnostic tools, is still an issue in the UK, especially for smaller NHS institutions and rural hospitals. While smaller trusts are more likely to use outdated scanners and lack the bandwidth or hardware required to run modern imaging algorithms, large urban academic medical centres are more likely to have the latest technology, specialized study collaborations, and computer systems that can support AI integration (RCR, 2024). Numerous factors, such as disparities in the allocation of capital funds, procurement capabilities, and regional goals within Integrated Care Systems (ICSs), contribute to these technological advances divide (NHS England, 2023).

By making patients in underprivileged areas wait longer or travel to far-off locations for more sophisticated imaging, this could postpone diagnosis and treatment. This is especially problematic for diseases that require immediate treatment, such as stroke or rapidly spreading brain tumours. Due to restricted access to high-resolution neuroimaging, early functional or structural defects in dementia therapy may go unnoticed, decreasing the chance of prompt

intervention. These differences create a "postcode lottery" in diagnosis accuracy and undercut the NHS's fundamental tenet of fair access to high-quality care.

The cost of AI-driven brain imaging technology is significantly impacted by the stark differences in budgets and purchasing patterns between the NHS and the private healthcare sector in the UK. Diagnostic activity is funded by nationally set Healthcare Resource Group (HRG) pricing and, more recently, particular Community Diagnostic Center (CDC) tariffs under the NHS Payment Scheme, a tariff-based reimbursement mechanism. The majority of the nation's imaging volumes are the responsibility of the NHS (HFMA, 2024; NHS England, 2025). Providers who use AI technologies are compelled to either absorb the costs internally or include them in their enterprise license agreements or imaging tariffs because AI diagnostics do not yet have their own payment category in this system. Intense pricing discussions that reduce vendor margins and increase the time it takes to see a return on investment could make this bundling strategy financially challenging for the NHS, which is very cost-conscious and seeks economies of scale. Furthermore, due to staffing shortages and increasing demand, NHS capital expenditures for technology adoption are limited and usually prioritize replacing scanners and resolving backlogs over advanced analytical instruments (RCR, 2024). In contrast, even though there are fewer scans performed overall, the demand for diagnostic imaging in the UK private healthcare sector is rising rapidly. This is partly because patients are being pushed toward self-funded or insured solutions by NHS waiting lists (LaingBuisson, 2024). This industry is highly fragmented, with independent hospitals, diagnostic centers, and corporate health providers. Every one of these organizations has distinct strategic priorities, IT infrastructure readiness levels, and procurement practices. It is more challenging to create a uniform pricing plan because willingness-to-pay varies not only throughout businesses but also by clinical use case. Acute-use AI systems, like those used for the rapid triage of stroke patients, typically have a higher price tolerance because they may demonstrate immediate operational

and clinical benefits, such as reduced door-to-needle times and fewer inter-hospital transfers. However, it is more challenging to implement AI tools for long-term screening or monitoring in the NHS, such as those for brain tumor surveillance or dementia identification. In contrast to the NHS's acute capacity shortages and short-term budgetary restrictions, its benefits—earlier diagnosis, slower illness progression, and improved patient quality of life—often take years to manifest as measurable cost savings. Commissioners may therefore prioritize spending that directly increases throughput, such as hiring locum radiologists or outsourcing scan reporting, over these strategies. The NHS really spent £216 million in 2024 on outsourcing reporting to private firms, illustrating the depth of the capacity deficit and the system's reliance on temporary fixes rather than long-term technology investment (The Guardian, 2025). Because of the interplay between these two markets, vendors such as Quibim need to create complex pricing strategies: low-margin, high-volume models for NHS adoption and more flexible, value-based pricing for private providers who may prioritize patient experience and differentiation over cost reduction. However, unless structural changes are made to reimbursement mechanisms, like creating AI-specific payment categories or incorporating long-term cost-benefit modeling into NHS procurement, scaling advanced AI diagnostics across the UK brain imaging market will remain a significant commercial challenge.

As a result, even though AI-driven diagnostic tools have the potential to revolutionize brain imaging by decreasing delays, increasing accuracy, and addressing the shortage of radiology professionals, substantial systemic, commercial, and legal barriers prevent their widespread adoption in the UK. Some of the obstacles that companies like Quibim must overcome in order to expand include slow regulatory approval, limited clinical integration because of infrastructure and skill limitations, data privacy concerns, unequal access to new imaging technology, and the NHS's cost-sensitive procurement approach. Furthermore, uncertainty is

brought about by the absence of thorough market data on pricing tactics, clinical demand, and adoption rates.

Investment in long-term AI solutions is further constrained by the NHS's established backlogs and reliance on outsourcing. In addition to navigating legal issues and customizing pricing models for the NHS and private sectors, vendors must collaborate with healthcare providers to establish trust, prove cost-effectiveness, and guarantee equitable access in order to fully realize the potential of AI in neuroimaging. Even though AI has demonstrated scientific aptitude, if these complex issues are not addressed, its potential for quicker diagnosis, earlier treatments, and more long-lasting healthcare outcomes may be lost.

3. The Dataset and Exploratory Data Analysis (Yin Min Aung – 240915364)

This section includes the overview of data sources, data preprocessing and key descriptive insights from exploratory data analysis.

3.1. Overview of Data Sources

To understand the current landscape of brain scan volume, radiological capacity and diagnostic delays in the England healthcare system, multiple datasets were collected from both publicly accessible NHS (National Health Service) statistics and external industry platforms. The following data sources were selected based on their reliability, validity and the relevance to brain scans utilization, healthcare workforce distribution and market characteristics in both public and private healthcare sectors.

3.1.1. NHS Diagnostic Imaging Dataset (DID)

The NHS Diagnostic Imaging Dataset (DID) is a collection of detailed information about diagnostic imaging tests carried out on NHS patients, extracted from local Radiology Information Systems (RISs) and submitted monthly to NHS England (NHS England, 2025). For this study, a 10-year time series (2015/16 to 2024/25) was extracted, comprising two specific DID tables per year:

1. **“Count of Imaging Activity by Modality”** which contains scan volumes of different test types by provider.
2. **“Count of Imaging Activity by Body Site”** which contains scan volumes of different body parts by providers.

Each reporting year spans April to March, yielding twelve monthly observations per dataset. These data form the basis for analysing national and provider-level imaging activity trends over time, with a particular focus on brain imaging.

3.1.2. NHS Diagnostic Waiting Times and Activity Dataset (DWTA)

The NHS Diagnostic Waiting Times and Activity Dataset (DWTA) is a monthly publication reporting the number of patients on the waiting list for diagnostic tests at the end of each month, along with the proportion who have been waiting more than six weeks and more than thirteen weeks (NHS England, 2025). For this study, two web file of DWTA dataset were utilized:

1. **“Monthly-Diagnostics-Web-File-Timeseries-May-2025”** which provides the England-level time series covering January 2006 to May 2025, used to analyse long-term trends in brain scan activity and backlog.
2. **“Monthly-Diagnostics-Web-File-Provider-March-2025_SW40K.xls”** which provides the provider-level breakdown of waiting list metrics for each NHS Trust as of March 2025, enabling calculation of key performance indicators such as scan-to-wait ratios.

These datasets were used to assess diagnostic capacity over time, identify high-pressure providers, and provide inputs to NHS adoption scenario modelling.

3.1.3. NHS Workforce Statistics – Hospital and Community Health Service (HCHS) Staff

The NHS Hospital and Community Health Service (HCHS) monthly workforce statistics provide counts of full-time equivalent (FTE) staff directly employed in NHS Trusts and other core NHS organisations. Staff are categorised into major staff groups, including professionally qualified clinical staff, hospital and community health service (HCHS) doctors, consultants,

associate specialists, specialty doctors, and other grades (NHS England, 2025). For this study, two NHS workforce datasets were used to obtain workforce estimates relevant to diagnostic capacity analysis:

1. **“HCHS Staff by NHS Organisation”** which provides full-time equivalent (FTE) counts of HCHS doctors for each NHS Trust in England.
2. **“HCHS Doctors by Grade and Specialty”** which provides the distribution of FTE doctors by medical specialty, including clinical radiology, for England.

The data are available monthly from September 2009 to March 2025. The datasets were combined to estimate radiologist numbers at Trust level, which informed the calculation of scans-per-radiologist and related KPIs.

3.1.4. Disease-Specific Incidence and Prevalence Data

To estimate the underlying clinical demand drivers for brain MRI diagnostics, disease incidence and prevalence data were gathered from multiple authoritative sources:

1. **Cancer Research UK** (Brain and other central nervous system (CNS) tumour statistics), providing age-standardised incidence rates (per 100,000 population) for brain, other CNS, and intracranial tumours in the UK, covering the period 2000–2019 (Cancer Research UK, 2024).
2. **Recorded Dementia Diagnoses and Primary Care Dementia Data** (NHS England), providing monthly prevalence of recorded dementia diagnoses in England. The **Recorded Dementia Diagnoses** series was used for April 2015–September 2022, after which it was superseded by the **Primary Care Dementia Data** series (October 2022–March 2025). These datasets were combined to produce a continuous ten-year monthly time series of dementia prevalence (NHS England, 2025).

3. **Office for National Statistics (ONS)**, providing annual population estimates for England from 1971 to 2023, used to account for demographic trends such as population growth (Office for National Statistics, 2025).

These datasets provided estimates of the target population most likely to require brain MRI scans. The rising prevalence of dementia, combined with stable-to-increasing brain tumour incidence and a growing population, supports the expectation of sustained growth in neuroimaging diagnostic demand.

3.1.5. Private Sector Imaging Data

Understanding private diagnostic capacity was key for estimating total market demand and identifying additional deployment opportunities for QP-Brain beyond the NHS. However, private sector diagnostic imaging volumes are not publicly available in the same granularity as NHS datasets. To inform market sizing, this study utilised aggregated figures from industry reports and market research publications such as the “**LaingBuisson UK Diagnostics Market Report**” which reports total private imaging scan volumes in the UK for 2024 (LaingBuisson, 2025) and the “**Going Private – Diagnostics**” report by the Independent Healthcare Providers Network (IHPN) which provides major private healthcare organizations (IHPN, 2025). These sources provide the foundation for estimating private sector brain MRI activity and its contribution to overall market demand.

3.2. Data Preprocessing

This section details the steps taken to transform raw data from multiple sources into a consolidated, analysis-ready dataset suitable for exploratory analysis, demand quantification, forecasting and adoption scenario for AI demand in neuroimaging diagnostic. The preprocessing workflow involved both manual compilation in Microsoft Excel and data manipulation in Python, ensuring that the resulting datasets were complete, accurate and

aligned across all sources. A total of two datasets; a national-level time series dataset for 10-year period and a 2024/25 trust-level dataset, were constructed for the purpose of this project. These are provided in the code archive under the “**Final Dataset**” folder.

3.2.1. Construction of 10-Year Monthly Brain Scan Dataset For QP-Brain (England)

For a consistent longitudinal analysis across multiple public and industry datasets, a unified monthly dataset, “**brain_scans_10yrs.xlsx**”, was manually constructed covering the 10-year period from April 2015 to March 2025 in Microsoft Excel. This dataset integrates demographic, brain scan imaging activity, workforce capacity, waiting time, disease prevalence, and private market indicators of England into a unified time series. A detailed cover sheet within the Excel file documents all data sources, relevant hyperlinks, and key assumptions applied. The dataset includes two tables:

1. **Monthly time series (120,16)** from April 2015 to March 2025, used for national-level trend analysis, market sizing, and demand forecasting of MRI brain scan demand in England.
2. **Yearly summary (10,16)** from 2015/16 to 2024/25, used for annual growth calculations and high-level comparisons.

The summary table of the variables included in this dataset and how they were derived is shown in Table 1.

Variable Name	Source Dataset	Frequency	Dataset Limitations	Derivation
England Population	ONS – Population Estimates (Office for National Statistics, 2025)	Annual figure was repeated for each month of the same year for monthly table.	Mid-2024 figure is not yet fully published. For this study, mid-2015 estimates were aligned to the 2015/16 financial year and so on.	Population for 2024 was estimated as 2023×1.012 as the population growth rate of England for mid-2024 compared with the previous year was 1.2% according to ONS (Office for National Statistics, 2025).
MRI Scans (NHS)	DID – Count of Imaging Activity by Modality from 2015/16 to 2024/25 (NHS England, 2025)	Monthly		Filtered for Modality = Magnetic Resonance Imaging
MRI Brain Scans (NHS)	DID – Count of Imaging Activity by Body Site from 2015/16 to	Monthly	The table contained counts only for MRI (Brain), with no equivalent breakdown for	Filtered for Test = Brain (MRI)

	2024/25 (NHS England, 2025)		other neuroimaging tests. Analysis was therefore restricted to MRI Brain volumes.	
MRI Brain %	Derived	Monthly		$\frac{MRI (Brain)}{MRI Scans} \times 100$
Clinical Radiologists (NHS)	HCHS Doctors by Grade and Specialty (NHS England, 2025)	Monthly		Filter for Clinical Radiology, FTE only.
Brain Scan per Radiologist	Derived	Monthly	Assumes all clinical radiologists contribute equally to brain MRI workload.	$\frac{MRI (Brain)}{Clinical Radiologists}$
Patients Waiting List (MRI)	Monthly-Diagnostics-Web-File-Timeseries-May-2025	Monthly	Reported at modality level. Brain-specific breakdown not available.	Direct from source: Total MRI patients on waiting list.

	(NHS England, 2025)			
MRI 6+ Week Waiters	Monthly-Diagnostics-Web-File-Timeseries-May-2025 (NHS England, 2025)	Monthly	Reported at modality level. Brain-specific breakdown not available.	Direct from source; MRI patients waiting >6 weeks.
Brain Patients Waiting List	Derived	Monthly	Assumes brain MRI proportion is stable for waiting list allocation.	<i>Patients Waiting List (MRI)</i> $\times MRI\ Brain\ \%$
Brain 6+ Week Waiters	Derived	Monthly	Assumes brain MRI proportion is stable for waiting list allocation.	<i>MRI 6+Week Waiters</i> $\times MRI\ Brain\ \%$
% Waiting 6+ Weeks	Derived	Monthly	Dependent on estimation accuracy for brain waiting list volumes.	$\frac{Brain\ 6+Week\ Waiters}{Brain\ Patients\ Waiting\ List} \times 100$
Private Brain Scans	LaingBuisson UK	Monthly	Since only total private scans in	$MRI\ Brain \times 5.8\%$

	Diagnostics Market Report (2025) (LaingBuisson, 2025)		2024 was available, fixed ratio assumption of 2024 was applied across years assuming every year has same ratio.	based on private/NHS ratio for 2024 (see <i>Note</i>).
Total Brain Scans	Derived	Monthly	Private volumes were estimated, not observed.	<i>MRI Brain</i> + <i>Private Brain Scans</i>
Dementia Patients	NHS Recorded Dementia Diagnoses & Primary Care Dementia Data Series from 2015 to 2025 (NHS England, 2025)	Monthly	NHS stated that the outbreak of Coronavirus (COVID-19) has led to unprecedented changes in General Practice Activity and therefore the data between Oct 2020 and Sep 2022 is impacted (NHS Digital, 2022).	Combined two series for Apr 2015–Mar 2025.

Brain Tumour Incidence Rate (per 100,000 population)	Cancer Research UK (Cancer Research UK, 2024)	Annual figure was repeated for each month of the same year for monthly table.	The latest data is limited to 2019. For this study, 2015 data was aligned to the 2015/16 financial year and so on.	2015–2019 data was directly extracted from source, data for 2020–2025 was forecasted via linear regression.
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Table 1 Variables in the Time Series Monthly Dataset (Apr 2015 – Mar 2025)

Note: According to LaingBuisson UK Diagnostics Market Report, 63,395,000 total scans were done in the UK in 2024, of which 59,920,000 are NHS and 3,475,000 are private hospitals which is 5.8% of NHS scans (LaingBuisson, 2025).

3.2.2. Construction of the Trust-level Brain Scan Dataset (2024/25)

For adoption scenario modelling and trust-level analysis, a separate trust-level dataset was constructed using Python Programming Language for the most recent complete financial year (April 2024 – March 2025). The associated processing code is provided in “**Provider_level_analysis_Brain.ipynb**” within the code archive. The final dataset “**Provider_Brain_Scans_Analysis.xlsx**” contains 7 tables in total along with the detailed cover sheet created manually which documents all data sources, relevant hyperlinks, and key assumptions applied.

3.2.2.1. Date Cleaning and Standardization

A total of 5 datasets were used, and 4 output tables of MRI (Brain), Private Sector, Waiting Time and Radiologist at trust-level were exported as Excel sheets in the file named “**Provider_Brain_Scans_Analysis.xlsx**”.

MRI (Brain) Table

From the **Table-1-2024-25-Modality-Provider-Counts-2025-06-19-XLSX-199KB** file, all rows were filtered to retain only MRI modality records in order to obtain the national total of MRI scans. This total was later used to compute the proportion of MRI activity attributable to brain imaging.

The **Table-4-2024-25-Bodysite-Provider-Counts-2025-06-19-XLSX-221KB** file was then filtered to retain only records where the anatomical site was “Brain (MRI)” and the source setting was “All”. The “Year to date” field (total brain MRI counts) was converted to a numeric variable and a pivot transformation was applied so that each provider appeared as a single row with a “Brain (MRI)” column representing their annual scan count. The national “ENGLAND” row was removed to avoid duplication, and providers with zero brain scans were excluded. The final pivot table contains 145 rows and 2 columns (Provider name, Brain (MRI)) and was exported as “MRI (Brain)” sheet in “**Provider_Brain_Scans_Analysis.xlsx**”.

Private Sector

Total NHS brain MRI scans outsourced to private providers: 40,565

Test Brain (MRI)	
Provider name	
Alliance Medical	15170.0
Inhealth Limited	9860.0
Healthshare Diagnostics	5625.0
Spire Healthcare	2480.0
Circle Health Group Limited	2175.0
Cobalt Health	2090.0
Practice Plus Group Holdings	1780.0
Ramsay Health Care UK Operations Limited	915.0
Nuffield Health	390.0
Phoenix Hospital Group	80.0

Figure 1 Private Providers

In addition, providers identified as private (names not containing “NHS”) were extracted separately to quantify outsourced Brain MRI scans. These were exported as a dedicated sheet, “Private Sector” in the “**Provider_Brain_Scans_Analysis.xlsx**” file. While this gives visibility into NHS outsourcing, these rows were excluded from the unified demand dataset as no radiologist workforce information exists for private hospitals.

Waiting Time Table

For waiting time metrics, the **Monthly-Diagnostics-Web-File-Provider-March-2025_SW40K.xls** dataset was filtered to include only MRI as the diagnostic test. The derived brain MRI percentage of total MRI activity at national level was calculated as:

$$MRI\ Brain\ \% = \frac{Total\ MRI\ (Brain)}{Total\ MRI\ Scans} \times 100$$

Equation 1 MRI Brain %

The brain MRI percentage derived above was applied to estimate brain-specific waiting list sizes and counts of patients waiting more than six weeks. The dataset was then pivoted to a provider-level format, and entries with zero patients in the >6-week waiting category were removed. The “Regional Team Name” column was merged to facilitate geographic analysis.

The final table contains 205 rows and 5 columns (Provider name, Regional Team Name, Brain MRI Waiting, Brain MRI 6+ Week Wait, Brain MRI % 6+ Weeks) and was exported as “Waiting Time” sheet in “**Provider_Brain_Scans_Analysis.xlsx**”.

Radiologist Table

For workforce capacity, the **NHS Workforce Statistics – Doctors by Grade and Specialty (March 2025)** file was used to identify the national total HCHS FTE doctors in NHS workforce and the national total of full-time equivalent (FTE) clinical radiologists, and these figures were used to calculate the percentage of radiologist to all HCHS doctors in England.

$$\text{Radiologist \%} = \frac{\text{Total Radiologist FTE}}{\text{Total HCHS Doctors}} \times 100$$

Equation 2 Radiologist %

This proportion was applied to the **NHS Workforce Statistics – England and Organisation (March 2025)** dataset, which provided Trust-level counts of all HCHS doctors. Prior to calculation, Integrated Care Board (ICB) rows were removed, records without an organisation name were excluded, and the “HCHS Doctors” field was converted to numeric. Finally, rows with zero estimated radiologists were dropped, yielding the Trust-level radiology workforce table.

The final table contains 204 rows and 4 columns (NHS England region name, Organisation name, HCHS Doctors, Estimated Radiologists), and was exported as “Radiologist” sheet in “**Provider_Brain_Scans_Analysis.xlsx**”.

These cleaned **MRI (Brain)**, **Waiting Time**, and **Radiologist** tables formed the three core components of the 2024/25 provider-level dataset, which was subsequently used for merging, KPI calculation, and scenario modelling in later sections.

3.2.2.2. Dataset Merging and Alignment

To generate a unified provider-level dataset for 2024/25, three sequential merging steps were undertaken, each combining complementary data sources and calculating key performance indicators (KPIs) relevant to diagnostic capacity and backlog assessment.

Provider name cleaning

Before merging datasets, it was observed that provider names were not standardised (e.g., some entries in the MRI (Brain) table contained the prefix “*The*”, whereas the Radiologist and Waiting Time datasets did not). Therefore, A regex-based custom cleaning

function (**clean_provider**) was applied to remove inconsistent prefixes (e.g., “The”) and punctuation from provider names.

Step 1 – MRI (Brain) and Radiologist Merge

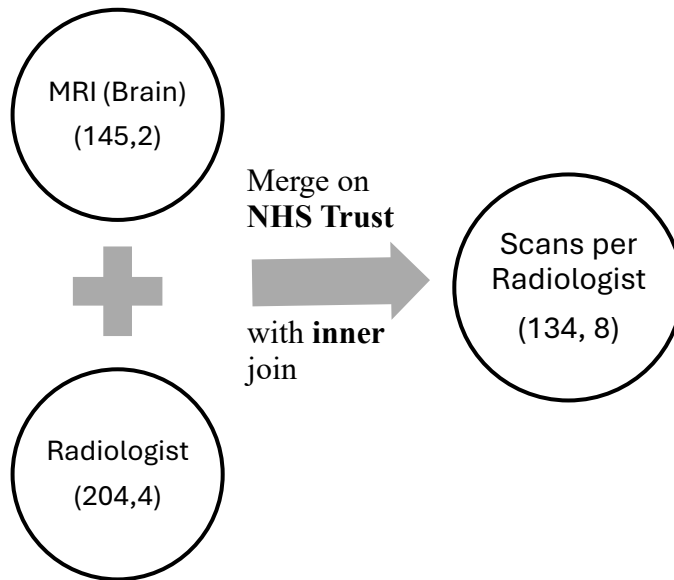


Figure 2 Step 1 – MRI (Brain) and Radiologist Merge

Before merging, the provider identifier columns (Provider name in the MRI table and Organisation name in the workforce table) were renamed to “NHS Trust” to ensure a consistent join key and the custom cleaning function (**clean_provider**) was applied to both datasets. An inner join was then performed on the “NHS Trust” column. The variables “Brain (MRI)” and “Estimated Radiologists” were converted to numeric format, and rows with missing values were removed. Three workload metrics were calculated:

1. $\text{Scans per Radiologist (Annual)} = \frac{\text{Brain (MRI)}}{\text{Estimated Radiologists}}$
2. $\text{Scans per Radiologist (Monthly)} = \frac{\text{Scans per Radiologist (Annual)}}{12}$
3. $\text{Scans per Radiologist (Daily)} = \frac{\text{Scans per Radiologist (Annual)}}{253 \text{ (number of working days in 2024/25)}}$

Equation 3 Scans per Radiologist

The final table is exported as “Scans per Radiologist” and added as a sheet in “**Provider_Brain_Scans_Analysis.xlsx**”.

Step 2 – MRI (Brain) and Waiting Time Merge

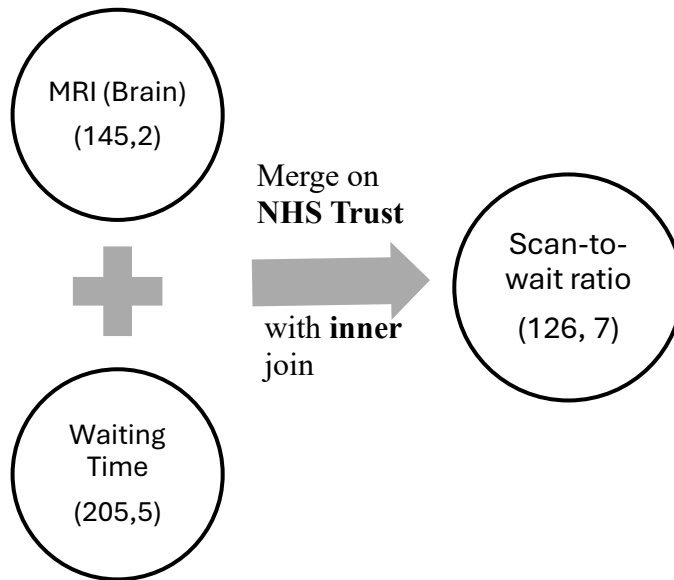


Figure 3 Step 2 – MRI (Brain) and Waiting Time Merge

The MRI (Brain) table was then merged with the Waiting Time table using the same name-cleaning function and merging steps as in Step 1. The dataset was then filtered to retain only providers with a non-zero count of patients waiting more than six weeks for a brain MRI scan. The backlog indicator was calculated as:

$$\text{Scan-to-6+Week-Wait Ratio} = \frac{\text{Brain (MRI)}}{\text{Brain MRI 6+Week Wait}}$$

Equation 4 Scan-to-6+week-wait Ratio

The final table is exported as “Scan-to-wait ratio” and added as a sheet in “**Provider_Brain_Scans_Analysis.xlsx**”.

Step 3 – Final Merge

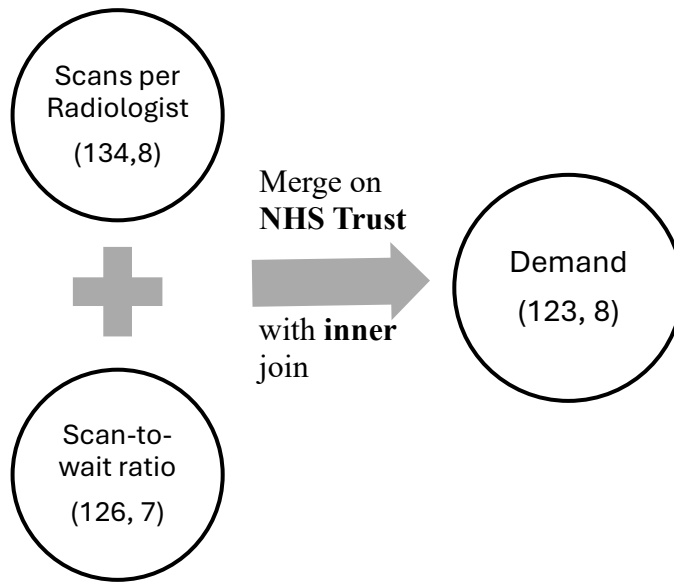


Figure 4 Final Merge

The output tables from Step 1 and Step 2 were merged on “NHS Trust”, retaining only the variables: Brain (MRI), Scans per Radiologist (Annual), Scan-to-6+Week-Wait Ratio, and Regional Team Name. Two ranking-based indicators were then computed: **Efficiency Rank** by ascending the rank of Scan-to-6+Week-Wait Ratio and **Workload Rank** by descending the rank of Scans per Radiologist (Annual). Then, these were summed to derive a **Demand Score**:

$$\text{Demand Score} = \text{Efficiency Rank} + \text{Workload Rank}$$

Equation 5 Demand Score

where lower demand score means higher pressure.

The final dataset is then exported as “Demand” table and added as a sheet in “**Provider_Brain_Scans_Analysis.xlsx**”.

3.2.2.3. Summary of Demand Table

The summary table of final dataset (Demand) is as follows:

Variable Name	Description	Type
NHS Trust	Name of NHS provider	Text
Regional Team Name	NHS England regional classification	Text
Brain (MRI)	Annual brain MRI scan volume (2024/25)	Count
Scans per Radiologist (Annual)	Annual brain scans per FTE radiologist	Count
Scan-to-6+Week-Wait Ratio	Ratio of annual brain MRI scans to patients waiting >6 weeks	Ratio
Efficiency Rank	Rank by scan-to-wait ratio (ascending, 1 = lowest ratio)	Rank
Workload Rank	Rank by scans per radiologist (descending, 1 = highest workload)	Rank
Demand Score	Sum of Efficiency Rank and Workload (lower = higher pressure)	Rank

Table 2 Summary table of final dataset (Demand)

3.3. Exploratory Data Analysis

This section explores the time series trend analysis and the provider-level analysis of England Diagnostics Imaging Activity of brain scans and other related data associated with market demand.

3.3.1. Time Series Trends Analysis

To understand how MRI brain scan demand of England has evolved over the past decade, we performed a trend analysis from April 2015 to March 2025 using “brain_scans_10yrs.xlsx” dataset. The associated processing code is provided in “Time_Series_Analysis.ipynb” within the code archive.

3.3.1.1. Yearly Trend Analysis

Brain MRI Growth

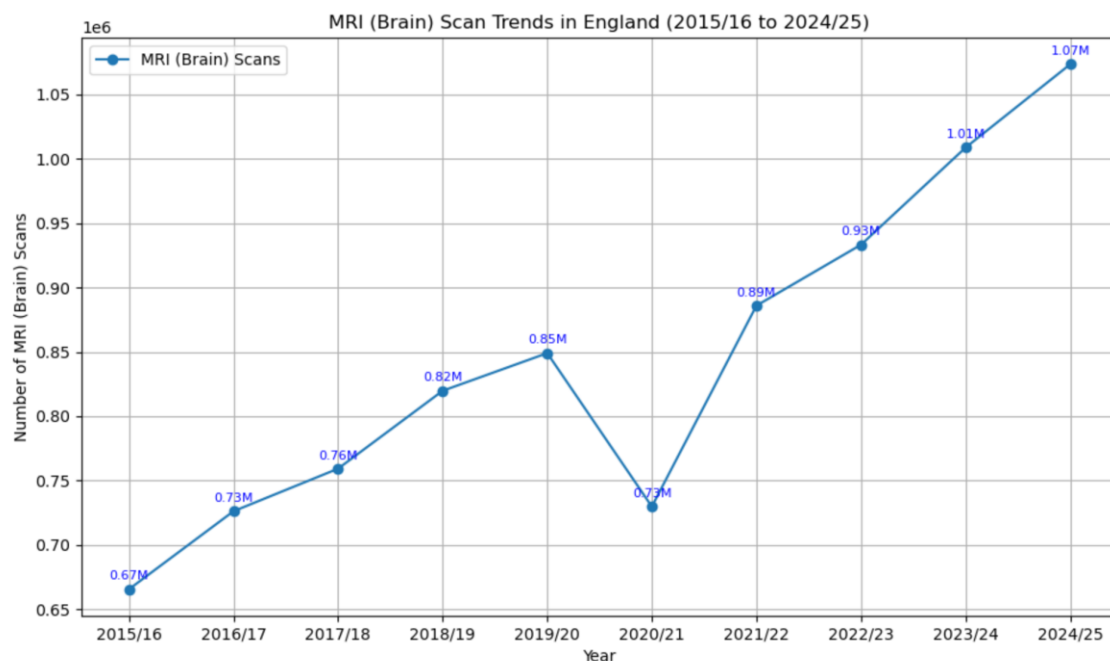


Figure 5 Brain MRI Growth

Between 2015/16 and 2024/25, annual MRI brain scans grew from 666,000 to 1.07 million, representing a 61.3% total growth with the compound annual growth rate (CAGR) of 4.9%. Activity dipped by 15% in 2020/21, likely due to service disruption during the COVID-19 pandemic. However, demand rapidly rebounded in 2021/22 and subsequently overshot pre-pandemic trends, reflecting a backlog recovery effect.

Population-adjusted trend

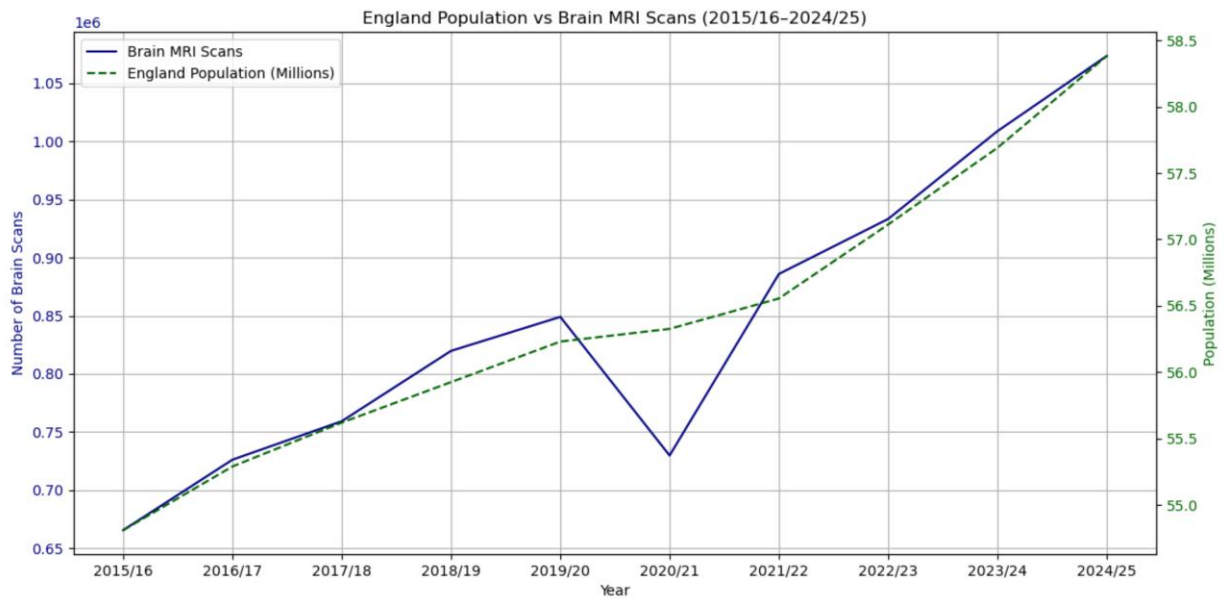


Figure 6 Population adjusted Trend

When comparing brain scan volumes against the growth of England's population, the results show that imaging demand has risen significantly faster than population. While the population grew by 6.5% over the same period, brain scan volumes increased by more than 60%. This indicates that growth in scan demand is being driven primarily by clinical need rather than demographics alone.

Comparison of the Growth Trend

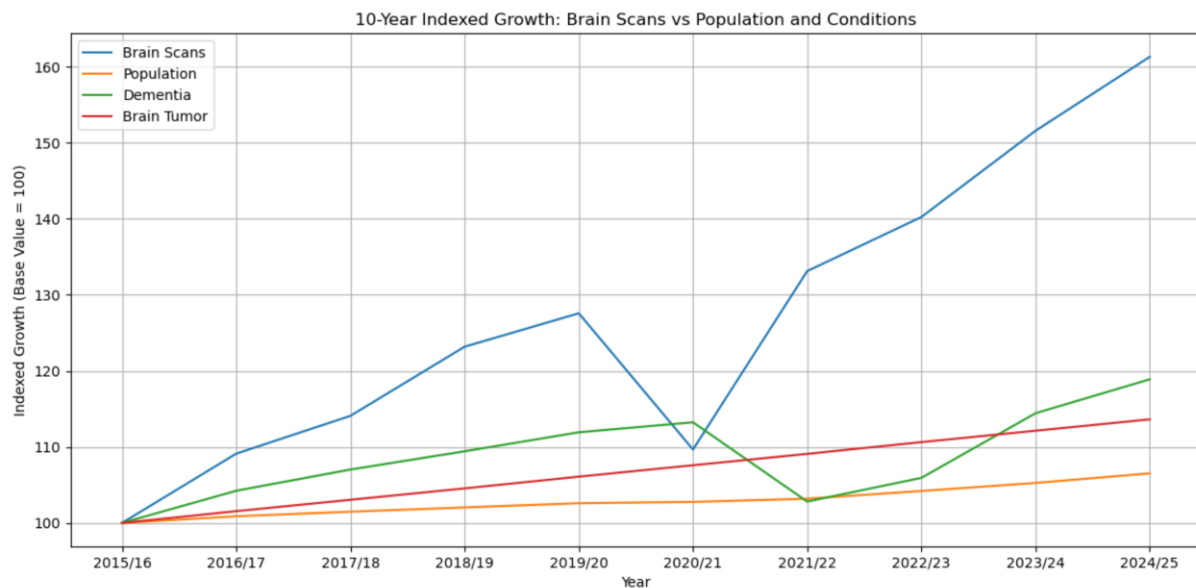


Figure 7 Comparison of Growth Trend

Indexed growth comparisons (base year = 100 in 2015/16) highlight the divergence between supply and disease prevalence. Brain scans rose by over 60%, while dementia diagnoses and brain tumour incidence increased more modestly (18–20%). This suggests that imaging demand is rising disproportionately relative to the underlying disease burden, likely reflecting changes in clinical pathways, earlier diagnostic emphasis, and increased access to MRI technology.

3.3.1.2. Monthly Trend Analysis

Monthly Trend of Total Brain Scans in England (10 Years)

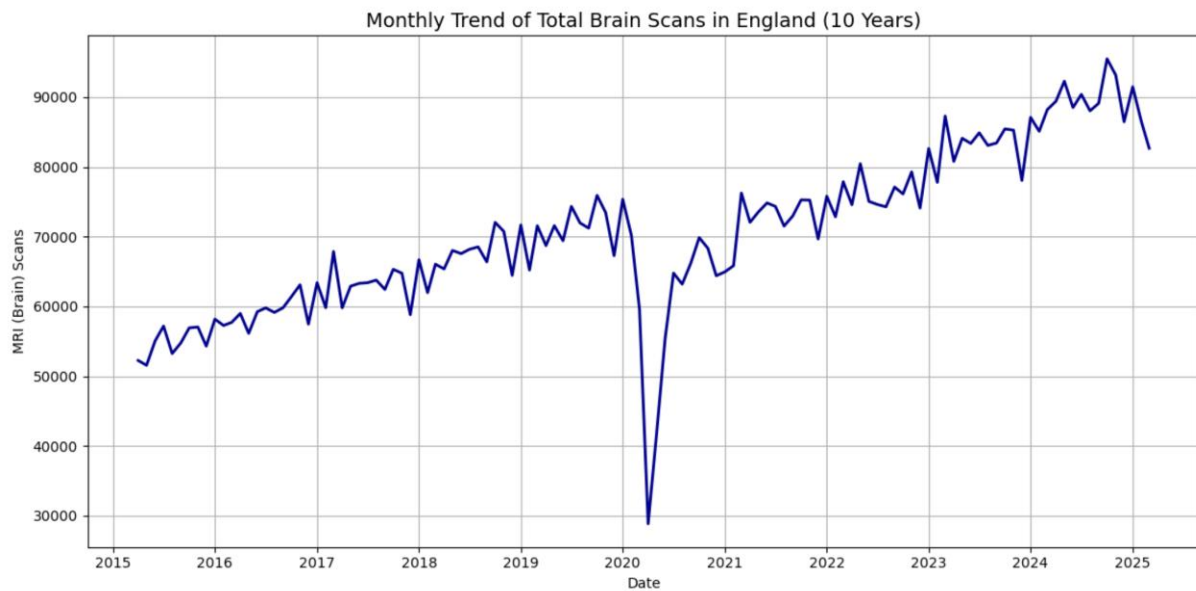


Figure 8 Monthly Trend of Total Brain Scans in England

The volume of brain MRI scans shows a clear upward trajectory from 2015 to 2025, with monthly activity rising from around 52,000 to over 90,000. A sharp decline is observed during early 2020, corresponding to the COVID-19 pandemic disruption, followed by a rapid rebound. From 2022 onwards, growth has accelerated, indicating sustained demand pressures.

MRI (Brain) Waiting List and % Delayed Over Time

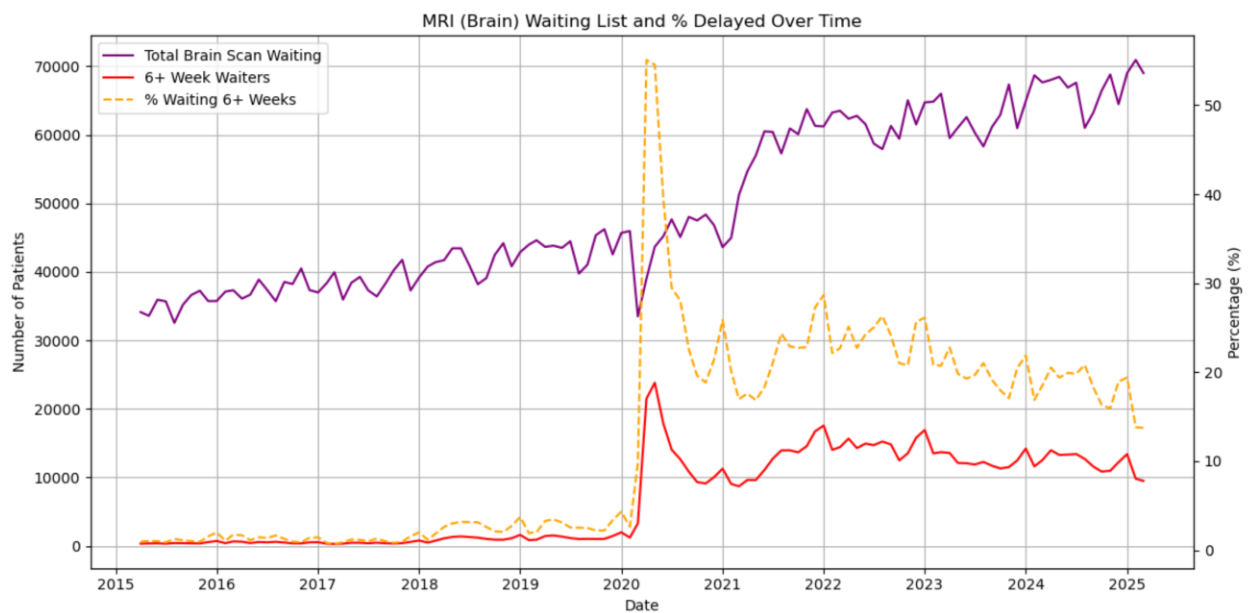


Figure 9 MRI (Brain) Diagnostic Waiting Time Activity

Waiting lists for brain MRI have grown steadily since 2015, with a clear surge in 2020 linked to deferred diagnostics during the COVID-19 pandemic. Although absolute volumes of patients waiting over six weeks remain high, the proportion has gradually declined since 2022. Still, by 2025 around 15–20% of patients experience delays beyond six weeks which is far exceeding the NHS Constitution’s threshold of $\leq 1\%$, and well above the interim 5% target set for May 2025 (NHS England, 2025). The persistent excess highlights a structural capacity gap and underscores the need for solutions such as QP-Brain to reduce diagnostic backlogs.

Radiologist Workforce Trends

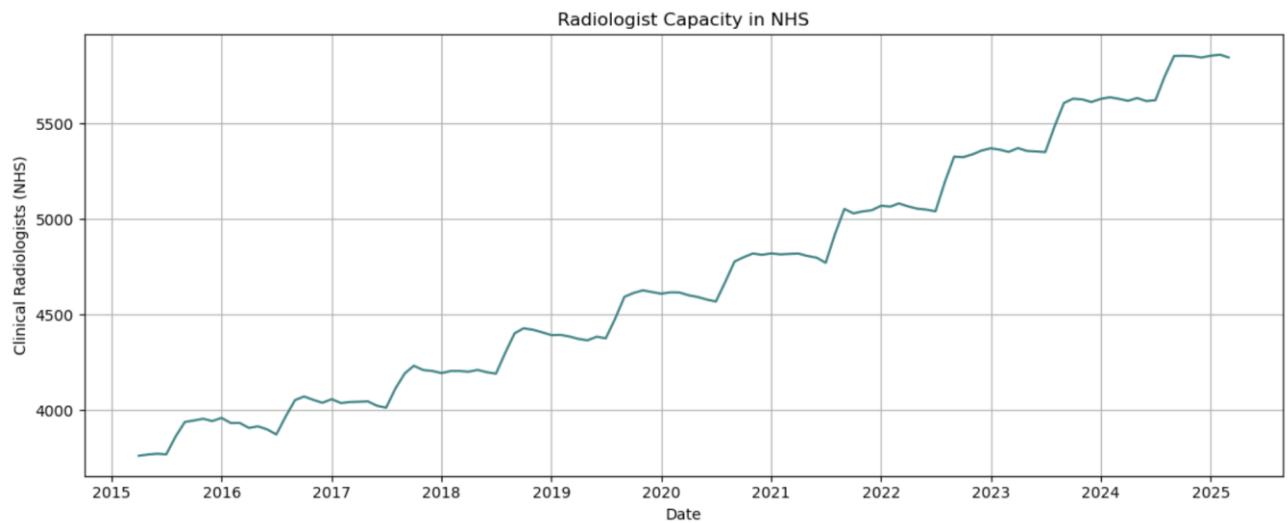


Figure 10 Radiologist Workforce Trends

The number of clinical radiologists in the NHS has increased steadily over the past decade, rising from around 3,750 FTEs in 2015 to nearly 5,900 FTEs by early 2025. While this represents significant workforce growth (approximately 48.7% over 10 years), it has not kept pace with rising imaging demand, which grew more than 60% in brain MRI alone during the same period. This widening imbalance between demand and workforce expansion underpins the persistent diagnostic backlog and highlights the structural capacity constraints facing NHS radiology services.

Brain MRI Demand vs Radiologist Capacity

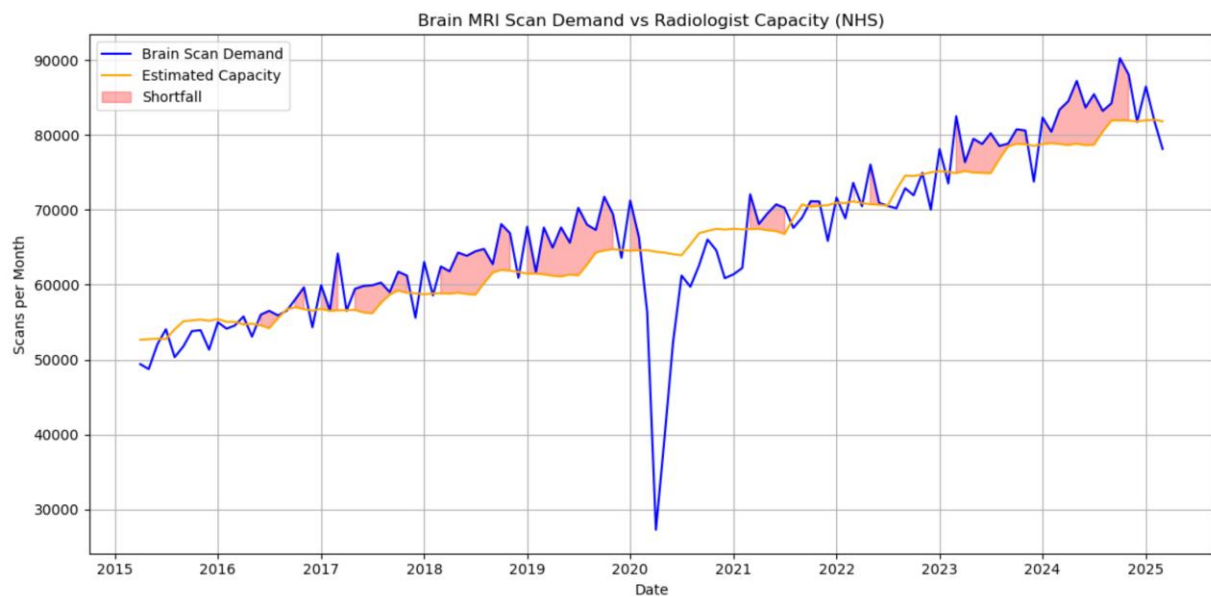


Figure 11 Radiologist Shortfall Trend

While plotting the radiologist shortfall trend, Radiologist capacity was estimated at 167 brain MRI scans per FTE per year (approximately 14scans/month), based on 2015–2025 averages. When comparing brain MRI demand against estimated NHS radiologist capacity, recurrent shortfalls are observed. These gaps widen post-2022 as demand outpaces workforce growth, highlighting systemic capacity pressures. The shaded shortfall area quantifies the excess demand unmet by current radiologist resources, underscoring the potential role for AI tools in addressing workforce bottlenecks.

Brain & CNS Tumour Incidence (UK)

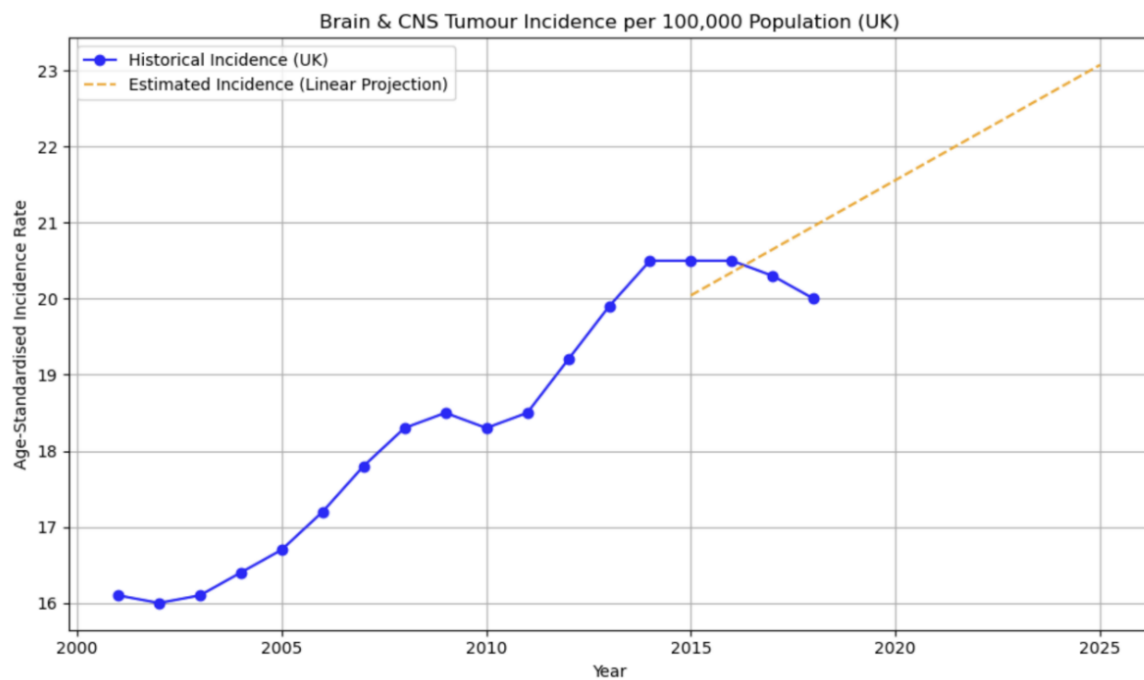


Figure 12 Brain & CNS Tumour Incidence (UK)

Historical incidence rates rose steadily between 2000 and 2015 and as the latest published data ends in 2019, a linear projection was applied for 2020–2025, yielding a modest continued increase. This underlines a persistent baseline demand driver for neuroimaging services.

Monthly Dementia Diagnoses in England

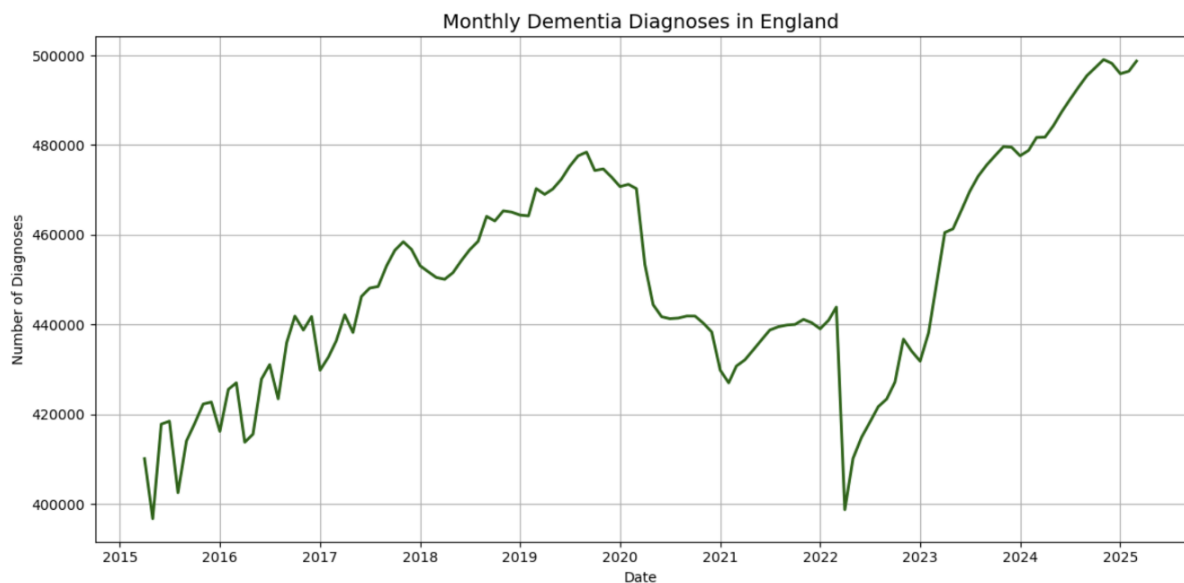


Figure 13 Monthly Dementia Diagnoses Trend

Dementia diagnoses rose steadily until 2019, but fell sharply in 2020–2022, reflecting COVID-19–related disruptions in primary care reporting rather than a genuine decline as NHS flagged “unprecedented changes in the behaviour of General Practices” during this period (NHS Digital, 2022). Diagnoses have since rebounded sharply, surpassing pre-pandemic levels by 2024. This reinforces dementia as a major and growing driver of brain MRI demand.

While this timeseries analysis highlights systemic pressures, understanding where these pressures are most acute requires moving to the provider-level analysis.

3.3.2. Provider-level analysis

To examine variation in diagnostic activity, workforce capacity, and waiting times across providers in 2024/25, we performed a trust-level analysis using the “**Provider_Brain_Scans_Analysis.xlsx**” dataset. The associated processing code is provided in “**Provider_level_analysis_Brain.ipynb**” within the code archive.

NHS Diagnostic Brain Scans Activity Analysis

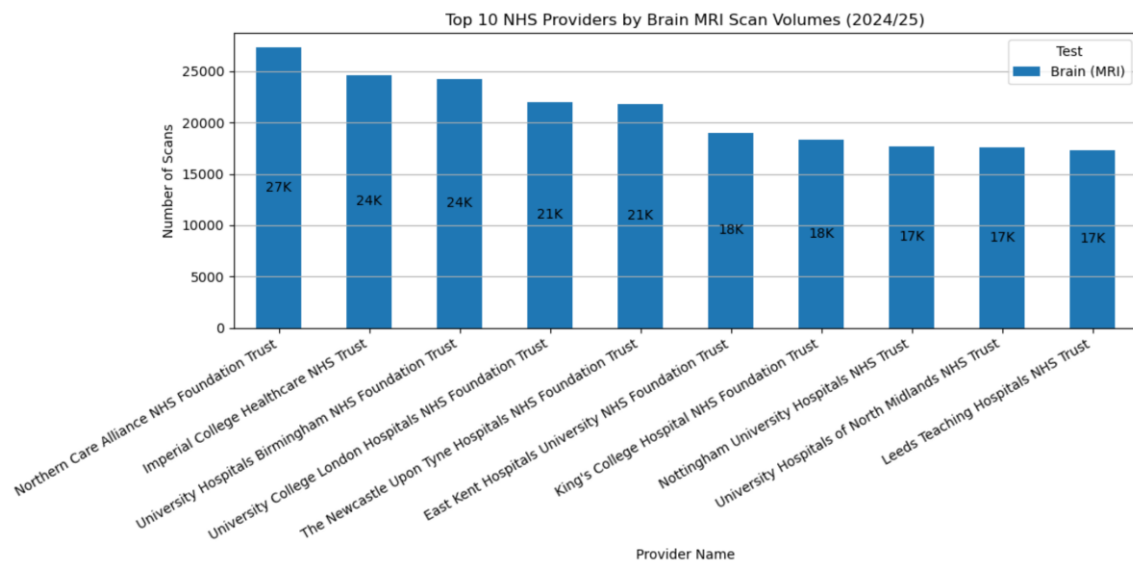


Figure 14 Top 10 NHS Providers by Brain MRI Scan Volumes (2024/25)

Brain MRI activity is highly concentrated, with the top 10 NHS providers each performing 17k–27k scans in 2024/25. Large teaching hospitals such as Northern Care Alliance, Imperial College, and Birmingham dominate volumes, reflecting their role as regional imaging hubs. This concentration highlights both centralised capacity and the pressure these large centres face in managing high demand.

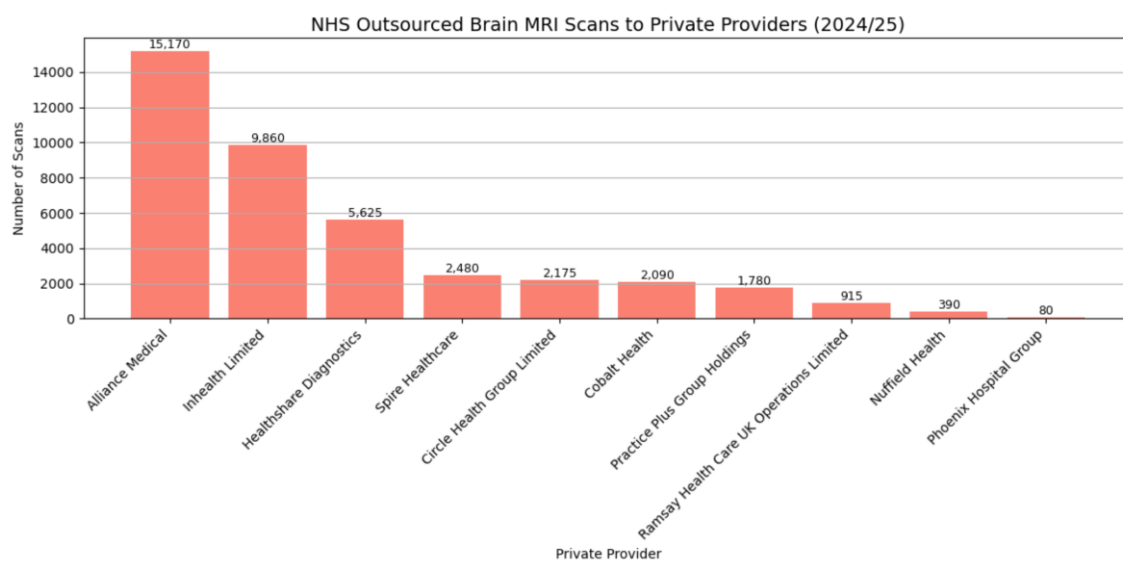


Figure 15 NHS Brain MRI Scans by Private Providers

According to Clinical Radiology Workforce Census by (The Royal College of Radiologists, 2025), NHS spent £216M on outsourcing to control the imaging reporting shortfalls in 2024. Among them, over 40,000 NHS-funded Brain MRI scans were outsourced to 10 private providers. This represents around 4% of all brain MRI activity, with 96% still delivered by the NHS. Reliance on outsourcing highlights NHS capacity shortfalls and demonstrate a clear opportunity for QP-Brain to help reduce diagnostic backlogs.

NHS Diagnostics Waiting Time Activity Analysis

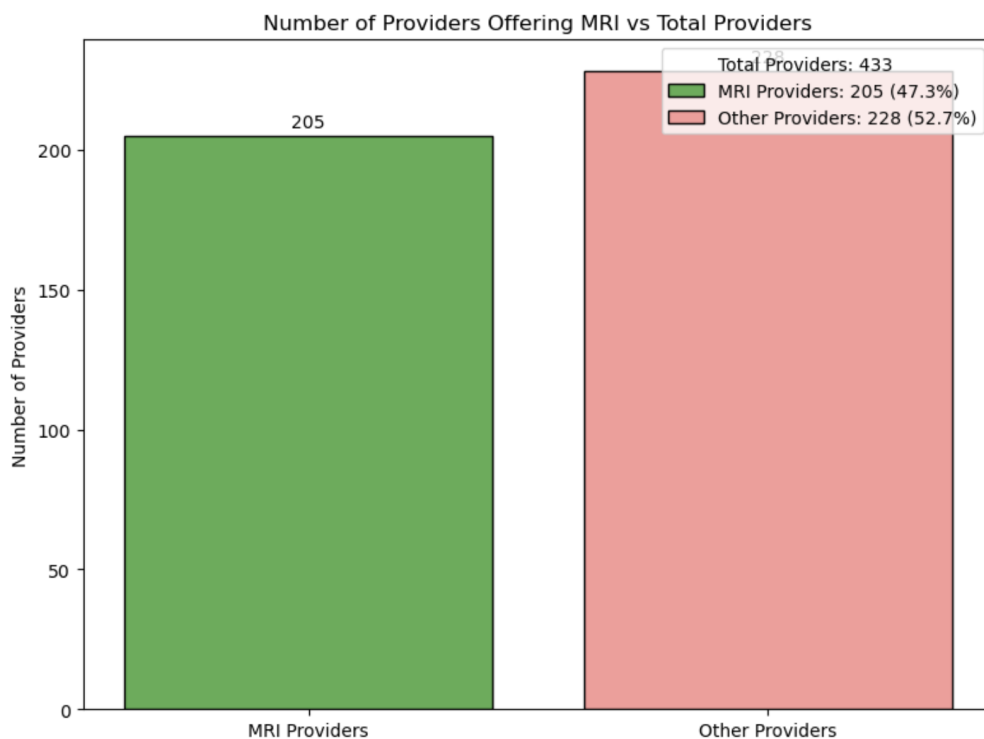


Figure 16 Number of Providers Offering MRI VS Total Provider in Waiting List

As of March 2025, the NHS Diagnostic Waiting List contained 433 providers, including NHS Trusts, private providers, and independent branches. Of these, only 205 (47.3%) offered MRI services. This indicates that MRI capacity is concentrated in fewer than half of all reporting providers, reinforcing the structural challenges in meeting rising neuroimaging demand.

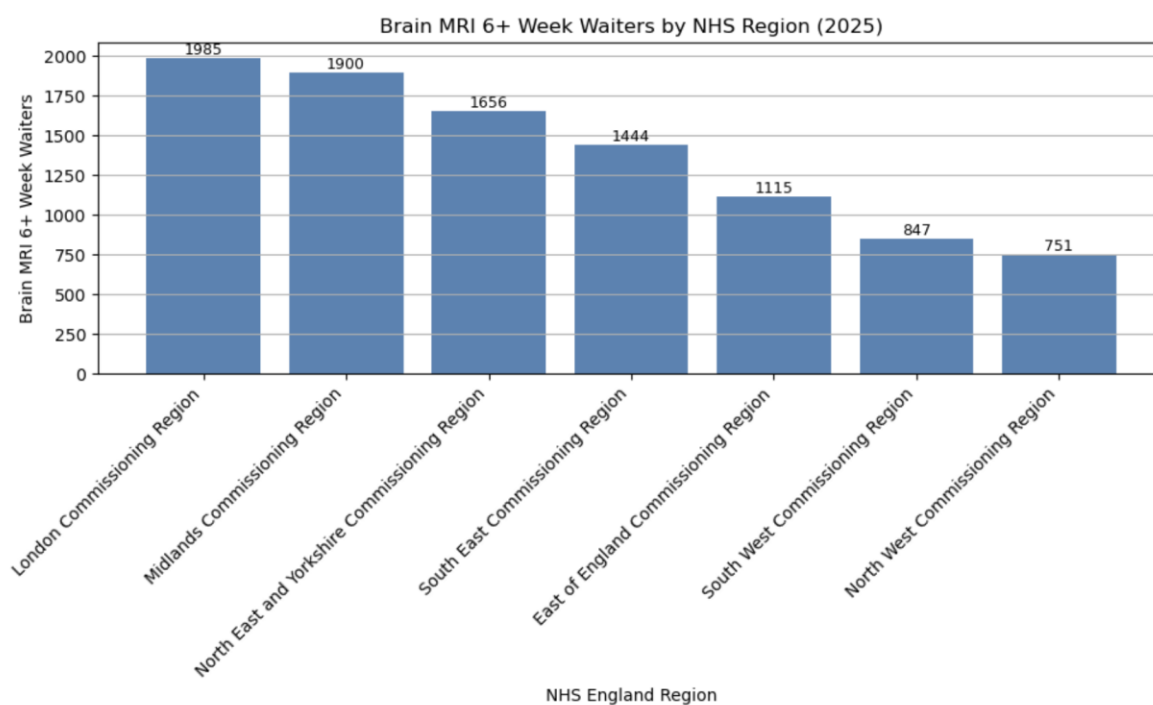


Figure 17 Brain MRI Delay by England Region

Long waits vary significantly by region. London and Midlands had the highest backlogs in March 2025 (1,985 and 1,900 patients respectively), while the North West had the lowest (751). This imbalance indicates uneven diagnostic capacity across NHS England regions, with northern and midland regions experiencing greater strain.

Estimated Brain MRI Backlog and Long Waits by NHS Provider (Interactive)

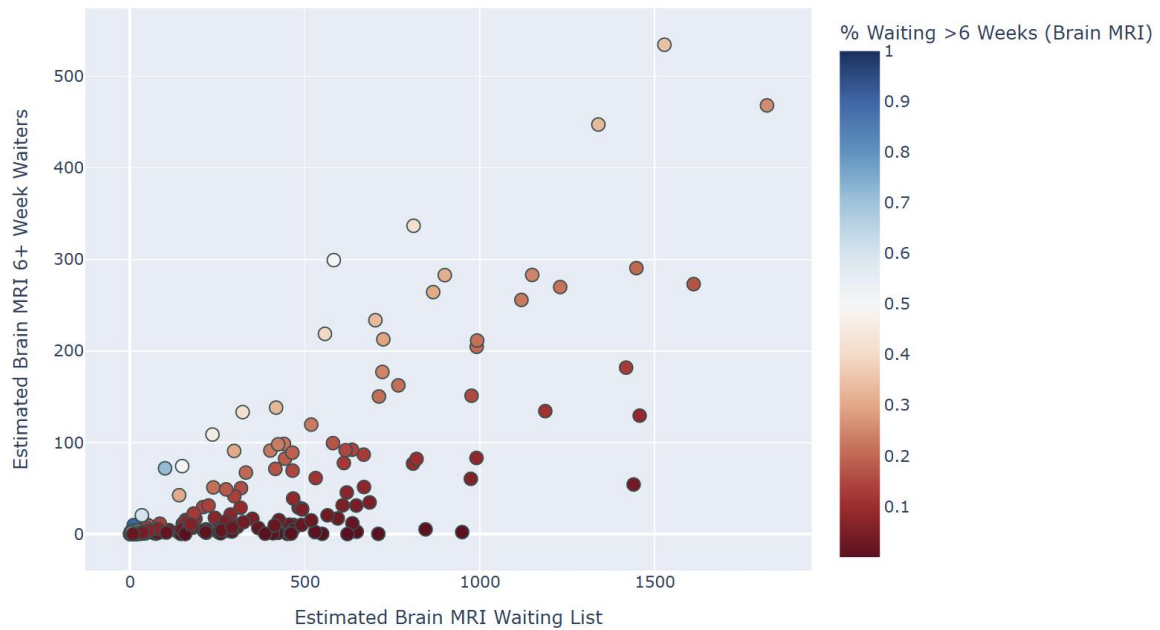


Figure 18 Scatter Plot of Waiting List VS Delayed Cases

Note: An interactive version of this plot is provided as 'brain_mri_waiting_scatter.html' in the code archive under "Interactive Plots" folder, where provider names can be viewed by hovering over each point.

At provider level, a scatter of results shows that both backlog size and percentage delayed contribute to access inequality. Some Trusts manage large waiting lists with relatively fewer long delays, while others struggle even with smaller backlogs. This heterogeneity highlights where targeted interventions such as AI support could deliver the biggest efficiency gains.

NHS Workforce Radiologists Analysis

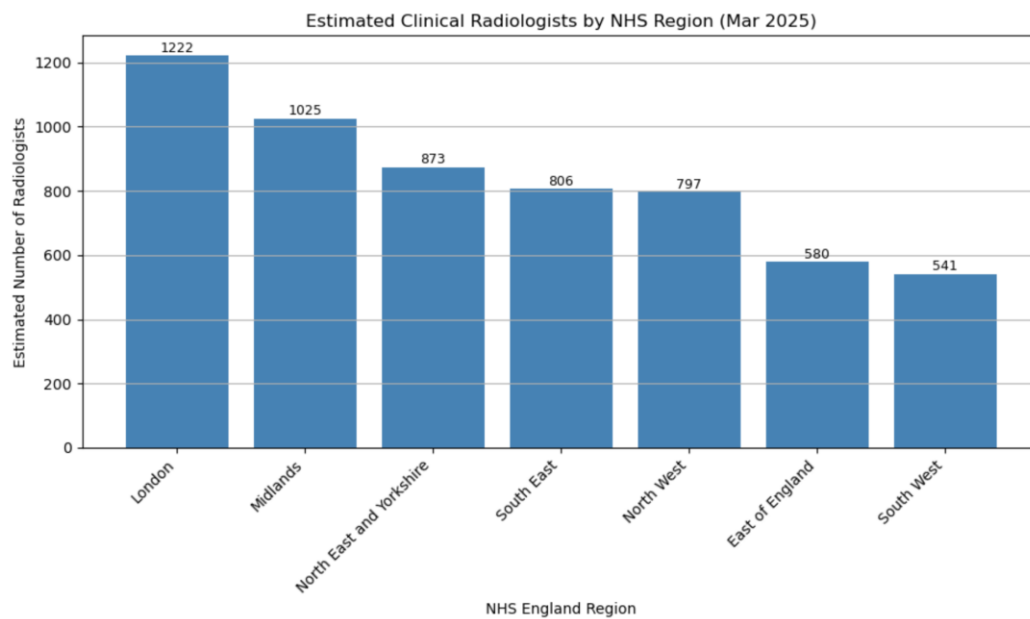


Figure 19 Radiologist by England Region

The London region has the largest radiology workforce (1,222 FTEs), while the South West has the fewest (541 FTEs). Regional variation mirrors broader NHS workforce inequalities, contributing directly to differing scan capacity and waiting list pressures.

Radiologist Workload Analysis (Brain Scans per Radiologist)

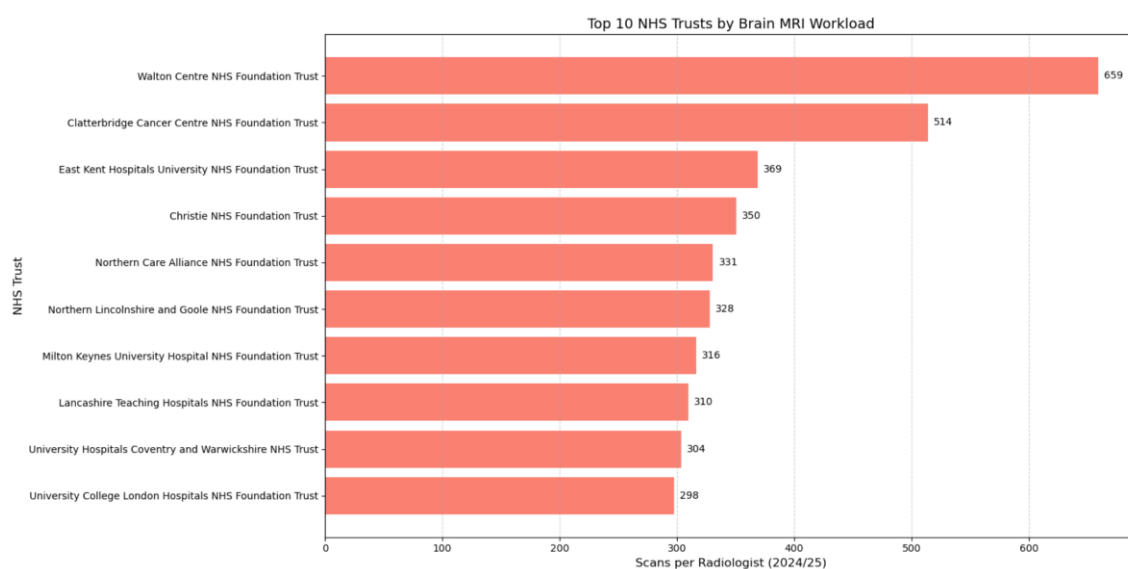


Figure 20 Top 10 NHS Trusts by Brain MRI Workload

Workload pressures at Trust level are stark. The Walton Centre and Clatterbridge Cancer Centre each exceed 500 scans per radiologist annually, more than double the national average. Such outliers reveal acute staffing mismatches that could accelerate burnout and make AI adoption more urgent in specialist centres.

Efficiency Analysis (Scans to 6+Week Waiting Ratio)

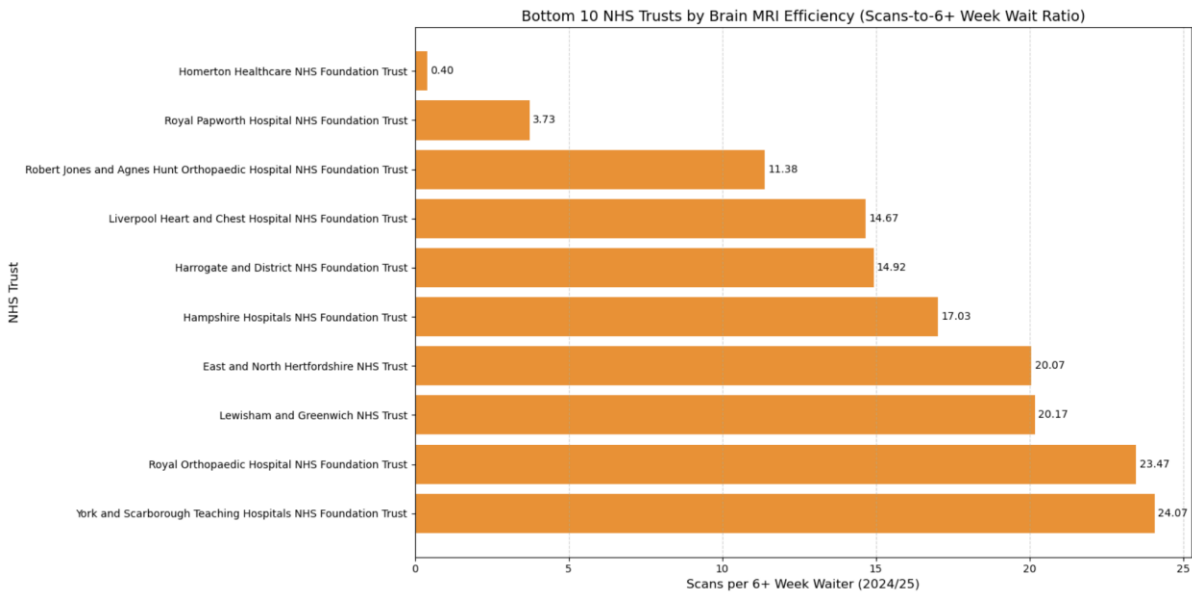


Figure 21 NHS Trusts with Lowest Efficiency

Trusts with the lowest scans-to-wait ratios include Homerton Healthcare (0.4) and Royal Papworth (3.73), signalling poor throughput relative to waiting list size. These Trusts face structural bottlenecks where improving diagnostic flow could yield substantial patient benefit.

Demand Analysis

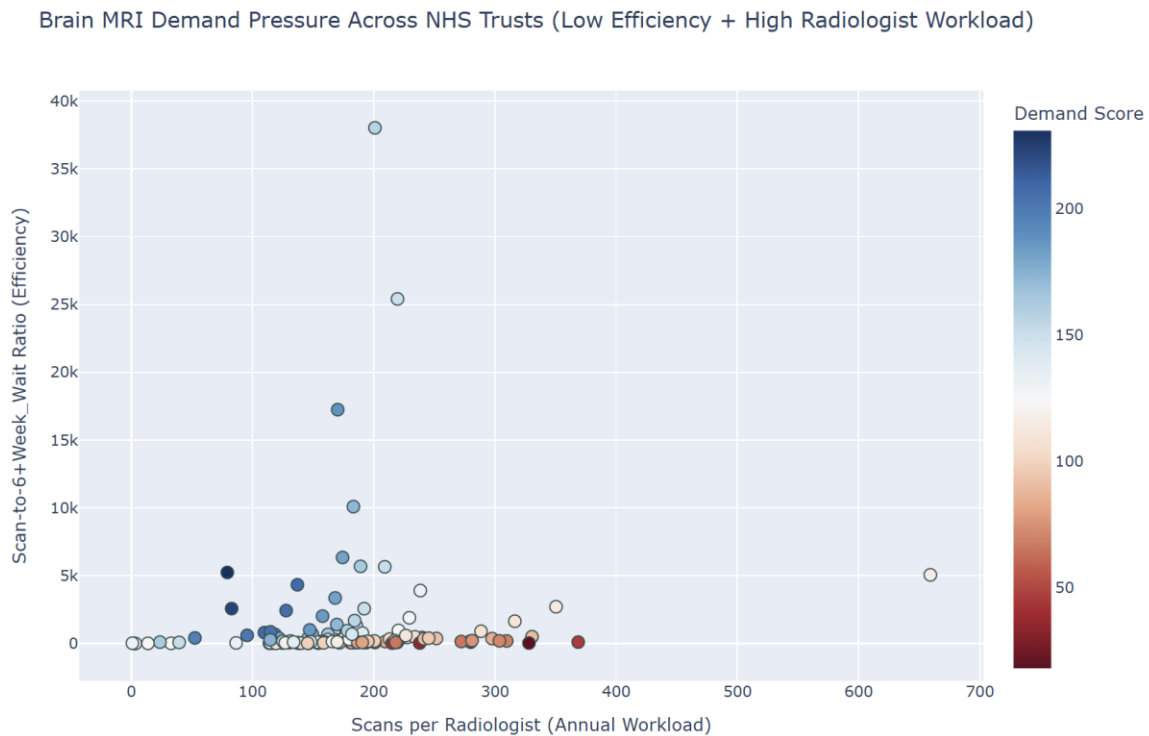


Figure 22 Scatter Plot of Demand Pressure

Note: An interactive version of this plot is provided as ‘demand_pressure_plot.html’ in the code archive under “Interactive Plots” folder, where provider names can be viewed by hovering over each point.

The combined scatterplot shows that demand pressures are not uniform. Trusts with both high workloads and low efficiency cluster as high-risk outliers, reflected in higher demand pressure (lower demand score). These hotspots represent the clearest opportunities for AI-enabled productivity gains like QP-Brain, where both backlog reduction and workforce support are most needed.

These provider-level insights highlight the structural imbalances in diagnostic activity, workforce capacity, and waiting times across NHS Trusts, providing the foundation for Section 4.

4. Analytical Framework and Methodology

This section outlines the analytical approach taken to address the project objectives. It is structured into three parts: quantifying England neuroimaging market, forecasting adoption scenarios, and evaluating pricing and market-entry strategies.

4.1. Quantifying England Neuroimaging Market (Yin Min Aung – 240915364)

This section discusses different methodologies, justification on methodology choice and how it was applied in neuroimaging market sizing.

4.1.1. Methodology

Market sizing is a critical step in quantifying demand and evaluating adoption scenarios for neuroimaging diagnostic. In market estimation, three principal modelling frameworks are commonly applied: Top-Down, Bottom-Up, and Hybrid approaches, each with distinct advantages depending on data availability and market maturity (Statista, 2025).

Top-Down Approach

The top-down approach begins with aggregated market figures (e.g., global or regional health expenditure, diagnostic imaging market size) which are then systematically broken down to country or segment levels using macroeconomic and industry-specific drivers such as GDP, healthcare spending, population demographics, or disease prevalence. This method ensures scalability and consistency, particularly when detailed local datasets are limited or fragmented (Statista, 2025).

Bottom-Up Approach

The bottom-up approach builds market size estimates from granular, localized inputs such as hospital activity data, number of MRI scanners installed, utilization rates, and clinical volumes. Then, these data are aggregated upwards to provide national-level estimates. This method captures local market nuances but is heavily dependent on the completeness and reliability of local datasets (Statista, 2025).

Hybrid Approach

A hybrid methodology integrates top-down scalability with bottom-up granularity. Broad market trends are established using high-level data, while local insights (e.g., Trust-level scan volumes or workforce data) refine and validate the estimates. This reconciles global consistency with local accuracy and is often applied in dynamic or fragmented healthcare markets but requires additional reconciliation effort and can introduce duplication risk if not carefully managed between the two data streams (Statista, 2025).

Justification for Methodology Choice

For this study, the Top-Down approach was chosen as the primary methodology because aggregate and system-level data, such as total MRI scans, brain MRI scans, and radiology workforce figures, are systematically collected and publicly available through NHS Digital and related sources. In contrast, detailed bottom-up data such as private hospital-level scan volumes, workforce capacity, or disease-specific diagnostic utilisation is more limited and therefore can lead to risk of underestimation. A hybrid or bottom-up approach, while valuable in other contexts, was therefore not feasible in this case due to the lack of sufficiently granular data.

4.1.2. Brain MRI Market Size Estimation (England, 2024/25)

Market sizing for England’s neurodiagnostic imaging was undertaken using a top-down approach, starting with national aggregates and progressively refining estimates for MRI brain scans. This method was adopted due to the availability of consistent national-level activity statistics (NHS England, 2025) and the absence of detailed private or disease-segmented breakdowns. The associated processing code is provided in “Brain_scan_market_sizing.ipynb” within the code archive.

Core MRI Brain Scan Volumes

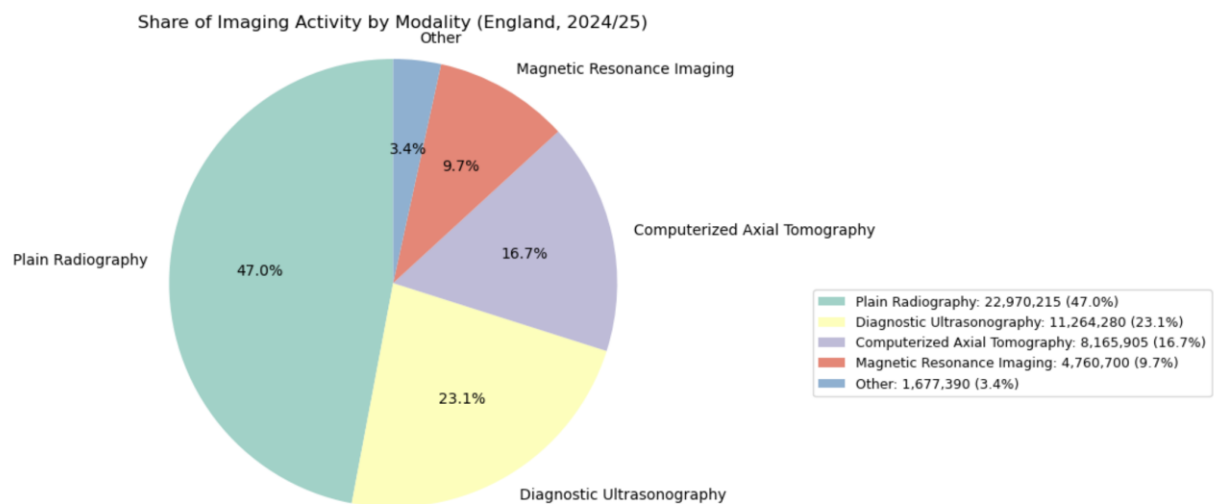


Figure 23 Diagnostic Imaging Activity by Modality (NHS England, 2025)

According to the NHS England Diagnostic Imaging Dataset, a total of 48.8 million scans were performed across all modalities in 2024/25. Of these, 4.76 million were MRI scans, representing 9.7% of all imaging.

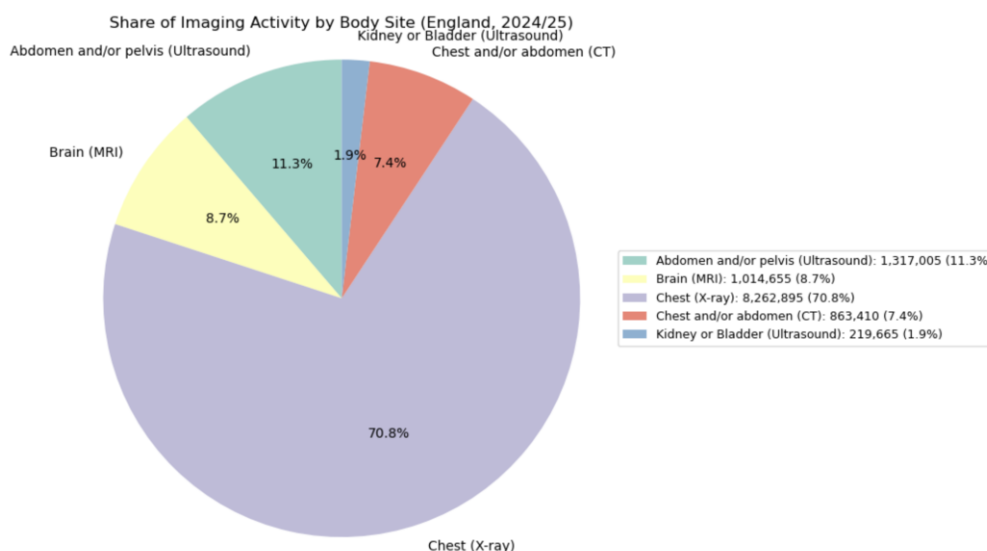


Figure 24 Diagnostic Imaging Activity by Body Site (NHS England, 2025)

The only body site-specific series consistently available in the NHS DID was Brain (MRI), which recorded 1,014,655 scans in 2024/25. This corresponds to 8.7% of all anatomical sites. It is important to note that analysis is restricted to MRI (Brain) because the NHS body site table reports count only for this category. Equivalent breakdowns for other neurological imaging (e.g., CT Brain) are not published. As a result, the estimated market size should be interpreted as the minimum quantifiable demand for neurodiagnostic MRI, recognising that broader neurological imaging activity is excluded.

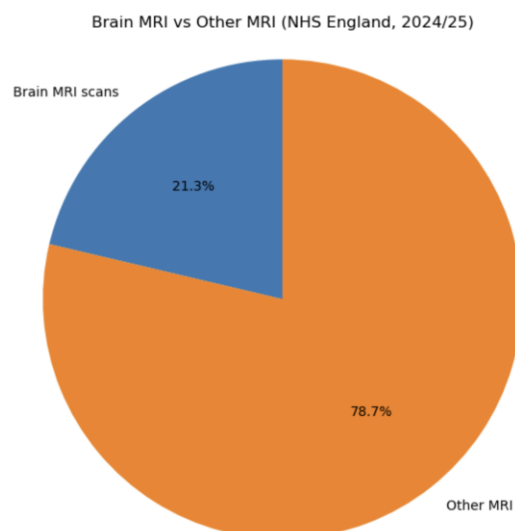


Figure 25 Brain MRI % of all MRIs

The brain MRI scans account for 21.3% of NHS total MRI activity in 2024/25.

Private Sector Activity

While the NHS accounts for the majority of diagnostic imaging, private provision plays a growing role. (LaingBuisson, 2025) estimates that 63,395,000 total scans were done in the UK in 2024, of which 59,920,000 are NHS and 3,475,000 are private scans, representing 5.8% of NHS activity. Applying this ratio to NHS brain MRI volumes suggests approximately 58,850 private brain MRIs, giving a combined total of 1,073,505 brain MRIs in England for 2024/25. Further analysis of the Private Sector Activity will be mentioned in Section 4.3.

Per Capita Demand

With an estimated population of 58.38 million in mid-2024 (Office for National Statistics, 2025), the per-capita utilisation of MRI was 86.3 scans per 1,000 people, of which 18.4 were brain MRIs. For benchmarking against international comparators, the 2017 data from (The Commonwealth Fund, 2020) was used to plot the comparison bar chart as shown in Figure below.

MRI Exams per 1,000 Population by Country (2017)

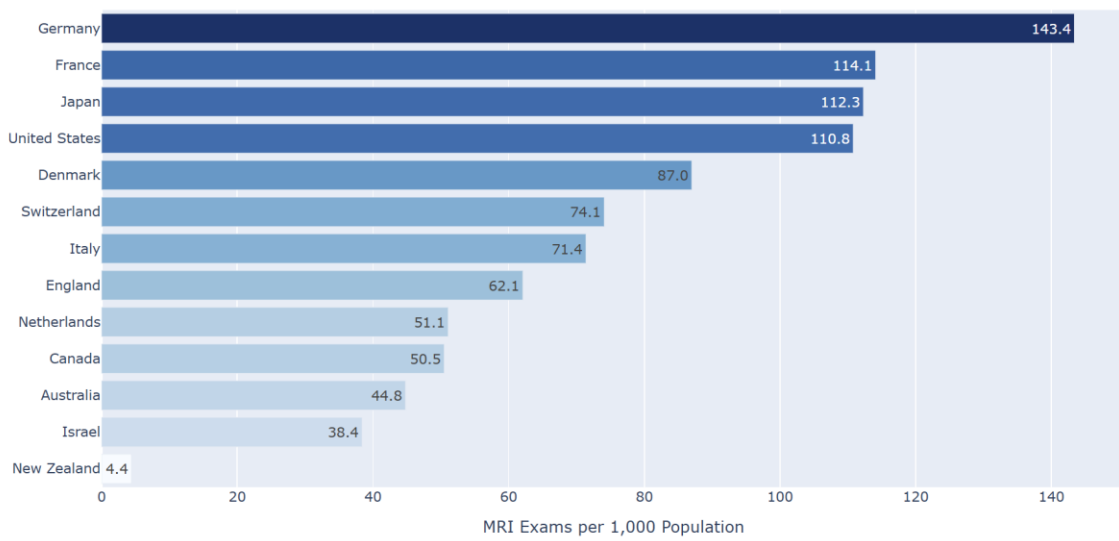


Figure 26 MRI Exams per 1,000 Population by Country (2017)

The figure shows that England performs substantially fewer MRI examinations per capita than leading nations as Germany conducts over 140 exams per 1,000 people, France and Japan exceed 110 per 1,000, while the US records 111 per 1,000 in 2017 (The Commonwealth Fund, 2020). England's rate of 86 per 1,000 in 2024 therefore remains well below international leaders, even after the growth over nearly decade.

Diagnostic Waiting Time Activity

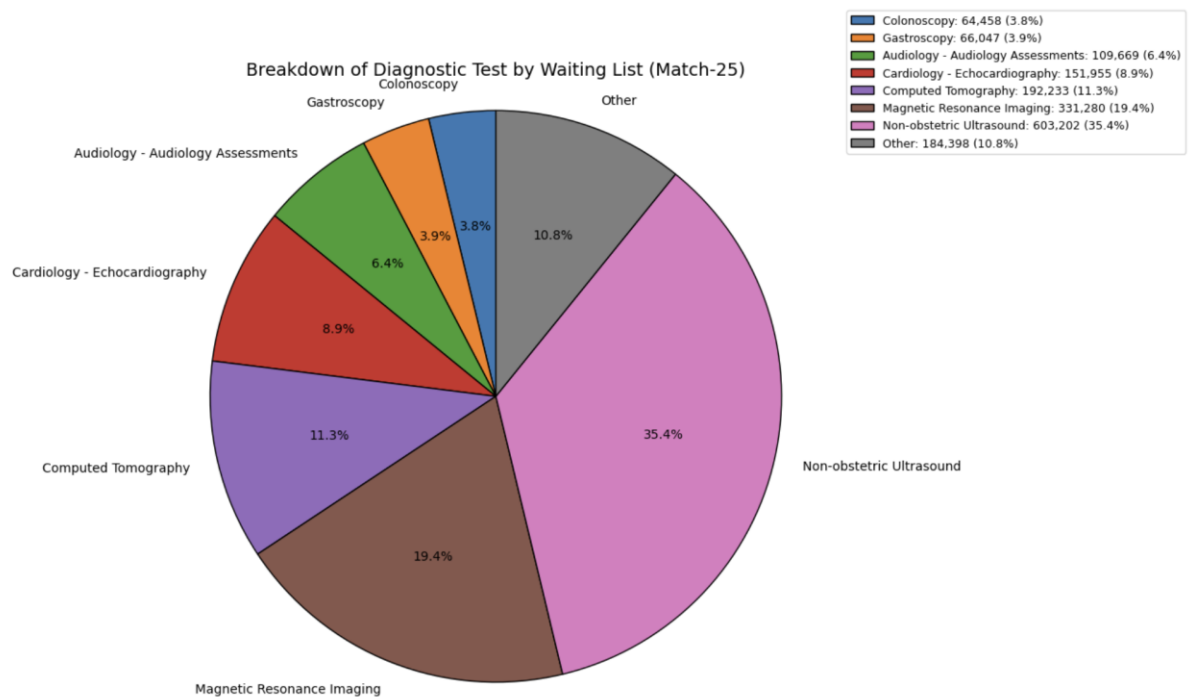


Figure 27 Breakdown of Diagnostic Test by Waiting List (NHS England, 2025)

According to NHS Diagnostic Waiting Time Activity Data (March 2025), there were over 1.7 million patients awaiting their diagnostic test, of whom around 331,300 (19.4%) were waiting for MRI test.

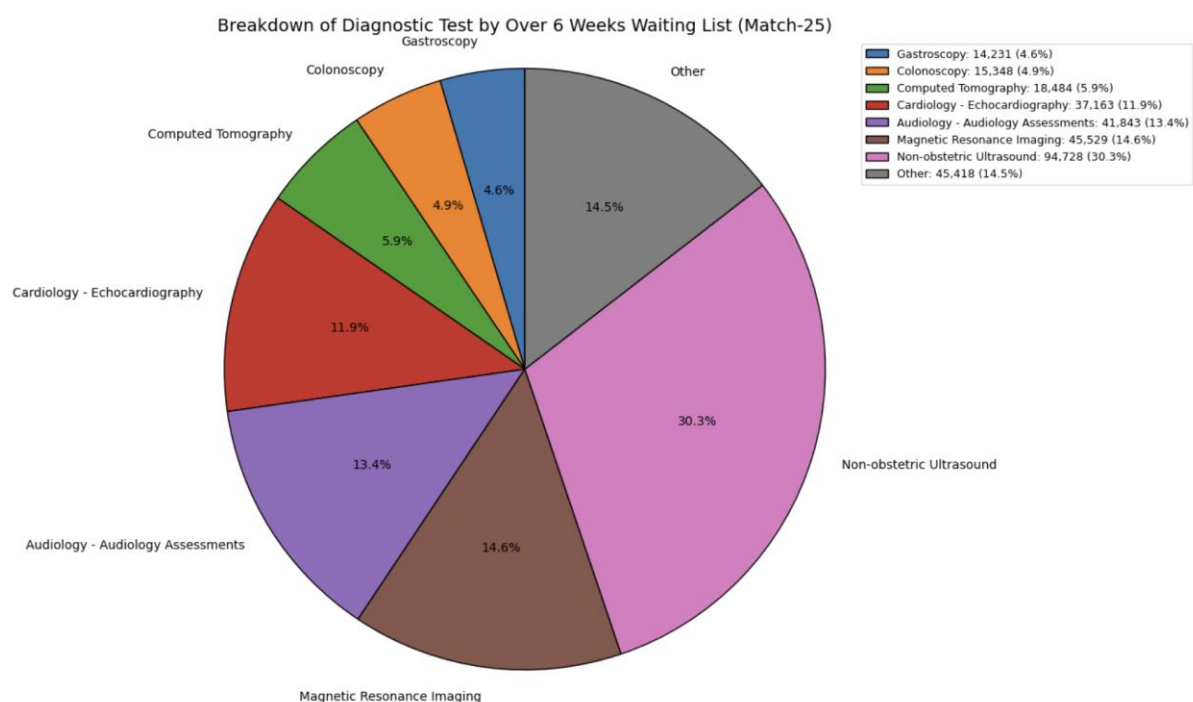


Figure 28 Breakdown of Diagnostic Test by Over 6 weeks waiting list (NHS England, 2025)

A key performance measure is whether patients receive their scan within six weeks of referral, as mandated by the NHS Constitution (NHS England, 2025). In March 2025, approximately 312,700 patients (18.4%) were waiting longer than six weeks. Within this group, around 45,500 patients (14.6%) were awaiting MRI scans. Applying the brain MRI share observed in DID activity data (21.3%) to waiting list figures suggests that approximately 70,563 patients were waiting for a brain MRI, of whom 9,698 (13.7%) had been delayed beyond six weeks. In other words, roughly 14 in every 100 brain MRI scans fail to meet the NHS Constitution's six-week target.

Performance against the 6-week wait target has plummeted since COVID-19

% of people receiving their diagnostic test within the six-week target

January 2010 - November 2024

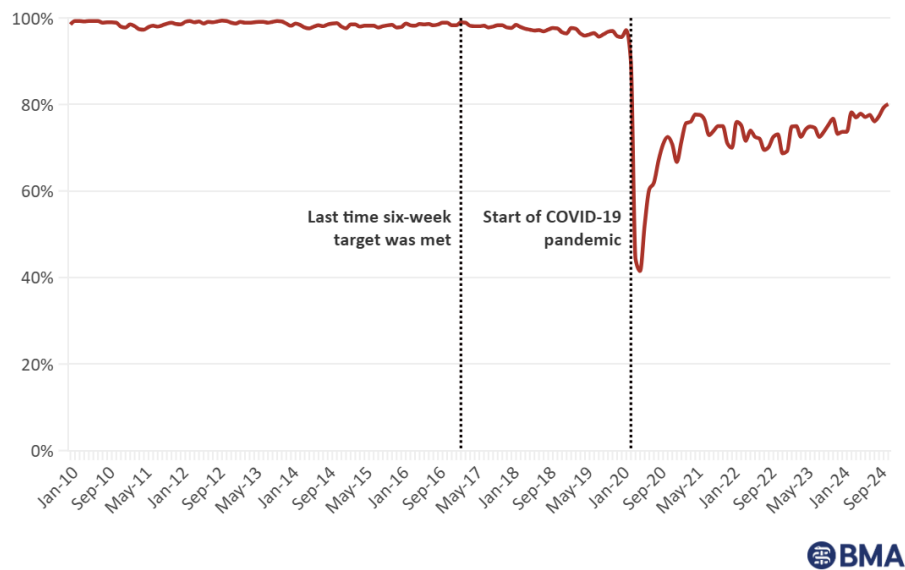


Figure 29 Performance against the NHS six-week diagnostic standard, January 2010–November 2024 (British Medical Association, 2025)

Moreover, analysis by the British Medical Association (2025) shows that the diagnostic waiting list has more than doubled over the past two decades, rising from 0.8 million in January 2006 to 1.6 million in November 2024. Importantly, the six-week standard for diagnostic tests has not been met nationally since February 2017, well before the onset of the COVID-19 pandemic (British Medical Association, 2025).

NHS Workforce Capacity

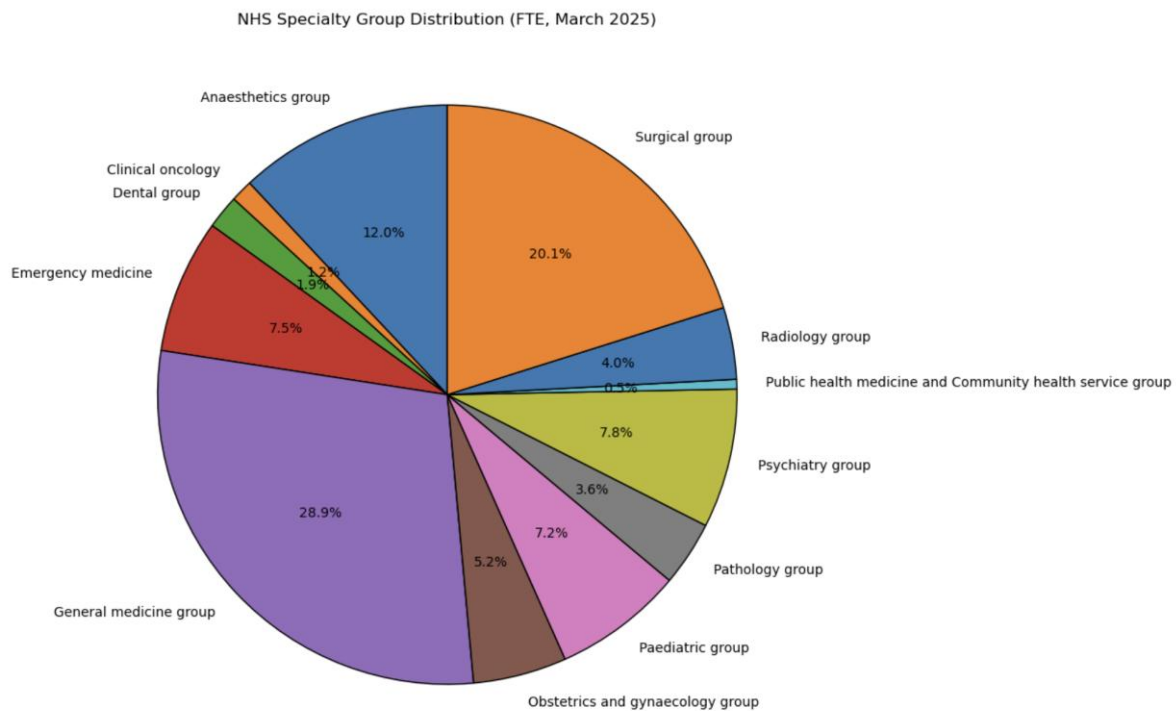


Figure 30 NHS Specialty Group Distribution (NHS England, 2025)

According to NHS workforce statistics (March 2025), there were 147,931 Hospital and Community Health Service (HCHS) doctor FTEs, of which 5,935 (4%) were within the radiology group. Among these, 5,845 were clinical radiology FTEs, representing 3.95% of the total medical workforce. Based on NHS Diagnostic Imaging Dataset (DID) activity, approximately 48.8 million scans were performed in 2024/25. This translates to an average reporting workload of 8,356 scans per radiologist annually (696 per month, 22 per day). When focusing on brain MRI activity specifically (around 1.1 million in 2024/25), the average distribution equates to 174 brain MRIs per radiologist annually (14.5 monthly, 0.7 daily).

Disease Prevalence Benchmarks

As of 31 March 2025, NHS England recorded 498,729 patients diagnosed with dementia, an increase of 2,258 patients since 28 February 2025 and 17,022 since 31 March 2024 (NHS Digital, 2025) and an estimated 13,300 new brain tumour cases annually (based on an incidence rate of 22.77 per 100,000 population) (Cancer Research UK, 2024). While not every patient undergoes MRI annually, these conditions alone account for a substantial proportion of neuroimaging demand.

4.1.2.1. Top-Down Market Sizing Flowchart for Brain MRI

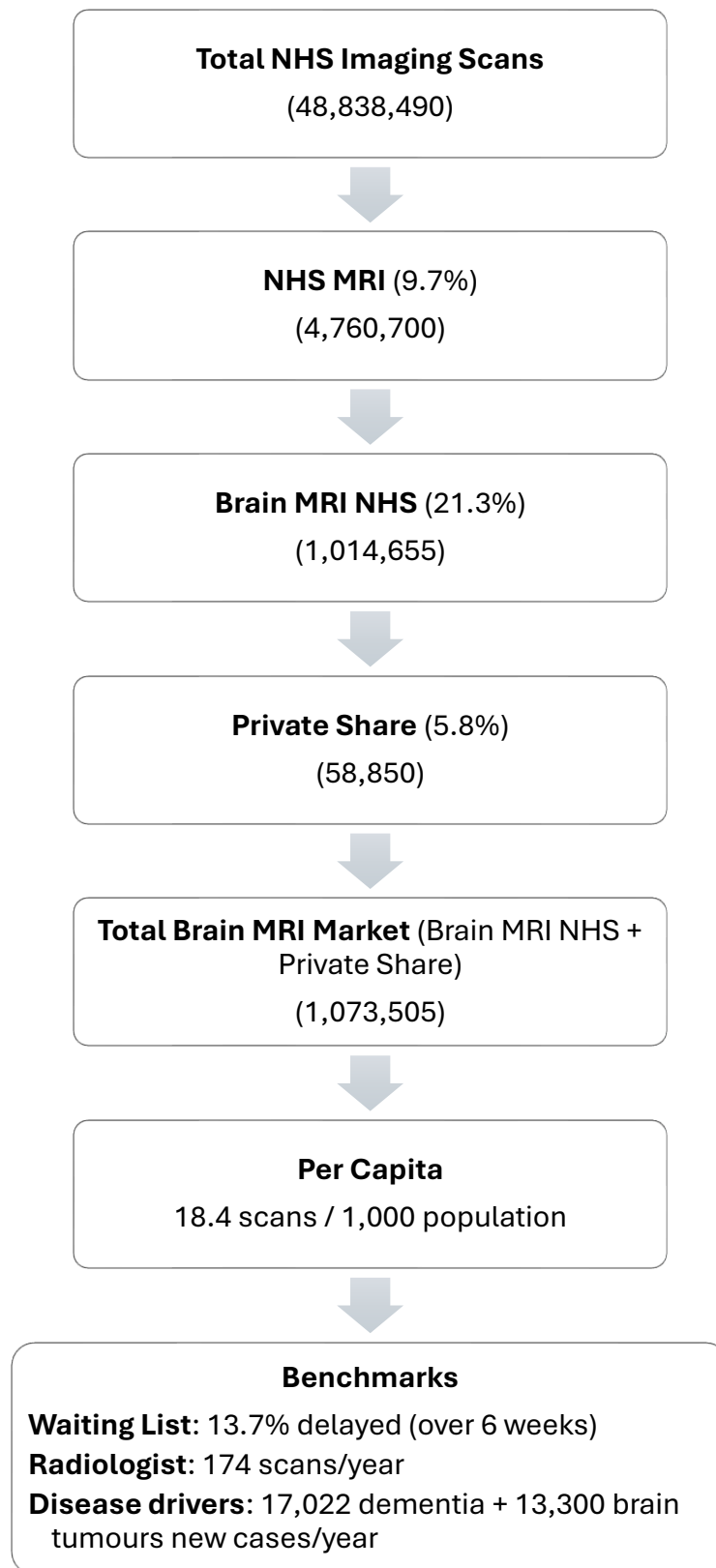


Figure 31 Top-Down Market Sizing Flowchart for Brain MRI in England (2024/25)

4.1.2.2. UK Brain MRI Market (Indicative Estimate)

Although this study is restricted to England due to data availability, an indicative UK-wide estimate can be derived by scaling results according to population share. Using the England market size (2024/25), UK Brain MRI Market can be calculated as follows:

$$UK\ Brain\ MRI\ Scans \approx \frac{England\ Brain\ MRI\ Scans}{England\ Population\ (2024)} \times UK\ Population\ (2024)$$

Equation 6 UK Brain MRI Scans

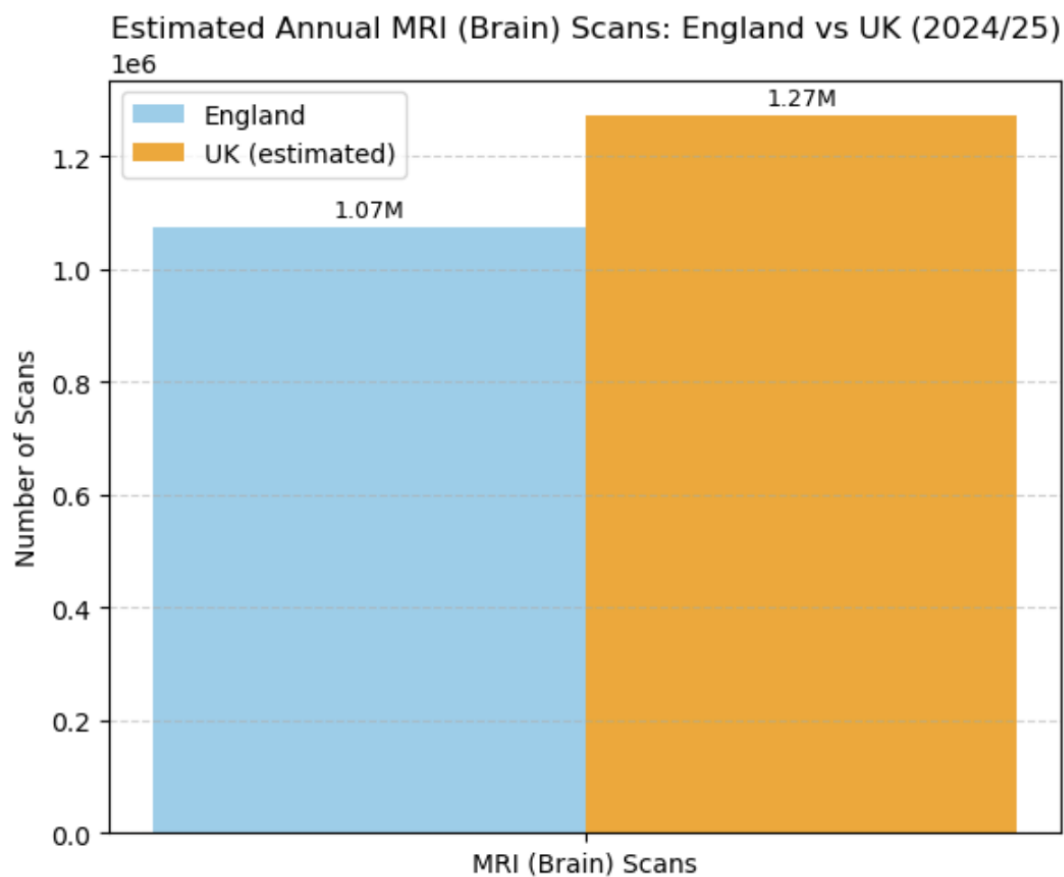


Figure 32 England vs. UK Brain MRI Market (2024/25, estimated)

This extrapolation suggests that the UK brain MRI market could reach around 1.27 million scans annually in 2024/25. However, this estimate should be treated as indicative only, since Scotland, Wales, and Northern Ireland may have distinct healthcare systems, and differ

significantly in diagnostic infrastructure and workforce distributions. Nevertheless, the calculation provides a broad sense of the national scale of opportunity if QP-Brain was adopted UK-wide rather than limited to England.

4.1.2.3. Market Sizing Calculation Table for Brain MRI (2024/25)

Component	Source/ Derivation	Value	Result
Total NHS Imaging Scans	Table-1-2024-25-Modality-Provider-Counts-2025-06-19-XLSX-199KB (NHS England, 2025)	—	48,838,490
Total NHS MRI Scans	Table-1-2024-25-Modality-Provider-Counts-2025-06-19-XLSX-199KB (NHS England, 2025)	—	4,760,700
Brain MRI (NHS)	Table-4-2024-25-Bodysite-Provider-Counts-2025-06-19-XLSX-221KB (NHS England, 2025)	—	1,014,655
Brain MRI Share of MRI	$\frac{\text{Brain MRI (NHS)}}{\text{Total NHS MRI Scans}} \times 100\%$	$\frac{1,014,655}{4,760,700} \times 100\%$	21.3%

Private Scan Share (all modalities)	$\frac{\text{Total Private Scans in UK}}{\text{Total NHS Scans in UK}} \times 100\%$	$\frac{3,475,000}{59,920,000} \times 100\%$ (LaingBuisson, 2025)	5.8%
Estimated Private Brain MRI	$\text{Brain MRI (NHS)} \times \text{Private Scan Share}$	$1,014,655 \times 0.058$	58,850
Total Brain MRI Market	$\text{Brain MRI (NHS)} + \text{Estimated Private Brain MRI}$	$1,014,655 + 58,850$	1,073,505
England Population (2024)	Population estimates for England and Wales: mid-2024 (Office for National Statistics, 2025)	–	58,382,600
Per Capita MRI	$\frac{\text{Total NHS MRI Scans}}{\text{England Population (2024)}} \times 1000$	$\frac{4,760,700}{58,382,600} \times 1000$	86.3 scans per 1,000 population
Per Capita Brain MRI	$\frac{\text{Total Brain MRI Market}}{\text{England Population (2024)}} \times 1000$	$\frac{1,073,505}{58,382,600} \times 1000$	18.4 scans per 1,000 population
Patients waiting for	Monthly-Diagnostics-Web-File-Provider-May-2025 (NHS England, 2025)	–	1,703,242 patients waiting and 331,280

Diagnostic Tests			(19.4%) are waiting for MRI test
Delay (Patients Waiting Over 6 Weeks)	Monthly-Diagnostics-Web-File-Provider-May-2025 (NHS England, 2025)	–	312,744 patients (18.4% of waiting list) are being delayed and 45,529 of them are delayed for MRI.
Estimated Brain Waiting	<i>Patients waiting for MRI</i> $\times$ <i>Brain MRI Share of MRI</i>	331,280 $\times$ 0.213	70,563
Delay (Brain) (Estimated Patients Waiting Over 6 Weeks for Brain MRI Test)	<i>MRI Waiting Over 6 weeks</i> $\times$ <i>Brain MRI Share of MRI</i>	$45,529 \times 0.213$	9,698 (13.7% are delayed)
Radiologist FTEs (NHS)	NHS Workforce Statistics – Doctors by Grade and Specialty (March 2025) (NHS England, 2025)	–	Out of 147931 HCHS doctors, 5,845 (3.95%) are Clinical Radiologists
Scans per Radiologist	$\frac{\text{Total NHS Imaging Scans}}{\text{Radiologist FTEs (NHS)}}$	$\frac{48,838,490}{5,845}$	8356 (696 monthly, 33 daily)

Brain MRI scans per Radiologist	$\frac{\text{Brain MRI (NHS)}}{\text{Radiologist FTEs (NHS)}}$	$\frac{1,014,655}{5,845}$	174 (14.5 monthly, 0.7 daily)
Brain Disease Growth Benchmarks	Primary Care Dementia Data, March 2025 (NHS Digital, 2025), Cancer Research UK (Cancer Research UK, 2024) $\frac{\text{Tumor Incidence Rate}}{100,000}$ $\times \text{England Population (2024)}$	New Brain Tumour case = $\frac{22.77}{100,000}$ $\times 58,382,600$	498,729 dementia patients (an increase of 17,022 cases from March 2024) and 13,300 new brain tumour per year
UK Population (2024)	(Data Commons, 2025)	–	69,226,000
UK Brain MRI Scans (Indicative Estimate)	$\frac{\text{England Brain MRI Scans}}{\text{England Population (2024)}}$ $\times \text{UK Population (2024)}$	$\frac{1,073,505}{58,382,600}$ $\times 69,226,000$	1,272,887

Table 3 Market Sizing Calculation Table for Brain MRI

4.2. Forecasting Global demand and Adoption Scenarios (Veena Poduru - 241051702)

4.2.1. Forecasting Global demand

Forecasting demand for brain MRI scans is a crucial step in anticipating future diagnostic pressures and identifying where technological innovations such as QP-Brain can provide the greatest value. By projecting how scan volumes are likely to evolve, it becomes possible to highlight which Trusts will face the highest demand in the coming years and therefore where radiology reporting capacity will be under the greatest strain. This not only informs NHS planning and resource allocation but also provides a targeted strategy for the adoption of AI-based diagnostic support.

In this study, forecasting was performed using ten years of monthly brain MRI activity, covering the period from April 2015 to March 2025. These forecasts form the evidence base for later adoption scenario analysis, where the focus shifts from demand to the readiness of Trusts to deploy AI solutions.

4.2.1.1. Data Source and Context

The dataset used for this analysis was obtained from the file *brain_scans_10yrs.xlsx*, introduced in Section 3.2.1. Specifically, the analysis relied on the sheet titled *Monthly*, which contains national-level totals of brain MRI scans performed across NHS England. The dataset spans the period April 2015 to March 2025, providing 120 consecutive monthly observations. This ten-year window is sufficiently long to capture both secular trends and seasonal fluctuations, making it well-suited to time-series analysis.

The dataset includes the variable *Total Brain Scans*, representing the monthly count of brain MRI examinations across all NHS providers in England. Before analysis, the *Date* column was parsed into a datetime format and set as the index, ensuring that the temporal ordering of observations was preserved. The series was then inspected for completeness and consistency. Visual exploration confirmed that the dataset was complete, with no missing months, though an unusual disruption was observed during 2020, reflecting the reduction in activity during the COVID-19 pandemic. Beyond this disruption, the series showed a clear upward trajectory, with volumes increasing steadily year on year, and distinct seasonal cycles where certain months exhibited systematically higher or lower activity.

These features—trend, seasonality, and occasional structural shocks—make brain MRI demand a classic case for time-series forecasting. Trend is important for capturing the long-term growth in demand, seasonality ensures that recurrent patterns such as winter peaks are properly reflected, and shocks such as COVID-19 illustrate the need for models that can accommodate irregular disruptions.

4.2.1.2. Methodology

The forecasting methodology followed a multi-stage approach to ensure both robustness and interpretability. The process began with structuring the dataset as a univariate time series, with monthly frequency specified explicitly. This allowed the models to detect seasonal cycles of twelve months and to align forecast horizons with calendar years.

To enable objective evaluation, the dataset was divided into two subsets: a training set comprising the first nine years of data (April 2015 – March 2024) and a test set comprising the final year (April 2024 – March 2025). The training set was used for model fitting, while the test set was withheld for validation, thereby allowing an out-of-sample assessment of forecast

accuracy. This approach ensured that the models were evaluated under realistic conditions, where they were required to predict unseen data rather than simply reproducing past patterns.

A **Seasonal Naïve model** was implemented first as a benchmark. This model predicts that each month will equal the value of the same month in the previous year. Although simple, it serves as an important comparator: if a more sophisticated model cannot outperform the Seasonal Naïve benchmark, it provides little value.

Four additional models were then estimated. The **Holt-Winters Exponential Smoothing model** was fitted with additive trend and additive seasonality, making it well-suited for data with both steady growth and recurring annual cycles. **ARIMA** (1,1,1) was tested as a non-seasonal autoregressive moving average model, designed to capture autocorrelations and differencing in the series. **SARIMA**, an extension of ARIMA that incorporates explicit seasonal terms, was implemented using automated parameter selection via the `auto_arima` function. Finally, **Prophet** was applied as a flexible model capable of handling structural breaks and nonlinear growth, configured with yearly seasonality and additive effects.

Each model generated forecasts for the twelve-month test horizon, and performance was evaluated using **Root Mean Squared Error (RMSE)** and **Mean Absolute Percentage Error (MAPE)**. RMSE measured the absolute scale of forecast errors, while MAPE provided a proportional measure of accuracy, which is particularly interpretable in a policy context. After validation, the best-performing model was refitted on the full dataset and used to generate a sixty-month projection extending to March 2030.

The implementation of this forecasting methodology was carried out in Python using the notebook **Forecasting.ipynb**, which is provided separately in the submitted archive **Team_5_Code.zip**. This file contains the complete preprocessing steps, model fitting, evaluation, and forecasting scripts corresponding to the results presented in this section.

4.2.1.3. Model Selection and Justification

The selection of models was based on both the statistical properties of the dataset and the practical requirements of healthcare forecasting. Because the data contained clear trend and seasonal patterns, models capable of handling these features were prioritised.

The **Seasonal Naïve model** was included as a baseline to provide a minimum standard of performance. **Holt-Winters** was chosen because it is designed for series with both trend and seasonality, and it provides an interpretable decomposition into level, trend, and seasonal components. **ARIMA** was included due to its strong historical record in time-series forecasting, though its lack of explicit seasonal terms made it less likely to capture annual cycles. **SARIMA** addressed this limitation by modelling seasonal autoregressive and moving average structures directly. **Prophet** was included as a modern alternative, capable of modelling nonlinear growth and structural breaks, which was relevant given the pandemic disruption in 2020.

By including models of differing complexity and theoretical underpinnings, the analysis ensured a comprehensive evaluation. This approach also reflected the need for interpretability, since forecasting in healthcare is not only a technical exercise but also a communication tool for policymakers and practitioners.

4.2.1.4. Model Comparison

The five models were evaluated against the test dataset, and their performance metrics are summarised in Table 4. As expected, the Seasonal Naïve model produced the weakest performance, confirming that brain MRI demand is not adequately explained by simple year-on-year repetition. ARIMA improved upon this baseline but underperformed in comparison

with models that explicitly accounted for seasonality. Prophet provided reasonable accuracy but tended to understate the regular seasonal fluctuations present in the data.

The **Holt-Winters** and **SARIMA** models delivered the best performance, closely tracking both the upward trend and the seasonal cycles. Holt-Winters achieved a MAPE of approximately **3.11%** and an RMSE of **3,866**, while SARIMA produced a MAPE of **3.33%** and an RMSE of **3,804**. The differences between the two were minor, but Holt-Winters was selected as the preferred model due to its interpretability, stability, and suitability for presentation to non-technical stakeholders.

Model	RMSE	MAPE (%)
Holt-Winters	3,866	3.11
SARIMA	3,804	3.33
ARIMA	3,934	3.41
Prophet	5,631	5.27
Seasonal Naive	12,972	13.83

Table 4: Model comparison

4.2.1.5. Forecasting Results

The **Holt-Winters model** was refitted on the full ten-year dataset to maximise the use of available historical information, and a **sixty-month forecast** was produced for April 2025 through March 2030. The forecasts indicate sustained growth in demand, with monthly scan

volumes projected to increase from approximately 80,000 in April 2025 to more than 100,000 by March 2030, representing an overall increase of about 30% over five years.

Seasonal fluctuations remain visible in the forecasts, with predictable peaks and troughs repeating annually. These cycles suggest that certain months will experience consistent strain on MRI capacity if planning is not adjusted to meet seasonal surges. The first twelve months of forecasts are summarised in **Table 5**, while the full five-year trajectory is illustrated in **Figure 33**, which presents both the historical series and the projected values.

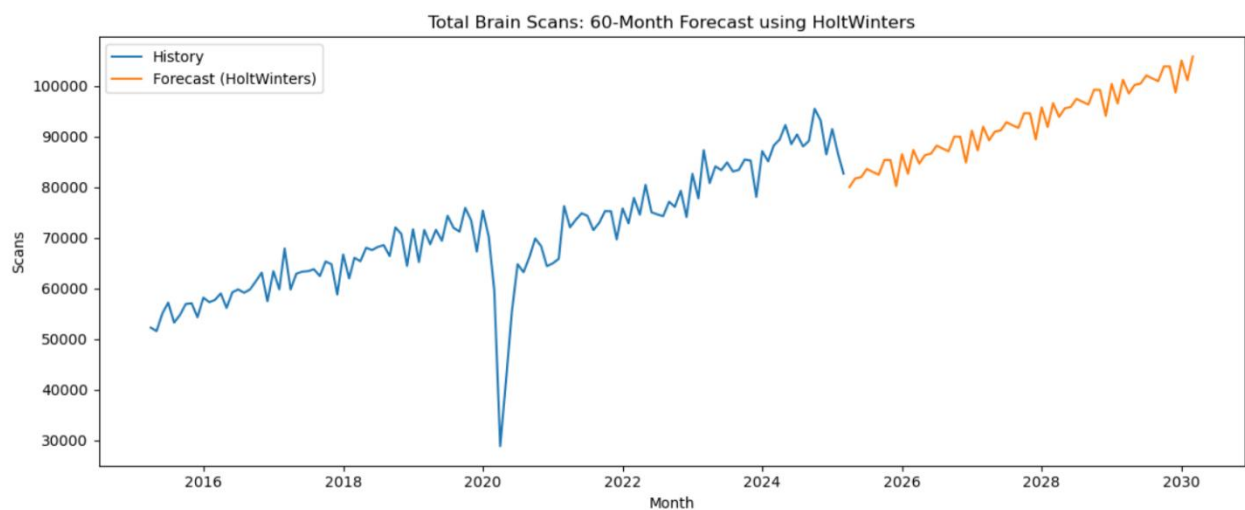


Figure 33: 5-year monthly forecast using Holt-winters exponential smoothing

Month	Forecasted Total Brain Scans
2025-04	80017
2025-05	81714
2025-06	81970

2025-07	83576
2025-08	83001
2025-09	82453
2025-10	85368
2025-11	85347
2025-12	80212
2026-01	86519
2026-02	82633
2026-03	87334

Table 5: 12-month forecast table

In conclusion, this forecasting exercise provides a comprehensive projection of brain MRI demand in England over the next five years. By applying a suite of time-series models to a robust ten-year dataset, and by systematically comparing their accuracy, the analysis demonstrates that the Holt-Winters model offers the most reliable basis for forecasting. The results indicate a sustained upward trajectory, with total monthly scan volumes expected to rise by nearly one-third by 2030, alongside persistent seasonal fluctuations. These findings confirm that diagnostic imaging services will face continued growth in demand and establish a robust evidence base for subsequent analyses. The forecasts generated here not only quantify the scale of expected pressures but also provide a structured platform for linking demand growth to future adoption scenarios, which are explored in the next section of this report.

4.2.2. Adoption scenarios

The adoption scenario analysis was designed to evaluate how NHS Trusts in England may respond to the introduction of a new AI-based imaging tool that specifically supports Brain MRI reporting. The motivation behind this analysis is rooted in the current pressures facing the NHS: a growing demand for imaging, significant radiology workforce shortages, and persistent waiting list backlogs for diagnostic procedures. In this context, AI tools represent not only an innovation but a potential necessity, offering the opportunity to optimize workflows, reduce delays, and improve patient access to timely diagnosis.

Because this is a new product, no real-world adoption data yet exist. To overcome this challenge, the analysis relies on five years of historical data (2020–21 to 2024–25). These datasets capture diagnostic activity levels, waiting times, and workforce availability at NHS Trust level. By drawing on these indicators, the analysis develops a proxy measure of adoption readiness, allowing us to forecast how uptake might occur across different scenarios.

A central part of this framework is the **Demand Score**, a composite indicator derived from provider-level Brain MRI volumes, waiting list backlogs, radiologist estimates, and backlog clearance efficiency. The Demand Score, described in detail in Section 4.2.2.1, provides the foundation for estimating adoption probabilities. Based on these scores, three scenarios were developed to represent plausible trajectories of adoption:

1. A **Conservative scenario**, where adoption progresses slowly, with only 0–5 new Trusts adopting the tool each year.
2. A **Medium scenario**, which reflects steady and balanced adoption, with 5–10 Trusts adopting annually.
3. An **Ambitious scenario**, which assumes rapid uptake, with 10–15 or more Trusts adopting per year, supported by strong policy, financial, and institutional backing.

These scenarios offer a structured way to explore different rates of diffusion, enabling us to compare how adoption might evolve under varying strategic, financial, and operational conditions. They are also visualised later in the form of **S-curve projections** (Figure 35), which represent the classic diffusion model for technological innovations.

4.2.2.1. Demand Score Calculation

The demand score is a composite indicator designed to evaluate the potential readiness of NHS Trusts in England to adopt AI-based Brain MRI diagnostic tools. Since this is a new technology and no direct adoption data currently exists, the score relies entirely on five years of historical diagnostic imaging data (2020–21 to 2024–25). By combining measures of scan activity, patient waiting times, radiologist workforce capacity, and backlog efficiency, the demand score provides a multi-dimensional assessment of where unmet needs and workload pressures are greatest. Trusts with lower demand scores represent those with both high diagnostic workload and poor backlog efficiency, making them the most likely candidates for early adoption of AI-enabled solutions.

Data Sources and Preparation

The demand score was calculated by consolidating information from **four sets of publicly available NHS datasets**, all covering the five-year period from 2020–21 to 2024–25. These datasets were sourced from NHS England’s official statistical portals and included the following:

1. **Diagnostic Imaging Dataset (DID)** – Ten datasets in total were used, comprising both *modality-level* (MRI, CT, X-ray, etc.) and *body-site-level* statistics for each year. These datasets provided detailed counts of imaging procedures carried out by NHS providers, enabling the extraction of Brain MRI activity at provider level.

2. **Monthly Diagnostics Waiting Times and Activity (DWTA)** – Five annual datasets (March census for each financial year) were obtained, each containing provider-level information on diagnostic waiting lists. These included the *total waiting list size*, *number of patients waiting more than six weeks*, and *percentage waiting longer than six weeks*. While DWTA does not directly disaggregate waiting times by body site, Brain MRI waiting figures were estimated by applying the national proportion of Brain MRI within total MRI activity (derived from the DID).
3. **NHS Workforce Statistics (March census)** – Five datasets were also obtained from NHS Workforce Statistics, covering the same period. These included counts of Hospital and Community Health Service (HCHS) doctors at provider level, which were used as a proxy for allocating national radiologist headcounts across Trusts. By proportionally distributing national radiologist totals (e.g., 5,845 radiologists in 2024–25) according to each Trust’s HCHS doctor numbers, provider-level radiologist workforce estimates were derived.

All datasets were imported into Python and processed using **pandas**. Data preparation involved standardising column names, cleaning provider identifiers (including trimming whitespace, removing prefixes such as “The,” and aligning formatting), and verifying completeness across years. Providers with no Brain MRI activity were removed from the analysis. To validate accuracy, national totals for England were cross-checked against published NHS summary tables. The combined outputs from these three data sources formed the foundation for building a multi-year, provider-level dataset, which was subsequently stored in **Provider_Scans_5_years.xlsx** for downstream analysis.

Brain MRI Activity

The first input into the demand score was the volume of Brain MRI scans conducted at Trust level. From the DID (Body Site) dataset, rows corresponding to “Brain (MRI)” under the “All” source setting were filtered and compiled into a consolidated dataset. This dataset captured, for each provider and year, the total number of Brain MRI scans performed.

To make the dataset usable for further analysis, annual results were combined into an Excel file titled **Provider_Scans_5_years.xlsx**, with a sheet named “MRI (Brain).” Each record included the year, provider name, and Brain MRI scan count. Table 6 shows a sample of this sheet. Alongside provider-level activity, a national summary was created to calculate the proportion of Brain MRI scans within total MRI activity.

Year	Provider name	Brain (MRI)
2024-25	Airedale NHS Foundation Trust	1410
2024-25	Alder Hey Children's NHS Foundation Trust	2290
2024-25	Alliance Medical	15170
2024-25	Ashford and St Peter's Hospitals NHS Foundation Trust	4345

Table 6: Sample of Provider_Scans_5_years.xlsx (MRI (Brain) sheet)

Calculating Brain MRI Percentage

To quantify the relative demand for Brain MRI scans, we calculated the percentage of Brain MRI scans out of total MRI scans performed nationally each year using the below code.

```
brain_mri_pct = (brain_mri_total / mri_total * 100) if mri_total else 0
```

This percentage helps us in understanding the importance of Brain MRI within the broader MRI activity and will be later used in the demand scoring methodology. Table 7 shows the

Brain MRI as a Percentage of Total MRI Scans in England. These percentages indicate a stable proportion of Brain MRI activity within the total MRI workload across England, highlighting the consistent relevance of this modality for assessing potential adoption of AI tools.

Year	Brain MRI	Total MRI	Brain MRI %
2024-25	1,029,265	4,837,820	21.3%
2023-24	953,500	4,415,465	21.6%
2022-23	882,125	4,084,055	21.6%
2021-22	837,395	3,845,155	21.8%
2020-21	689,780	3,008,970	22.9%

Table 7: Brain MRI as a Percentage of Total MRI Scans (England)

This forms the first, foundational step of the **demand score calculation**, which feeds into the adoption scenario analysis. By combining five years of historical Brain MRI activity at the provider level, we now have a robust dataset that can be used to score Trusts by potential demand for an AI-assisted imaging tool.

Waiting Time Analysis

The second component involved integrating waiting list pressures using the DWTA datasets. For each year between 2020–21 and 2024–25, provider-level MRI waiting list data were extracted, focusing on three indicators which are **Total waiting list** – patients yet to receive an

MRI, **Number waiting over six weeks** – patients breaching the six-week standard and **Percentage waiting over six weeks** – share of the waiting list delayed.

Since the Diagnostics Waiting Times and Activity dataset does not provide body-site detail, the national Brain MRI proportion was applied to provider-level MRI waiting lists as shown in the code snippet below, thereby estimating Brain MRI waiting figures. These included **Brain MRI Waiting**, **Brain MRI 6+ Week Wait**, and **Brain MRI % 6+ Weeks**, which were stored in the “Waiting Time” sheet of *Provider_Scans_5_years.xlsx* (Table 8).

```
pct = brain_mri_pct_lookup.get(year, 0) # default 0 if missing
wait_total['Brain MRI Waiting'] = wait_total["Total Waiting"] * pct
wait_6plus['Brain MRI 6+ Week Wait'] = wait_6plus["Waiting 6+ Weeks"] * pct
wait_percent['Brain MRI % 6+ Weeks'] = wait_percent["% Waiting 6+ Weeks"]
```

Providers with no patients waiting over six weeks were identified as efficient, while those with large backlogs represented sites of unmet need. This step ensured that the demand score incorporated not just activity, but also patient access pressures.

Year	Region	Provider	Brain MRI Waiting	Brain MRI 6+ Week Wait	Brain MRI % 6+ Weeks
2024- 25	North East and Yorkshire Commissioning Region	Airedale NHS Foundation Trust	138.876	2.343	0.016871166

2024-25	South East Commissioning Region	Ashford and St Peter's Hospitals NHS Foundation Trust	482.232	28.968	0.060070671
2024-25	London Commissioning Region	Barking, Havering and Redbridge University Hospitals NHS Trust	647.094	31.311	0.048387097
2024-25	London Commissioning Region	Barts Health NHS Trust	1338.705	447.3	0.334128878

Table 8: Sample of Waiting Time sheet in Provider_Scans_5_years.xlsx excel file

Radiologist Workforce and Workload Estimation

Radiologist capacity was then integrated into the analysis. Since provider-level radiologist headcounts are not published, estimates were derived by allocating national totals proportionally to each Trust using their HCHS doctor counts. For example, in 2024–25, England reported **5,845 radiologists nationally**, which were distributed across providers according to their share of HCHS doctors.

By combining these estimates with Brain MRI scan counts, three workload measures were computed for each provider which are **Annual scans per radiologist**, **Monthly scans per radiologist** and **Daily scans per radiologist** (assuming 253 working days per year) as shown in code snippet below.

These metrics highlighted providers where radiologists were under particular strain. For example, **Manchester University NHS Foundation Trust** was estimated to have 143 radiologists in 2024–25, each facing thousands of Brain MRI scans annually. The results were

compiled in the “Radiologists” and “Scans per Radiologist” sheets of *Provider_Scans_5_years.xlsx* (Table 9).

=== Calculate Scan per Radiologist ===

scan_radio["Scans per Radiologist (Annual)"] = scan_radio["Brain (MRI)"] /

scan_radio["Estimated Radiologists"]

scan_radio["Scans per Radiologist (Monthly)"] = scan_radio["Scans per Radiologist (Annual)"] / 12 # FTE

scan_radio["Scans per Radiologist (Daily)"] = scan_radio["Scans per Radiologist (Annual)"] / 253 # assuming working days each year = 253 days

This allocation provides a consistent approximation of radiologist capacity at Trust level, which can be used in workload and efficiency analysis.

Year	NHS England region name	Organisation name	HCHS Doctors	Estimated Radiologists
2024-25	North West	Manchester University NHS Foundation Trust	3616.729	143.1448
2024-25	London	Guy's and St Thomas' NHS Foundation Trust	3093.357	122.4305
2024-25	London	Barts Health NHS Trust	3077.523	121.8038

Table 9: Sample of Radiologists sheet in *Provider_Scans_5_years.xlsx* excel file

Scan-to-Wait Ratio

The next dimension was backlog efficiency, captured by the **Scan-to-6+Week-Wait Ratio** as shown in the code snippet below. This metric compared the annual volume of Brain MRI scans completed to the number of patients waiting more than six weeks.

```
# calculate scan-to-6+week-wait ratio
scan_wait = scan_wait[scan_wait['Brain MRI 6+ Week Wait'] > 0].copy() # only keep the trusts who has 6+ weeks waiting patients
scan_wait["Scan-to-6+Week-Wait Ratio"] = scan_wait["Brain (MRI)"] / scan_wait["Brain MRI 6+ Week Wait"]
```

Trusts with **high ratios** demonstrated efficient backlog management, while those with **low ratios** exhibited persistent bottlenecks which can be seen in the Table 10. For instance, **Airedale NHS Foundation Trust** achieved a ratio exceeding 600, suggesting strong throughput relative to backlog. In contrast, **Barts Health NHS Trust**, despite high scan volumes (over 14,000 in 2024–25), recorded a ratio of just 31.7, indicating severe waiting list pressures. These variations illustrate how workload and efficiency differ across providers, even among high-activity Trusts.

Year	Provider	Brain (MRI)	Region	Brain MRI Waiting	Brain MRI 6+ Week Wait	Brain MRI % 6+ Weeks	Scan-to-6+Week-Wait Ratio
2024-25	Airedale NHS Foundation Trust	1410	North East and Yorkshire Commissioning Region	138.876	2.343	0.016871	601.7926

2024-25	Ashford and St Peter's Hospitals NHS Foundation Trust	4345	South East Commissioning Region	482.232	28.968	0.060071	149.9931
2024-25	Barts Health NHS Trust	14190	London Commissioning Region	1338.705	447.3	0.334129	31.72368

Table 10: Sample Scan-to-wait ratio sheet in *Provider_Scans_5_years.xlsx* excel file.

Demand Score Integration

Once both *Scans per Radiologist* and the *Scan-to-Wait Ratio* had been derived, the final step was the calculation of a **Demand Score**. The purpose of this score was to combine two critical dimensions:

1. **Workload intensity** – captured as the *annual scans per radiologist* for each Trust. High values indicate staff are under greater pressure.
2. **Efficiency of backlog clearance** – reflected by the *Scan-to-6+Week-Wait Ratio*. Low ratios suggest that even though scans are being delivered, backlogs of patients waiting longer than six weeks are not being reduced effectively.

By integrating both factors into a single score, the analysis could highlight those Trusts where radiology teams are under sustained pressure and patients are facing delays, making them the most likely candidates for AI-based support.

The calculation process involved merging the "Scans per Radiologist" dataset with the "Scan-to-wait ratio" dataset on the common fields of *Year* and *Provider*. Once merged, two rankings were applied for each year as the following:

1. **Efficiency Rank:** Trusts were ranked by their *Scan-to-6+Week-Wait Ratio* in ascending order. A lower ratio meant worse performance (i.e., fewer scans per waiting patient), so a low ratio was assigned a low rank number.
2. **Workload Rank:** Trusts were ranked by their *Scans per Radiologist* in descending order. A higher workload per radiologist was considered worse, so high workloads received a low rank number.

These two ranks were then summed to form the **Demand Score**. Lower Demand Scores correspond to Trusts under both high workload and low efficiency pressures, representing areas of greatest unmet demand. This calculation process is shown in the code snippet below.

```
# Normalize within each year: low efficiency = worse, high workload = worse
demand_df['Efficiency Rank'] = demand_df.groupby('Year')['Scan-to-6+Week-Wait Ratio'] \
    .rank(ascending=True)

demand_df['Workload Rank'] = demand_df.groupby('Year')['Scans per Radiologist (Annual)'] \
    .rank(ascending=False)

# Combine into a demand score (lower = worse = higher demand)
demand_df['Demand Score'] = demand_df['Efficiency Rank'] + demand_df['Workload Rank']
```

The resulting dataset was saved in the "Demand" sheet of *Provider_Scans_5_years.xlsx* which mainly contains MRI Brain scans, Scans per Radiologist(Annual), Scan-to-6+Week-Wait Ratio, Efficiency Rank, Workload Rank and Demand Score for each year, trust and Region. This Demand Score sheet served as the final input layer for the adoption scenario analysis, providing a quantitative basis to evaluate which Trusts face the strongest drivers for adopting AI solutions.

Lower scores correspond to higher demand pressures, reflecting both overburdened staff and persistent backlogs. The resulting dataset was recorded in the “**Demand**” sheet of *Provider_Scans_5_years.xlsx* (Figure 34). This dataset serves as the final input for adoption scenario analysis.

	Year	Provider	Brain (MRI)	Scans per Radiologist (Annual)	Region	Scan-to-6+Week-Wait Ratio	Efficiency Rank	Workload Rank	Demand Score
0	2024-25	Airedale NHS Foundation Trust	1410	95.757408	North East and Yorkshire Commissioning Region	601.792574	87.0	110	197.0
1	2024-25	Barking Havering and Redbridge University Hosp...	12775	242.873245	London Commissioning Region	408.003577	79.0	18	97.0
2	2024-25	Barts Health NHS Trust	14190	116.498838	London Commissioning Region	31.723675	14.0	104	118.0
3	2024-25	Bedfordshire Hospitals NHS Foundation Trust	11370	215.295058	East of England Commissioning Region	146.247347	50.0	32	82.0

Figure 34: Sample data in the demand sheet in *Provider_Scans_5_years.xlsx* excel file

The demand score thus integrates Brain MRI activity, patient waiting list backlogs, radiologist workforce capacity, and backlog clearance efficiency into a single quantitative measure. By grounding the analysis in five years of NHS operational data, the demand score provides a transparent and evidence-based method for identifying NHS Trusts most likely to adopt AI-based diagnostic tools. This measure forms the foundation for the adoption scenario modelling presented in the next section.

4.2.2.2. Methodology for Adoption Scenario Analysis

The methodology underpinning the adoption scenario analysis was developed to provide a systematic and transparent framework for forecasting how NHS Trusts might adopt AI-based Brain MRI tools such as QP-Brain under varying conditions. While the demand forecasting exercise (Section 4.2.1) established the overall scale and timing of brain MRI activity, this stage sought to translate those projections into adoption probabilities, recognising that Trusts differ in their readiness, resource availability, and incentives to adopt new technologies. The

framework therefore combines **demand-driven probability modelling with scenario-based simulation**, ensuring that projected adoption pathways are firmly grounded in NHS operational realities while also reflecting uncertainty in future diffusion patterns.

Data Preparation

The modelling process began with the provider-level demand dataset compiled in *Provider_Scans_5_years.xlsx*, which contained brain MRI activity, waiting list backlogs, radiologist workforce estimates, and the derived Demand Score for each Trust. This dataset covered the five financial years from 2020–21 to 2024–25. Because raw demand scores varied significantly across providers and years, a **min–max transformation** was applied to rescale values between 0 and 1. Trusts experiencing the most acute demand pressures were assigned values close to one, while those under lighter pressure were closer to zero. This normalisation step ensured comparability across providers and established a consistent foundation for probability modelling.

Probability Modelling

To convert demand scores into adoption likelihoods, a **logistic transformation** was applied. Logistic functions are commonly used in diffusion modelling because they capture the non-linear nature of adoption decisions: at lower levels of demand pressure, probability of adoption rises only modestly, but as pressure intensifies, the likelihood increases sharply. This ensured that Trusts under extreme demand conditions faced disproportionately higher probabilities of adopting QP-Brain, while still allowing for variation across the system.

In addition, a **decay factor** was introduced to reflect persistence across years. Adoption readiness is not determined by a single year's data but by the accumulation of pressures over time. A decay factor of 0.5 was applied, meaning that half of the probability weight for any

year carried over from the previous year's value. This provided continuity while still allowing more recent demand conditions to influence readiness to adopt. The code snippet below shows the decay factor and logistic transformation application.

```
# Set Decay Factor
decay_factor = 0.5 # weight of previous year's probability
# Logistic transformation of scaled Demand Score
current_probs = 1 / (1 + np.exp(-year_df['Demand_Score_scaled']))
current_probs.index = year_df['Provider']
# Combine with previous year probabilities
combined_probs = decay_factor * prev_probs + (1 - decay_factor) * current_probs
```

Scenario Definition

Once probabilities were established, three **adoption scenarios** were defined to reflect alternative diffusion pathways. These scenarios were based on the annual number of Trusts expected to adopt the AI tool as follows:

1. The **Conservative scenario** simulated between 0 and 5 adoptions per year, representing a cautious and resource-limited environment where adoption spreads slowly.
2. The **Medium scenario** simulated between 5 and 10 adoptions per year, reflecting a balanced trajectory where adoption gradually becomes mainstream.
3. The **Ambitious scenario** simulated between 10 and 15 or more adoptions per year, corresponding to a rapid rollout supported by strong policy intervention, financial incentives, or widespread clinical endorsement.

These scenarios, previously introduced in Section 4.2.2, provided structured bounds within which adoption simulations were carried out. They ensured that the model captured not only variability in provider readiness but also uncertainty in system-level adoption rates.

Simulation of Adoption

Adoption was simulated year by year under each scenario. For every simulated year, the model randomly selected a number of Trusts to adopt, constrained within the scenario's defined range (e.g., 0–5 for Conservative). However, selection was not uniform: the probability of a Trust being chosen was weighted by its calculated adoption likelihood. This ensured that providers experiencing greater demand pressures were consistently prioritised, while still allowing for stochastic variation.

This probability-weighted sampling generated realistic adoption patterns in which high-demand Trusts dominated early uptake but were not the sole adopters. Importantly, this step replicated the inherent uncertainty of real-world diffusion, where adoption does not follow a rigid order but is influenced by multiple interacting factors.

Forecasting and Trajectory Analysis

The methodology extended beyond the historical dataset to forecast adoption in future years. Initial probabilities for each Trust were drawn from their most recent demand scores, and then updated annually using the decay-weighted mechanism alongside simulated adoption events. Trusts with consistently high pressures retained elevated probabilities across the forecast horizon, while those with lower demand conditions had correspondingly lower chances of adoption unless systemic changes occurred.

At the system level, cumulative adoption was aggregated and plotted as S-shaped diffusion curves under each scenario. These curves (Figure 35) illustrate the contrasting dynamics of slow, moderate, and rapid uptake. The Conservative scenario shows a gradual trajectory with limited penetration by the end of the forecast horizon, the Medium pathway displays steady

incremental adoption consistent with mainstream diffusion, and the Ambitious case demonstrates accelerated early adoption leading to near-saturation in a relatively short period.

The use of S-curves provides an intuitive visual representation of adoption patterns and allows easy comparison across scenarios. It highlights how policy interventions, financial support, or strategic leadership could move the system from a slower to a faster adoption pathway. This is consistent with the broader innovation diffusion literature, which finds that new technologies in healthcare typically follow an S-curve trajectory as early adopters are followed by the majority and eventually laggards.

The adoption scenario modelling was implemented in Python using the notebook **Demand_Score_Adoption_Scenarios.ipynb**, included in the submitted archive **Team_5_Code.zip**. This file contains the demand score calculation, probability modelling, scenario simulation, and multi-year forecasting procedures described above.

4.2.2.3. Adoption Scenario Insights

The adoption scenario analysis provides a set of insights into how NHS Trusts are likely to adopt AI-based Brain MRI tools under different conditions and over time. By combining provider-level demand scores, probability modelling, and scenario-based simulations, the analysis highlights which Trusts emerge as frontrunners, how adoption is likely to diffuse across the system, and what strategic lessons can be drawn for implementation planning. The results are presented at three levels: insights for the current year (2024–25), projections for the first forecast year (2025–26), and long-term trajectories over the following five years.

Insights for 2024–25

In the baseline year, adoption likelihood varies depending on the scenario assumed. Under the Ambitious scenario, the analysis identifies predominantly large teaching and foundation Trusts

as the most likely early adopters. Institutions such as University Hospitals Dorset NHS Foundation Trust, Royal National Orthopaedic Hospital NHS Trust, and University Hospitals of Morecambe Bay NHS Foundation Trust consistently appear at the top of the rankings. These Trusts are characterised by their relatively large diagnostic capacity, stronger infrastructure, and higher readiness to adopt innovative solutions. Their early adoption reflects the tendency of large, resource-rich institutions to take the lead in technology diffusion when policy support and funding are strong.

By contrast, the **Medium scenario** produces a more mixed group of likely adopters. Alongside large institutions such as **Cambridge University Hospitals NHS Foundation Trust**, the results also highlight mid-sized providers such as **Dorset County Hospital NHS Foundation Trust** and **Epsom and St Helier University Hospitals NHS Trust**. This pattern suggests that when adoption progresses at a steady but not overly aggressive pace, diffusion is not confined to the largest Trusts. Instead, a broader set of organisations participates, including those with intermediate capacity but sufficient demand pressures to justify adoption.

The **Conservative scenario** paints a different picture, highlighting smaller or regionally focused Trusts such as **Croydon Health Services NHS Trust**, **Calderdale and Huddersfield NHS Foundation Trust**, and **Harrogate and District NHS Foundation Trust**. Their presence indicates that even under conditions of slow adoption, organisations experiencing acute demand pressures may still be among the first to take up the technology. In these cases, the drivers are less about institutional size and more about immediate operational challenges, such as significant waiting list backlogs or limited radiologist availability.

The results for 2024–25 are summarised in **Table 11**, which presents the top five Trusts under each adoption scenario. While differences exist across scenarios, a key insight is that certain providers, notably **University Hospitals Dorset** and **Dorset County Hospital**, appear across

multiple pathways. Their repeated presence suggests that they may represent “core early adopters,” whose readiness to adopt is relatively robust regardless of whether adoption is rapid, moderate, or cautious.

Another important finding is the clustering of adoption likelihood scores. Across all scenarios, values fall within a relatively narrow range (approximately 0.298–0.343). This indicates that although some Trusts are marginally more likely to adopt, there are no overwhelming outliers. Adoption patterns are therefore sensitive to external factors—such as funding allocations, vendor outreach, and regional policy initiatives—which could decisively tip the balance in favour of one Trust over another.

Scenario	Provider	Adoption_Likelihood
Ambitious	University Hospitals Dorset NHS Foundation Trust	0.342762
Ambitious	Royal National Orthopaedic Hospital NHS Trust	0.330378
Ambitious	University Hospitals of Morecambe Bay NHS Foundation Trust	0.330378
Ambitious	Dorset County Hospital NHS Foundation Trust	0.327845
Ambitious	Epsom and St Helier University Hospitals NHS Trust	0.327845
Conservative	Croydon Health Services NHS Trust	0.338360
Conservative	Calderdale and Huddersfield NHS Foundation Trust	0.306977

Conservative	Harrogate and District NHS Foundation Trust	0.304301
Conservative	Barking Havering and Redbridge University Hospitals NHS Trust	0.298369
Medium	University Hospitals Dorset NHS Foundation Trust	0.342762
Medium	Cambridge University Hospitals NHS Foundation Trust	0.333394
Medium	Dorset County Hospital NHS Foundation Trust	0.327845
Medium	Epsom and St Helier University Hospitals NHS Trust	0.327845
Medium	Hull University Teaching Hospitals NHS Trust	0.314395

Table 11: Top five Trusts per scenario for current year (2024-25)

Insights for 2025–26

The first forecast year (2025–26) shows adoption likelihoods increasing across all scenarios, reflecting the cumulative effect of persistent demand pressures and the compounding mechanism of the probability model.

In the **Ambitious scenario**, large regional Trusts emerge as the most likely adopters. **Royal Surrey County Hospital NHS Foundation Trust**, **Royal Berkshire NHS Foundation Trust**, and **Medway NHS Foundation Trust** are highlighted as frontrunners. Their prominence demonstrates how accelerated adoption tends to favour larger providers with the capacity to integrate new technologies quickly, particularly when strong systemic support is assumed.

The **Medium scenario** again displays a balanced picture, with both large and mid-sized providers in the rankings. For instance, **Northumbria Healthcare NHS Foundation Trust** and **Dorset County Hospital NHS Foundation Trust** appear prominently, with adoption likelihood scores above 0.65. These results suggest that under moderate adoption conditions, Trusts from different tiers of capacity remain active participants in early diffusion.

In the **Conservative scenario**, smaller and mid-sized providers continue to play a role, with **Dartford and Gravesham NHS Trust** highlighted as a likely early adopter. Although overall probabilities remain lower than in more aggressive scenarios, the results indicate that slow adoption does not preclude smaller providers from making the transition, particularly where operational pressures are acute.

These findings are summarised in **Table 12**, which lists the top five Trusts under each scenario for 2025–26. Together with Table 12, the results provide a comparative perspective between the baseline year and the first forecast year. They show that while the set of frontrunner Trusts evolves, certain institutions—such as **Dorset County Hospital**—maintain their position across years and scenarios, reinforcing their status as consistent early adopters.

Scenario	Provider	Adoption Likelihood
Ambitious	Royal Surrey County Hospital NHS Foundation Trust	0.635806
Ambitious	Medway NHS Foundation Trust	0.633548
Ambitious	Royal Berkshire NHS Foundation Trust	0.628253
Ambitious	Chesterfield Royal Hospital NHS Foundation Trust	0.626350

Ambitious	East Cheshire NHS Trust	0.600476
Conservative	Dartford and Gravesham NHS Trust	0.614826
Medium	Northumbria Healthcare NHS Foundation Trust	0.655386
Medium	Dorset County Hospital NHS Foundation Trust	0.653334
Medium	Blackpool Teaching Hospitals NHS Foundation Trust	0.622886
Medium	South Tyneside and Sunderland NHS Foundation Trust	0.590957
Medium	Worcestershire Acute Hospitals NHS Trust	0.565916

Table 12: Top 5 Trusts per Scenario (2025–26)

4.2.2.4. Five-Year Trajectories and S-Curve Dynamics

Beyond single-year insights, the cumulative adoption trajectories reveal how adoption might unfold across the NHS system over the next five years. The results, illustrated in **Figure 35**, show three distinct diffusion pathways corresponding to the Conservative, Medium, and Ambitious scenarios.

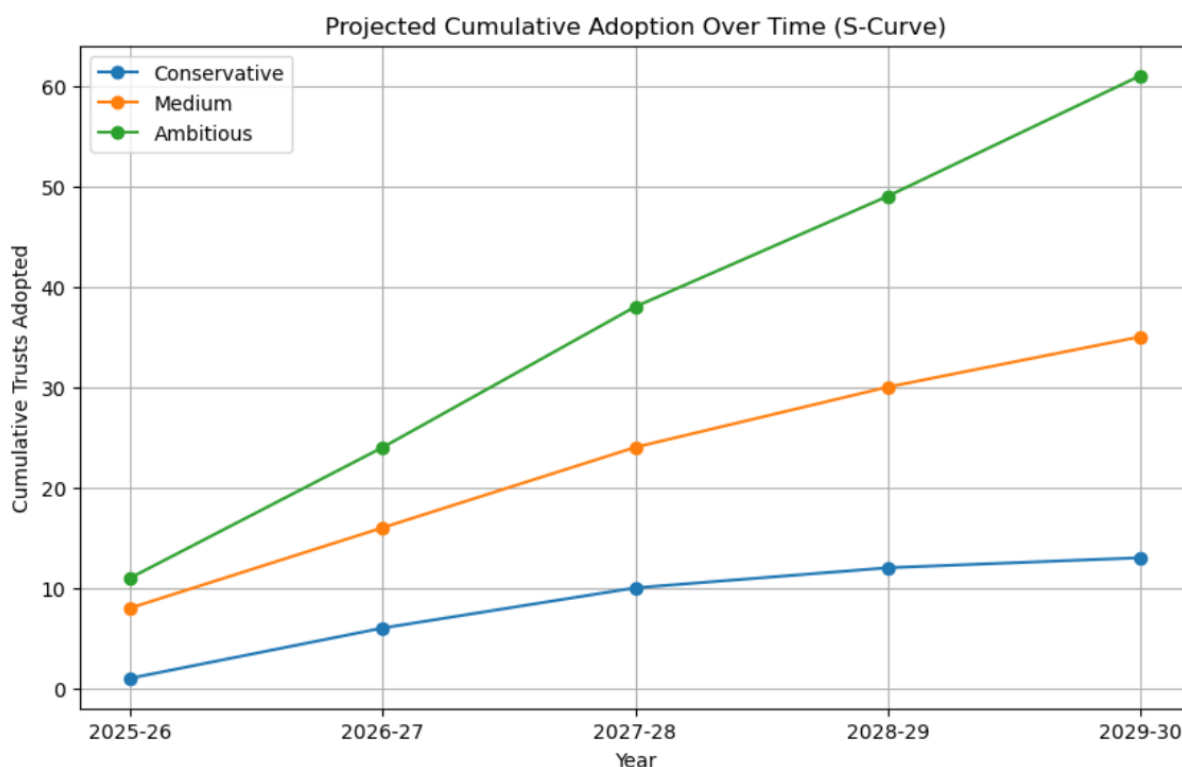


Figure 35: S-curves for next 5 years from 2025-26 to 2029-30

Under the **Conservative scenario**, adoption proceeds slowly, with only a limited number of Trusts adopting each year. The cumulative trajectory rises gradually, producing a shallow S-curve that suggests widespread diffusion could take many years to achieve. This scenario reflects a risk-averse environment where barriers such as funding constraints, scepticism about AI tools, or limited infrastructure slow the pace of adoption.

The **Medium scenario** demonstrates steady incremental growth. Adoption occurs at a consistent pace, producing a smoother and steeper curve than in the Conservative pathway, but without the rapid acceleration of the Ambitious scenario. This trajectory reflects a balanced environment where moderate incentives and policy support encourage adoption while allowing providers to adapt at a manageable pace.

The **Ambitious scenario** shows rapid early uptake, producing a steep S-curve that approaches saturation within the forecast period. This pathway assumes strong enabling conditions, such

as national policy mandates, targeted funding programmes, or widespread clinical advocacy. The results suggest that if such drivers are present, adoption could accelerate dramatically, leading to early system-wide benefits in terms of reduced waiting times and workload relief.

4.2.2.5. Strategic Implications

The scenario insights carry several implications for policymakers, vendors, and NHS decision-makers. First, the identification of **core early adopters**—Trusts that consistently appear across multiple scenarios and years—provides clear targets for pilot projects and early implementation efforts. For example, **Dorset County Hospital** and **University Hospitals Dorset** stand out as strong candidates for early engagement. Their consistent presence in the results suggests that their adoption is less dependent on external conditions, making them reliable entry points for initial rollout.

Second, the clustering of adoption likelihood scores indicates that external interventions could play a decisive role. Since differences between Trusts are relatively narrow, targeted engagement, vendor outreach, or policy-driven incentives could tip adoption towards particular providers. This implies that proactive strategy, rather than passive expectation, will be necessary to shape adoption trajectories.

Third, the differences across scenarios highlight the role of systemic factors in determining adoption speed. Without strong policy or financial support, adoption is likely to proceed cautiously, as in the Conservative scenario. With moderate support, adoption will spread steadily, while with strong intervention, adoption could accelerate rapidly. These dynamics underscore the importance of aligning institutional readiness with system-wide enablers to achieve widespread diffusion.

Finally, the S-curve analysis underscores the cumulative nature of adoption. Early adopters play a critical role in shaping system-wide diffusion by demonstrating feasibility, generating evidence, and reducing uncertainty. Trusts identified as early adopters can therefore serve as champions for broader rollout, helping to build momentum and reduce barriers for later adopters.

In summary, the adoption scenario analysis provides a structured view of how NHS Trusts may take up AI-based Brain MRI tools under different conditions. By combining demand scores with probability modelling and scenario simulations, it identifies likely early adopters and illustrates the pace of diffusion across Conservative, Medium, and Ambitious pathways. This section therefore establishes the groundwork for understanding adoption dynamics and informs the subsequent stages of analysis.

4.3. Analysing Private Market (George Dreyer-Wilton – 240669537)

Unlike the NHS, where MRI activity, adoption trends, and procurement processes are more regularly documented through publicly available data, information on private healthcare providers in England is far more limited. This lack of transparency makes it difficult to apply the same level of quantitative modelling used for the NHS. As a result, the analysis presented in this section draws on secondary sources, high-level estimates, and comparative benchmarks with NHS data to build a reasoned picture of the private market. The discussion will mirror the structure of the NHS-focused analysis in Sections 4.1–4.2 by considering market size and demand for MRI services, and potential adoption pathways for Quibim’s AI tool. However, given the limited availability of private-sector data, this section provides a less in-depth and more high-level assessment compared to the detailed NHS analysis.

4.3.1. Market Sizing

Earlier analysis estimated that approximately 58,850 brain MRI scans were conducted in England’s private sector in 2024/25, based on LaingBuisson’s finding that private activity represents around 5.8% of total imaging volumes (LaingBuisson, 2025). While this provides a useful starting point, the true size of the private neuroimaging market remains difficult to establish due to the limited transparency of private providers.

An alternative benchmark is offered by the Independent Healthcare Providers Network (IHPN), which reported that over 1.1 million private diagnostic scans were delivered across the UK in 2024, based on data from the nine largest independent providers (IHPN, 2025). While IHPN did not publish modality-level breakdowns, applying the proportion of MRI activity typically seen in NHS diagnostics (10–15% of total scans, as referenced in prior analysis) provides an estimated range of 110,000–165,000 MRI scans (NHS England, 2023). Assuming one-fifth of

MRIs are brain-related, in line with NHS modality distributions (NHS England, 2023), this equates to approximately 22,000–35,000 private brain MRI scans annually across the UK. Adjusting for England’s population share ($\approx 84\%$ of the UK total) yields a range of 19,000–29,000 scans (ONS, 2023).

This IHPN-derived estimate is notably lower than the LaingBuisson extrapolation of 58,850 scans. The discrepancy reflects important methodological differences: LaingBuisson’s approach scales private demand proportionally from NHS volumes, while IHPN’s data captures activity from a limited number of major providers and reports at a UK-wide level (LaingBuisson, 2025; IHPN, 2025). Taken together, these estimates suggest that the true level of private brain MRI activity in England likely lies within a broad range of 20,000 to 60,000 scans annually.

Beyond these national figures, the market is regionally concentrated, with London and other large metropolitan centres hosting the majority of private MRI capacity through providers such as HCA, Spire, Nuffield Health, and Circle Health Group (IHPN, 2025). Independent diagnostic chains such as Alliance Medical and InHealth also operate community diagnostic centres and mobile fleets, further supplementing provision (Alliance Medical, 2024; InHealth Group, 2024). However, coverage outside major cities remains patchy compared with the NHS, underscoring the uneven distribution of private MRI capacity.

Crucially, the private baseline is not static. Demand for private diagnostics has been growing at double-digit rates, largely in response to NHS backlog pressures (IHPN, 2025). If this trend continues at 10–15% annually, a baseline of around 60,000 private brain MRIs could expand towards 100,000 within five years (LaingBuisson, 2024). This dynamic indicates a progressively larger market opportunity for AI-enabled neuroimaging solutions such as Quibim Brain.

4.3.2. Drivers of Private Demand

The prolonged delays in NHS diagnostics are a major force fuelling growth in private MRI usage. NHS backlogs remain significantly elevated, even by late 2023, elective waiting lists contained approximately 7.6 million incomplete pathways nationally. Diagnostic waiting lists, which include MRI, rose from around 1.0 million in 2019 to approximately 1.6 million by the end of 2023, marking a 55% increase (Warner and Zaranko, 2024).

By contrast, private diagnostic providers are reporting strong growth: more than 1.1 million scans were delivered in 2024, with some providers experiencing demand growth of 20% or more (IHPN, 2025). The principal draw is access. Private services frequently offer MRI appointments within 48 hours and provide rapid turnaround of results, an assurance of timeliness and convenience unavailable in the NHS (IHPN, 2025).

A central feature of this growth is the expansion of the self-pay market. Research by the IHPN indicates that more than one in five people in England would now consider paying privately **for** diagnostics if they faced a long NHS wait (“Going Private”, 2025). The trend is particularly marked among working-age patients who cannot afford delays in care, as well as those seeking reassurance and control over the diagnostic pathway. Private brain MRI demand is therefore not solely driven by affluent patients but increasingly reflects a reaction to systemic NHS delays, with self-pay emerging as the dominant funding route for private imaging (LaingBuisson, 2024).

For Quibim Brain, this dynamic presents a compelling opportunity. Private providers facing rising volumes of self-paying patients are under pressure to differentiate on speed, accuracy, and patient experience; factors that align closely with the value proposition of AI-enabled neuroimaging.

4.3.3. Provider Distribution

Private diagnostic imaging capacity in England is unevenly distributed, with activity concentrated in London and other major urban hubs. Large hospital groups including HCA Healthcare UK, Spire Healthcare, Nuffield Health, and Circle Health Group account for the majority of hospital-based MRI provision, often embedded within wider inpatient services (IHPN, 2025). Alongside these groups, independent imaging specialists such as Alliance Medical and InHealth play a central role, operating community diagnostic centres (CDCs) and mobile scanner fleets that serve both private-pay patients and NHS contracts. Outside London, coverage is patchier, with fewer dedicated private facilities and more reliance on outsourced or mobile models. This regional imbalance suggests that early adoption of advanced imaging technologies such as Quibim Brain is likely to cluster initially in metropolitan centers where patient demand, infrastructure, and investment capacity are greatest.

4.3.4. Demand Drivers

Several factors underpin rising private demand for brain MRI in England. NHS waiting lists remain the most significant push factor: elective backlogs stood at 7.6 million incomplete pathways in late 2023, with diagnostic waiting lists expanding by 55% since 2019 (Warner & Zaranko, 2024). Long waits for MRI in particular have motivated patients to seek faster alternatives.

At the same time, self-pay growth has transformed private healthcare. The IHPN (2025) reported that one in three private patients now self-fund diagnostics, with working-age groups especially likely to pay for rapid imaging access. This reflects not only affluence but also the value placed on convenience and reassurance.

Private medical insurance remains a secondary driver: insurers fund a substantial share of MRI demand and increasingly encourage providers to adopt efficiency-enhancing technologies. Finally, demographic pressures add a structural driver. England's ageing population, with over 650,000 dementia patients and around 13,000 new brain tumor cases annually, generates steadily increasing demand for neuroimaging (Alzheimer's Society, 2024; Cancer Research UK, 2024). Together, these dynamics point to sustained expansion of the private brain MRI market.

4.3.5. Demand Barriers

Despite strong growth signals, several barriers may inhibit widespread adoption of Quibim Brain in the private imaging market. Affordability remains a major constraint: private MRI scan prices in the UK range from £199 to over £1,500, depending on the scan type and provider, with typical brain MRI costs between £299–£750 (GetScanned, 2025; Practice Plus Group, 2023). Layering extra AI processing fees may push total costs beyond what many self-pay patients are willing or able to bear. The fragmented structure of private healthcare is another challenge. Unlike the centrally managed NHS system, private diagnostic centres and hospital chains independently negotiate and invest, complicating scale-up and slowing coordinated adoption of new tools. Capacity constraints further limit growth. While private providers have expanded diagnostic units, they still operate significantly fewer MRI scanners than the NHS. Rising volumes risk outpacing available slots and radiologist capacity unless accompanied by equipment and workforce investment. Beyond structural issues, clinician acceptance and regulatory hesitancy pose real barriers. Radiologists may be cautious about medico-legal responsibility when employing algorithmic tools without robust oversight, and many providers await clearer regulatory guidance, such as MHRA approvals or professional endorsements, before integrating AI into routine workflows. This clinical and regulatory uncertainty may delay adoption even when demand and technical readiness are high.

4.3.6. Market Growth

Current evidence suggests rapid expansion. The private diagnostics market is growing at 10–15% annually, with some providers reporting year-on-year demand growth exceeding 20% (IHPN, 2025). In 2024, a record 1.1 million private diagnostic scans were delivered across the UK, demonstrating strong momentum in both self-pay and insurance-funded activity. Applying these growth rates to brain MRI volumes indicates significant upside potential. If current estimates of ~60,000 private brain MRIs annually in England grow at 10–15% per year, the market could approach 100,000 scans within five years. This would nearly double the current opportunity space.

For technology providers such as Quibim, these dynamics are highly favorable. Continued expansion not only increases the addressable market but also creates pressure on providers to adopt tools that enhance efficiency and differentiate their services. AI-based analysis platforms like Quibim Brain are therefore positioned to benefit directly from the sector's projected trajectory.

4.4. Adoption Within The Private Sector (George Dreyer-Wilton – 240669537)

While the NHS dominates diagnostic imaging in England, the private sector plays a growing and strategically important role in shaping early adoption of innovative technologies. For Quibim’s AI-enabled brain MRI solution, the private market offers a distinct pathway. It is smaller in overall size but often faster-moving, less bureaucratic, and more commercially flexible than the NHS. This makes private providers a critical testing ground and launchpad for establishing clinical credibility, patient acceptance, and demonstrable value.

Unlike the NHS-focused analysis in Section 4.2, which employs quantitative forecasting based on historical data, the private market lacks consistent, publicly available time-series data. Consequently, this section adopts a qualitative, scenario-based approach, drawing on secondary sources and broader adoption trends. The analysis focuses on three dimensions: which providers and regions are most likely to adopt early, the drivers and barriers that will influence adoption, and possible adoption scenarios and timelines over the rest of the decade. Together, these insights highlight how Quibim might navigate the smaller but opportunity-rich private landscape, and how private uptake may complement and accelerate eventual NHS adoption.

4.4.1. Potential Early Adopters

Given the concentration of MRI capacity within a relatively small number of large private providers, early adoption of AI-enabled tools such as Quibim Brain is likely to begin among the most resource-rich and innovation-oriented groups. These providers combine scale, established diagnostic infrastructure, and exposure to both self-pay and insured patient flows, which makes them well-positioned to experiment with technology that enhances efficiency and reporting quality. Adoption is also likely to cluster in metropolitan hubs, like London, where private diagnostic volumes are highest and competition for patients is most intense.

Among hospital-based providers, HCA Healthcare UK operates the largest network of private hospitals in London, with extensive diagnostic facilities and a track record of adopting advanced technologies (HCA Healthcare UK, 2024). Spire Healthcare, one of the UK's largest private hospital groups, has demonstrated a clear willingness to invest in diagnostic innovation, exemplified by the introduction of a new £1.4 million MRI scanner at Spire St Anthony's Hospital. This system incorporates AI-based reconstruction software to deliver faster, higher-quality scans (Spire Healthcare, 2023). This suggests both financial capacity and organizational readiness to integrate AI into diagnostic workflows, positioning Spire as a likely early adopter of neuroimaging AI such as Quibim Brain. Nuffield Health and Circle Health Group also represent significant hospital networks with substantial diagnostic activity and a strategic focus on improving patient experience, making them logical candidates for early uptake (Nuffield Health, 2024; Circle Health Group, 2024).

Independent imaging providers also play a critical role. Alliance Medical operates one of the largest imaging fleets in Europe, providing both fixed-site and mobile MRI services while balancing NHS contracts and private self-pay demand (Alliance Medical, 2024). Similarly, the InHealth Group runs an extensive network of community diagnostic centres and mobile services, offering flexibility in deploying new tools across diverse care settings (InHealth, 2024). Both organisations, given their scale and operational agility, are well-positioned to trial AI-enabled solutions that deliver measurable efficiency gains.

4.4.2. Adoption Drivers

Adoption of AI-enabled imaging in the private sector will be shaped by several reinforcing drivers. The most immediate is the pressure created by NHS backlogs, which has redirected patients toward private providers that can provide faster diagnosis. By late 2023, elective waiting lists contained 7.6 million incomplete pathways nationally, while diagnostic waiting

lists rose by 55% since 2019, a trend which has pushed more patients to self-pay for quicker access (Warner and Zaranko, 2024; IHPN, 2025). While this surge has fueled self-pay growth, continued migration could eventually strain private capacity, making efficiency-enhancing tools like Quibim Brain a preventative investment.

A second driver is competitive differentiation. Private groups must demonstrate value to both self-pay and insured patients, where speed, accuracy, and quality of reporting act as decisive factors. Research by the Independent Healthcare Providers Network shows that competition in urban hubs, particularly London, is pushing providers to invest in technology that enhances diagnostic capability and patient experience (IHPN, 2025).

Insurers are also likely to shape adoption patterns. With cost pressures rising, private medical insurance providers have an incentive to support AI tools that can reduce reporting time and radiologist workload, thereby lowering overall claim costs while maintaining quality (LaingBuisson, 2024).

Finally, structural demographics provide a steady baseline driver of demand. NHS England projects that the proportion of people aged 65 and over will grow substantially over the next decade, increasing the prevalence of dementia, stroke, and tumor cases requiring neuroimaging (NHS England, 2023). Together, these forces provide a strong rationale for early private-sector experimentation with AI diagnostics, particularly in high-demand metropolitan centers.

4.4.3. Adoption Limiters

Despite clear incentives, several barriers may slow the uptake of Quibim Brain in the private sector. The most immediate is affordability. Self-pay MRI scans typically cost between £400 and £700 in the UK, and layering additional AI-related fees could restrict uptake, particularly outside affluent groups, unless insurers absorb or offset the cost (LaingBuisson, 2024).

Return on investment also remains an uncertainty. Many providers are cautious about committing capital to AI tools without robust evidence of efficiency gains, diagnostic accuracy improvements, or revenue impact. Studies highlight that integration into existing workflows, particularly with Picture Archiving and Communication Systems (PACS) and Radiology Information Systems (RIS), often requires additional investment and training, which can limit early adoption (Pesapane et al., 2018).

The fragmented structure of private healthcare further complicates scaling, with decisions made independently by hospital groups, diagnostic firms, and smaller imaging centres rather than through unified procurement pathways. This contrasts with the NHS, where national initiatives can drive systemic adoption.

Finally, the absence of widespread regulatory endorsement or standardisation for AI in diagnostics reinforces hesitation. The UK's Medicines and Healthcare products Regulatory Agency (MHRA) is still in the process of updating guidance on Software as a Medical Device (SaMD), including AI algorithms, with a new framework due by 2025 (MHRA, 2023). Until clearer standards are widely implemented, clinicians may be reluctant to adopt tools without formal validation, particularly in high-stakes neuroimaging applications.

4.4.4. Adoption Scenarios

Conservative Adoption: In this scenario, only a small number of early adopters pilot Quibim Brain between 2025 and 2026, primarily among innovation-oriented groups in London. Adoption remains limited through 2030, with around 10% of private brain MRI scans incorporating AI as an auxiliary tool. This outcome reflects a cautious environment, where regulatory uncertainty and cost considerations slow diffusion despite proven technical feasibility (MHRA, 2023).

Moderate Adoption: Under this trajectory, early pilots demonstrate efficiency gains and reductions in reporting delays, encouraging broader uptake. By 2027, many private hospital groups and diagnostic chains begin integration, with insurers playing a role in incentivizing adoption. By 2030, AI-enabled scans could encompass 40–50% of private brain MRI volumes. This trajectory aligns with the NHS Long Term Plan and the England 10-Year Health Plan, both of which highlight AI as a strategic lever to address workforce shortages and backlog pressures (NHS England, 2019; Azmed, 2025).

Accelerated Adoption: In this optimistic case, systemic pressures, including insurer incentives, self-pay patient demand, and spillover effects from NHS partnerships, converge with regulatory momentum. The UK government has prioritized scaling AI-enabled imaging under the AI Life Sciences Vision and NHS AI Lab funding schemes, which could accelerate validation and adoption (Office for Life Sciences, 2021; NHS AI Lab, 2023). With regulatory clarity and demonstrated efficiency, Quibim Brain and similar tools could achieve 70–80% penetration of private brain MRI scans by 2030, particularly in metropolitan hubs where competitive pressures are strongest.

4.5. Determining Ideal Pricing and Value-Based Proposal – Competitor Analysis, Pricing Strategy, Market-entry Recommendation (Nisha Sunariya – 240847906)

4.5.1. Quibim’s Market Strategy: Competitor Insights, Pricing Models & NHS Entry Pathways

4.5.1.1. Introduction

Artificial Intelligence (AI) is transforming medical imaging, moving beyond the experimental stage to become a vital tool for improving diagnostic accuracy, operational efficiency, and patient outcomes. Nowhere is this shift more critical than in the United Kingdom, where the National Health Service (NHS)—the world’s largest single-payer healthcare system—faces growing demand for imaging alongside a chronic shortage of radiologists. In this context, AI solutions for MRI, particularly in prostate cancer detection and brain imaging, are emerging as game changers.

Quibim, a company specialising in advanced MRI image quantification and imaging biomarker extraction, is well positioned to address these challenges. Its technology can uncover sub-visual insights in oncology and neurology, improving detection, reducing unnecessary procedures, and enhancing reporting efficiency. For adoption in the UK, however, success will depend on more than technological strength. Quibim must demonstrate how its MRI-based solutions deliver measurable value to the NHS by reducing unnecessary prostate biopsies, improving turnaround times in brain and prostate MRI, enhancing consistency across radiologists, and ultimately generating a strong return on investment (ROI) for both trusts and private providers.

At the same time, Quibim must navigate the realities of the NHS procurement environment, which is shaped by fixed diagnostic tariffs, strict cost-effectiveness requirements, and high

expectations for regulatory compliance. Demonstrating UKCA marking and alignment with the NHS Digital Technology Assessment Criteria (DTAC) will be essential to reduce buyer risk. Equally, fitting within existing IT infrastructures, such as PACS and national AI marketplaces, will be key to smooth integration and wider adoption.

This report provides a structured analysis to support Quibim’s strategy for adoption in England. It begins with a review of the competitor landscape, examining how established vendors such as Brainomix, Icometrix, Aidoc, Qure.ai, Oxipit, Lucida Medical, and ScreenPoint have positioned themselves in the UK market. It then analyses pricing benchmarks using NHS tender documents, NICE guidelines, and competitor evidence to define realistic per-study and subscription ranges for MRI AI. Building on this, the report proposes tailored pricing models for NHS and private providers, alongside potential market entry pathways through AI marketplaces and private hospital partnerships. Finally, it outlines a value-based proposition that articulates the clinical, operational, and financial benefits Quibim can deliver, ensuring its MRI solutions are positioned as both credible and sustainable in the UK healthcare system.

By anchoring its strategy around MRI scans—where demand and clinical impact are greatest—Quibim can present itself as a differentiated partner to both NHS and private providers, aligning technological innovation with the system’s pressing clinical and financial needs.

4.5.1.2. UK Market Context: Why AI Imaging is a Strategic Priority

a. NHS Pressures and Diagnostic Gaps

The NHS is facing one of the most severe diagnostic backlogs in its history. By May 2024, more than 1.7 million patients were waiting for a diagnostic test, a record-breaking figure that highlights the scale of the challenge. The situation is particularly acute for MRI, where in many regions patients are now waiting more than 12 weeks for a scan. These delays cause considerable anxiety for patients and, more critically, increase the likelihood that conditions

such as prostate cancer or neurological disorders are detected at more advanced, harder-to-treat stages.

This challenge has been building for years. According to NHS England, the number of patients waiting longer than six weeks for a diagnostic test has doubled between 2019 and 2022, with MRI services among the most heavily affected. This is not just an equipment issue; it is above all a workforce crisis. The 2023 Royal College of Radiologists (RCR) census revealed that 97% of clinical leads do not have enough consultant radiologists to provide safe and effective services. Looking ahead, the RCR warns of a 30% shortfall in consultant radiologists by 2030 if training and recruitment are not significantly accelerated. With demand for scans growing far faster than the workforce, traditional solutions such as hiring more staff or buying more machines are unlikely to be sufficient.

This mismatch between capacity and demand creates what many experts describe as a “perfect storm” for diagnostic services. Without new approaches, waiting lists will continue to climb, patients will face harmful delays, and the NHS will be forced to spend more on costly late-stage treatments.

In this environment, the adoption of AI-driven imaging tools is no longer just an option—it has become essential. AI can help radiology departments cope with rising demand by speeding up reporting, prioritising urgent cases, and standardising interpretation across sites. Importantly, it can also provide data-driven insights that improve diagnostic accuracy and consistency. The NHS itself has recognised this potential: policy initiatives such as the NHS Long Term Plan and the AI in Health and Care Award scheme explicitly support the use of AI in diagnostics, highlighting a clear willingness to fund and adopt technologies that can address these systemic gaps.

For vendors like Quibim, which specialise in MRI quantification in oncology and neurology, this moment represents both a challenge and an opportunity. By demonstrating robust clinical validation, seamless IT integration, and measurable efficiency gains, Quibim can position its AI platform not as a luxury, but as a necessary partner for the NHS in tackling one of the most pressing operational crises in modern healthcare.

b. The Role of NICE and Procurement Transparency

In many markets, pricing and procurement can be hard to track. In the UK, though, organizations like NICE and frameworks like G-Cloud help make things more transparent. NICE shares evaluation summaries with cost-effectiveness models, and NHS tenders sometimes reveal vendor pricing. This transparency helps set realistic benchmarks for AI imaging solutions.

For instance, Lucida Medical's prostate AI pricing is documented within NHS procurement materials, giving competitors a rare insight into the per-study and subscription ranges accepted by NHS buyers. Similarly, NICE guidance on AI solutions like Paige Prostate provides per-use costs that can be extrapolated to adjacent applications. These references form an invaluable foundation for Quibim's pricing strategy.

c. Private Sector Demand for Imaging AI

Even though the NHS controls the majority of healthcare in the UK, the private sector is a significant avenue for AI adoption. Diagnostic centers are operated by private companies including Nuffield Health, Spire Healthcare, LivingCare, Vista Health, Circle Health, and Alliance Medical, and they accept both NHS contracts and self-paying patients. These organizations frequently adopt new technologies faster than the NHS and have a tendency to be more agile in their procurement. Before expanding into NHS frameworks, they provide a trial ground for AI suppliers to demonstrate their worth.

Early private-sector alliances might help Quibim prove ROI in a real-world UK context, producing case studies and enhancing its reputation, all of which would bolster its NHS offer.

d. AI Marketplaces as a Critical Enabler

In recent years, AI imaging marketplaces have emerged as key intermediaries for NHS adoption. Platforms such as **Sectra Amplifier, CARPL.ai, DeepcOS, and Nuance AI Marketplace** allow hospitals to access multiple AI solutions through a single integration, streamlining IT burden, procurement, and compliance. For vendors, these marketplaces reduce overhead, accelerate onboarding, and lend credibility through association with trusted intermediaries.

In fact, many NHS trusts now prefer to access AI through marketplaces rather than contracting directly with individual vendors. This makes marketplace participation not optional but strategic. For Quibim, joining one or more marketplaces could represent the fastest route to national visibility.

To structure the report, we focus on three central strategic questions :

1. Who are our competitors?

What do they offer, and how do their characteristics differ from Quibim?

2. What should be our target pricing?

Which pricing modalities are available, and what benchmarks can we rely on?

3. What are the optimal market entry channels?

Which marketplaces or distributors provide the best access to the NHS and private sector?

Each section of the report is designed to provide both evidence and interpretation, culminating in a final recommendation for Quibim's UK market strategy.

e. The Regulatory Gateway: UKCA Marking and DTAC

Alongside commercial strategy and clinical validation, no company can succeed in the UK health system without clearing the regulatory hurdles that govern adoption. Since leaving the EU, the UK has introduced its own standard for medical devices: the UKCA (UK Conformity Assessed) mark, which now replaces the CE mark for products entering the British market. For Quibim, this means ensuring that its platform and individual MRI algorithms are fully certified under UKCA requirements before approaching NHS procurement teams.

Equally important is the NHS Digital Technology Assessment Criteria (DTAC). Increasingly, NHS trusts use this framework as a common yardstick when evaluating digital health technologies. The DTAC sets out five core requirements: clinical safety, data protection, technical security, interoperability, and usability. A completed DTAC self-assessment has quickly become a standard request during procurement, and technologies that fail to meet these benchmarks rarely make it past the evaluation stage.

For Quibim, proactive compliance with DTAC is not just a regulatory formality—it is a way of building confidence with NHS decision-makers. It demonstrates that the company is prepared to meet the same stringent standards as any other trusted supplier in the UK health system. By addressing UKCA certification and DTAC requirements early, Quibim can reduce perceived risk, shorten procurement timelines, and strengthen its reputation as a credible, ready-to-scale partner for both NHS and private providers.

4.5.2. Competitor Landscape in the UK AI Imaging Market

4.5.2.1. Why Competitor Mapping Matters

i. Brainomix

Brainomix is one of the most recognisable UK-based AI imaging companies, best known for its *e-Stroke* platform. This solution has been widely adopted across stroke networks and is endorsed in NHS stroke pathways. Its main advantage is quickly spotting large vessel blockages and bleeding in the brain, which helps doctors make faster treatment decisions and shortens the time to treatment for stroke patients.

Brainomix has secured both CE and UKCA approvals and has established a significant presence within the NHS, with deployments across multiple trusts. Its main strength lies in a strong brand association with stroke AI, where it is deeply embedded within NHS stroke pathways. However, its focus remains relatively narrow, concentrating primarily on stroke with limited expansion into other organ systems. For Quibim, the Brainomix example highlights the importance of aligning AI solutions with national clinical priorities. By positioning its prostate MRI solution in line with the NHS ambition of early cancer detection, while also emphasising its broader multi-organ platform, Quibim can demonstrate both immediate clinical relevance and long-term strategic value.

ii. Icometrix

Icometrix is known in neurology for its work in monitoring multiple sclerosis using brain MRI scans. The *icobrain* platform helps neurologists and radiologists measure lesion load and brain atrophy, making it easier to track disease changes over time.

Icometrix has achieved both CE marking and FDA clearance, establishing its credibility as a clinically validated AI provider. Within the UK, its NHS presence remains limited to pilot

projects, though the company benefits from strong academic credibility. Its key strength lies in the delivery of high-quality quantitative biomarker extraction for brain imaging, making it a trusted solution in neurology. However, its narrow focus on neurological applications limits its suitability for oncology workflows or broader radiology integration. For Quibim, this presents an opportunity to differentiate by positioning its brain MRI solutions within oncology and across multiple domains, rather than restricting applications to neurological monitoring alone.

iii. **Aidoc**

Aidoc is perhaps the most globally visible competitor. Its solutions span multiple acute care scenarios, including pulmonary embolism, intracranial haemorrhage, and cervical spine fractures. Aidoc's strength lies in its enterprise-wide integration, embedding AI directly into PACS and radiology workflows.

Aidoc holds multiple CE and FDA certifications, giving it strong international credibility. In the UK, its NHS presence is still limited to early pilot projects, with wider adoption held back mainly by its higher costs. The company's main strengths are the wide range of conditions it covers, smooth workflow integration, and a strong global reputation. However, its high cost structure and enterprise-level focus do not always fit well with the NHS's tighter budget constraints. For Quibim, this opens up an opportunity to stand out by offering a more focused and cost-efficient solution, particularly for prostate and brain MRI pathways, where affordability is just as important as performance.

iv. **Qure.ai**

Qure.ai, originally from India, has gained traction globally by focusing on affordable triage solutions for chest X-rays and head CT scans. Its solutions are often used in emergency and low-resource settings, but it has also begun pilot projects with the NHS.

Qure.ai is CE certified and has begun engaging with the NHS through limited pilot projects, although its adoption in the UK is still at an early stage. The company is known for its affordable pricing, scalability, and strong track record in low- and middle-income countries (LMICs), which has helped build its reputation globally. At the same time, its focus is narrower, with less emphasis on advanced imaging areas like MRI and limited depth across multiple organs. For Quibim, this highlights an opportunity: while Qure.ai shows that affordability can be a strong differentiator, Quibim's strength lies in more sophisticated imaging modalities such as prostate MRI, enabling it to target a higher-value niche within the NHS.

v. Oxipit

Oxipit has pioneered autonomous chest X-ray reporting, with products like *ChestEye Quality* and *ChestEye CAD*. Its ambition to deliver fully autonomous reporting is bold, but adoption within the NHS remains limited.

Oxipit is CE certified, but its NHS presence remains very limited. The company is recognised for its innovative approach to autonomous reporting and its strong focus on quality control. However, its scope is narrow, as it works exclusively with chest X-rays and has not gained much traction in more advanced imaging areas. For Quibim, the key takeaway is that while Oxipit proves that autonomy in reporting is technically possible, the NHS remains cautious about fully autonomous AI. This creates an opportunity for Quibim to position itself differently—by emphasising augmented intelligence that supports and empowers radiologists, rather than aiming to replace them.

vi. Lucida Medical

Lucida is arguably Quibim's most direct competitor in the UK. Its *Prostate Intelligence (Pi)* software has been validated in NHS clinical trials and is explicitly listed in NHS procurement

frameworks. Importantly, Lucida's pricing is publicly documented through NHS G-Cloud tenders, with per-study rates around £15–£20 and subscription tiers also available.

Lucida has already secured CE and UKCA approvals and has built a strong presence within the NHS, being actively involved in multiple clinical studies and procurement frameworks. Its main strengths lie in its specific focus on prostate MRI, deep engagement with the NHS, and published NICE evaluations that reinforce its credibility. However, its offering is somewhat limited by a narrow organ focus, lacking the broader multi-organ capabilities that some competitors are developing. For Quibim, this positions Lucida as the key benchmark competitor. To compete effectively, Quibim will need to at least match Lucida's pricing model while standing out through its wider multi-organ coverage, more advanced biomarker capabilities, and stronger integration into oncology workflows.

vii. **ScreenPoint Medical**

ScreenPoint works only in breast cancer screening, and its *Transpara* solution is already used in Europe and the UK. Although it does not compete directly with Quibim's focus on prostate and brain MRI, its success with the NHS shows how niche vendors can thrive by proving clinical value and becoming part of screening programs.

ScreenPoint has obtained both CE and FDA approvals and is already deployed within the NHS in breast cancer screening settings. Its strengths lie in a strong focus on breast imaging and robust clinical validation, which add credibility and trust to its offering. However, its scope is limited to breast imaging, meaning it is not a direct competitor to Quibim's broader platform. For Quibim, the key takeaway is the importance of clinical validation—ScreenPoint's success highlights how crucial strong evidence is. Quibim should mirror this approach by generating solid data from prostate cancer detection trials that can later support submissions to NICE.

Comparative Competitor Table

Vendor	Focus Area	Regulatory Status	NHS Presence	Strengths	Limitations
Brainomix	Stroke AI	CE, UKCA	Strong	NHS stroke alignment, brand credibility	Stroke-only, narrow scope
Icometrix	Brain MRI (MS)	CE, FDA	Limited pilots	High-quality biomarkers, neurology focus	Narrow domain, limited oncology focus
Aidoc	Multi-condition	CE, FDA	Limited pilots	Breadth, workflow integration, global brand	High cost, enterprise focus
Qure.ai	Chest X-ray, CT	CE	Limited pilots	Affordable, scalable, LMIC success	Limited MRI and oncology focus
Oxipit	Chest X-ray	CE	Minimal	Autonomous reporting, QA focus	Chest-only, limited traction
Lucida Medical	Prostate MRI	CE, UKCA	Strong	NHS engagement, pricing transparency	Narrow organ scope
ScreenPoint Med.	Breast Imaging	CE, FDA	Deployed in NHS	Screening programme validation	Not overlapping with prostate/brain MRI

Table 13 Comparative Competitor Table

4.5.3. Quibim's Competitive Position and SWOT Analysis

From this detailed mapping, three strategic insights emerge for Quibim:

i. MRI and Multi-Organ as a Key Differentiator: The competitor landscape is made up of many organ-specific solutions, with most focusing on non-MRI methods like CT (Aidoc, Qure.ai) or X-ray (Oxipit). Quibim stands out by specializing in advanced MRI quantification, especially for oncology and neurology, which helps solve complex clinical problems that rely

on quantitative data. Its platform also delivers insights for several organs, such as the prostate, brain, and liver, all in one place. This makes it easier for hospitals by reducing IT complexity and simplifying procurement, compared to working with several single-use vendors.

ii. Lucida Anchors Pricing, but Paige Defines the Ceiling: Lucida Medical's published NHS pricing (~£15–£20 per prostate MRI study) provides an undeniable market anchor. Ignoring this benchmark risks exclusion from tenders. However, Paige Prostate demonstrates that with strong evidence, pricing can reach £25–£40 per use. Quibim's strategy must therefore be to align its entry-point pricing near Lucida's range while relentlessly building the clinical validation necessary to justify a premium, value-based price point through subscription and hybrid models.

iii. Validation and Integration are Non-Negotiable: Competitors that have gained significant NHS traction, such as Brainomix and ScreenPoint, succeeded through strong clinical validation aligned with national priorities (e.g., stroke networks, breast screening). For Quibim, producing early, robust validation evidence within UK prostate MRI pathways and brain imaging applications will be the single most critical factor for adoption. Furthermore, a seamless integration strategy, preferably through established AI marketplaces, is essential to overcome NHS IT constraints.

SWOT Analysis

Strengths: Superior multi-organ quantitative platform; advanced biomarker expertise; strong research pedigree.

Weaknesses: Limited UK-specific clinical validation; less brand recognition than incumbents; no current UKCA mark.

Opportunities: NHS demand for efficiency tools; gap in the market for a validated multi-organ platform; private sector agility for pilot studies.

Threats: Lucida’s first-mover advantage in prostate; potential price wars; long and complex NHS procurement cycles.

4.5.4. Pricing Models & NHS Benchmarks

Why Pricing is Strategic in AI Imaging

Pricing in the healthcare AI market is not simply about covering costs or undercutting competitors; it is about signalling value to hospitals, aligning with NHS procurement expectations, and ensuring sustainable adoption. For the NHS in particular, pricing strategies must balance affordability at scale with clear evidence of return on investment.

Vendors rarely publish list prices, leaving hospitals and trusts to negotiate directly. However, in the UK, NHS procurement frameworks, NICE reports, and tender documents provide rare but valuable insights into real-world cost structures for AI imaging. For example, Lucida Medical’s prostate AI pricing (~£15–£20 per study) and Paige Prostate’s higher per-use pricing (£25–£40) set a useful benchmark for organ-specific imaging.

When it comes to brain MRI, the baseline scan cost is significantly higher—typically £200–£350 in the NHS and £250–£750 in the private sector. Against this backdrop, an AI add-on priced at £15–£30 per scan would represent only a small fraction of the total cost, but could unlock significant value by reducing reporting times, standardising results, and improving early detection. Positioning brain MRI AI pricing within this range not only mirrors prostate benchmarks but also signals proportional value in relation to the higher base scan costs.

These insights form the foundation for Quibim’s pricing strategy: align with NHS expectations by staying within the £15–£30 range per study, while tailoring offers for private providers and high-volume trusts through subscription or value-based models.

Pricing Modalities in Medical AI

Vendors typically use one of four primary pricing modalities, often blending them depending on the size and needs of the healthcare provider:

i. Per-Study Pricing (Pay-Per-Use):

- Hospitals are charged for each scan processed by the AI.
- Benefits: Simple, scalable, attractive for smaller hospitals or pilot projects.
- Risks: High per-scan costs may deter large-volume adoption.

ii. Annual Subscription/Licensing:

- Flat fee for unlimited (or high-volume capped) usage over a period.
- Benefits: Predictable cost for hospitals, revenue stability for vendors.
- Risks: May be unaffordable for smaller NHS trusts; requires strong proof of ROI.

iii. Tiered Hybrid Models(Try before you buy):

- Combination of per-study and subscription pricing. For example, a base subscription plus reduced per-study fees above a threshold.
- Benefits: Balances affordability with scalability; aligns well with NHS procurement expectations.
- Risks: Complexity of contracts can slow procurement negotiations.

iv. Outcome-Based/ Value-Based Pricing:

- Fees tied to measurable outcomes, such as reduced reporting times or fewer missed cancers.
- Benefits: Strongly aligned with NHS focus on value-based healthcare.

- Risks: Complex to implement; requires real-world validation and trust.

For Quibim, the most viable approach will likely be a **tiered hybrid model**, aligning with Lucida’s disclosed benchmarks while also differentiating through a value-based ROI proposition.

4.5.5. NHS Benchmarks and Publicly Available Data

4.5.5.1. Lucida Medical – The Key Benchmark

Lucida’s *Prostate Intelligence (Pi)* solution provides the clearest publicly available insight into NHS-accepted AI pricing. According to its published G-Cloud 14 tender document (2024), Lucida offers highly competitive per-study pricing, starting as low as £15 per prostate MRI scan for high-volume deployments. It also provides subscription-based tiers for trusts seeking unlimited access, giving NHS buyers flexibility to match their purchasing preferences.

This transparency is rare in the AI imaging market and gives Quibim a clear price anchor. Entering the NHS market significantly above Lucida’s per-study rate would risk exclusion from tenders. However, Quibim can justify modestly higher pricing if it demonstrates added value—such as broader multi-organ coverage (prostate and brain) or measurable efficiency improvements in reporting and early detection.

4.3.5.2. The Role of NHS Tariffs and Reimbursement

A critical factor in NHS procurement is the National Tariff Payment System, which sets fixed reimbursement prices for thousands of healthcare activities, including diagnostic imaging. For instance, a multi-parametric prostate MRI is reimbursed at approximately £200–£300, while brain MRI scans are typically reimbursed in a similar range (£200–

£350 depending on trust and scanner type). These tariffs must cover all associated costs: staff, equipment, overheads, and, increasingly, digital tools like AI.

Therefore, any AI cost must be framed in one of two ways:

- **Efficiency-neutral:** absorbed within the existing tariff by generating measurable savings (e.g., reduced reporting time, avoidance of unnecessary biopsies).
- **Value-adding:** justified as an additional cost that improves outcomes so significantly that it saves the system money downstream (e.g., earlier detection of brain tumours or reduced late-stage treatment costs).

For Quibim, this means positioning its AI not as an “extra expense” but as a **strategic investment** that strengthens tariff sustainability and delivers financial and clinical value at scale.

4.3.5.3 NICE Guideline DG57 (CT Scan Benchmarks)

The NICE DG57 guideline on AI tools for chest CT provides useful benchmarks, listing vendor-reported costs in the range of £20–£50 per scan depending on deployment model. While this evidence is CT-specific, it strongly suggests that MRI AI pricing should align with or slightly exceed CT benchmarks, given the higher clinical complexity of MRI pathways.

For prostate MRI, this reinforces the £15–£30 per-scan expectation seen in NHS tenders. For brain MRI, where baseline scan costs are higher and efficiency gains may be even more impactful, positioning within the £20–£35 range is both realistic and defensible.

Other NHS Cost References

- The cost of a **prostate MRI** scan in the NHS: **£200–£300** (depending on trust and scanner type).
- The cost of a **brain MRI** scan in the NHS: **£200–£350**, with private providers charging £250–£750.
- AI per-scan costs in the **£15–£30 range** therefore represent only **5–10% of the total scan cost**—a proportion that is generally acceptable if the AI demonstrably delivers efficiency, consistency, and clinical value.

4.5.6. Innovative Pricing Models in MedTech

Beyond traditional pricing, vendors and consultants (e.g., IDR Medical) have advocated for **innovative medtech pricing models**, including:

- **Risk-sharing agreements:** Vendors only receive full payment if performance targets are achieved.
- **Per-patient episode bundles:** AI priced as part of a broader diagnostic bundle (e.g., prostate MRI+AI reporting).
- **Enterprise “all-you-can-use” licenses:** Larger hospital groups or private providers pay a fixed fee to access all AI modules across imaging modalities.

These models are not yet widespread in the NHS but may represent an opportunity for Quibim to **differentiate its offer**, especially when engaging with large private groups such as Alliance Medical or Spire.

From this analysis, three pricing lessons emerge:

Lesson	Insight	Implication for Quibim
Benchmark: Lucida Sets the Baseline	Lucida’s NHS pricing (~£15–£20 per prostate MRI study) is the current reference point.	Quibim should align entry-level pricing in this range for NHS acceptance.
Flexibility: Multiple Models Needed	NHS trusts vary in size and purchasing preferences.	Quibim should offer both per-study pricing and annual subscription tiers .
Value-Based Differentiation	NHS accepts higher pricing (Paige £25–£40 per case) when strong ROI is demonstrated.	If Quibim shows measurable ROI (e.g., fewer biopsies, faster reporting), it can justify pricing at £25–£30 per scan .

Table 14 Key Pricing Lessons for Quibim

4.5.6.1. Recommended Pricing Strategy for Quibim

The Need for a Balanced Pricing Approach

Pricing in the AI imaging market is not simply about covering costs; it is a strategic tool that influences adoption speed, NHS procurement decisions, and vendor credibility. For Quibim, designing the right pricing strategy in the UK is especially important because the NHS is highly cost-sensitive, with procurement teams demanding both affordability and strong evidence of clinical value.

Competitor benchmarks have already shaped expectations. **Lucida Medical** has set the lower bound with prostate MRI pricing at around **£15–£20 per scan**, while premium solutions such as **Paige Prostate** have demonstrated that the NHS is prepared to pay **£25–£40 per scan** when strong validation shows clear benefits. In parallel, **NICE’s DG57 guidance** and other CT

benchmarks reinforce that AI for imaging is generally expected to fall within the **£20–£50 range per study**.

For **brain MRI**, which is a more complex modality and where efficiency gains (e.g., faster reporting, earlier tumour detection) may be even more impactful, a **£20–£35 per scan range** is both realistic and defensible. This ensures Quibim is aligned with NHS cost thresholds while also signalling added value beyond basic prostate applications.

In contrast, **private providers** (such as Spire Healthcare, Nuffield Health, or Alliance Medical) are less constrained by tariff systems and more willing to pay premium prices if technology directly improves patient experience and workflow efficiency. This creates room for higher-margin opportunities alongside NHS-focused pricing.

Proposed Pricing Models for Quibim

Quibim should structure its offering around **three main pricing modalities**, ensuring that each aligns with market expectations:

- i. **Per-Study (Pay-Per-Use) Model (£18–£22 for prostate MRI; £20–£35 for brain MRI)** : This entry-level model is designed for **small NHS trusts or hospitals piloting AI solutions** with relatively low annual scan volumes (fewer than 2,000 prostate or brain MRIs). By aligning closely with Lucida Medical’s benchmark (£15–£20 for prostate) but signalling additional value through advanced biomarkers and multi-organ capability, Quibim strikes a balance between affordability and differentiation. The benefit of this model lies in its **low barrier to entry**. Trusts that are hesitant to commit to large upfront investments can adopt AI on a per-use basis, gaining confidence in the technology before scaling. For Quibim, this model supports faster adoption and

creates a pathway to transition satisfied pilot sites into larger subscription contracts over time.

- ii. **Annual Subscription (£50,000–£70,000 per trust):** An annual subscription model provides **unlimited access to Quibim’s prostate and brain MRI AI tools** across an entire trust or hospital group. This model is particularly suitable for **large NHS trusts, imaging networks, and private providers** (e.g., Spire Healthcare, Alliance Medical, Nuffield Health), which deal with high imaging volumes and require predictable budgeting. The key advantage here is **simplicity and scalability**. NHS procurement teams often prefer fixed annual costs over variable per-study pricing, as it makes budget planning easier. For Quibim, subscriptions deliver **stable recurring revenue**, allowing the company to forecast growth and allocate resources more effectively. Framing the subscription cost as comparable to “the annual salary of one consultant radiologist” also strengthens the ROI narrative, since the software can deliver equivalent productivity gains.
- iii. **Hybrid Model (Base £20,000 annually + £10–£12 per scan beyond 1,500 scans):** This flexible approach combines the predictability of a base subscription with **volume-based scalability**. It is well-suited to **mid-sized NHS trusts and regional imaging centres** that sit between pilot-level adoption and full enterprise deployment. By charging a lower flat annual fee, the hybrid model ensures affordability for trusts with modest volumes. At the same time, adding a reduced per-scan fee for usage above 1,500 scans incentivises higher adoption without runaway costs. This model mirrors NHS expectations of “**fair and scalable**” pricing structures and signals Quibim’s willingness to adapt to the operational realities of different trusts. It also creates a **stepping stone** for

mid-volume sites to eventually upgrade to full subscriptions once their usage grows.

- iv. **Value-Based Pilots (Negotiated “pay-for-results” contracts):** This innovative pricing tier links payments directly to measurable **clinical or operational outcomes**, such as reductions in unnecessary biopsies, faster reporting turnaround times, or earlier cancer detections. It is most attractive to **research-focused NHS teaching hospitals and early adopters** that are motivated by innovation and evidence generation. By positioning contracts as **“pay for results”**, Quibim demonstrates confidence in the value of its technology while aligning with the NHS’s broader shift towards **value-based healthcare**. This model also strengthens credibility: positive outcomes from value-based pilots can be leveraged as evidence to convince other trusts, creating momentum for wider adoption. For Quibim, while short-term revenues may be variable, the long-term reputational and evidence-building benefits are significant.

By adopting a flexible pricing framework that accommodates both NHS and private sector expectations, Quibim can enter the UK market competitively while preserving opportunities for value-based differentiation. Aligning prostate MRI pricing to Lucida benchmarks while positioning brain MRI within the £20–£35 range ensures affordability, credibility, and scalability. Over time, demonstrating ROI through efficiency gains, biopsy avoidance, and earlier cancer detection will allow Quibim to justify moving closer to premium benchmarks like Paige, particularly in private and research-led settings

4.5.7. Illustrative ROI Calculations

Pricing must always be backed by a **return on investment (ROI) story**. Below are illustrative examples Quibim can use in NHS proposals:

Example 1: Prostate MRI Biopsy Avoidance

The average cost of an unnecessary prostate biopsy is around £1,500–£2,000. If Quibim’s AI can help reduce these unnecessary procedures by just 10% across 1,000 patients each year, the resulting savings would amount to £150,000–£200,000 annually. With an annual cost of £20 per scan—around £20,000 in total—this translates to a return on investment of roughly 7–10 times, making a strong economic case for adoption.

Example 2: Radiologist Time Savings

The average reporting time for a prostate MRI is typically 20–30 minutes, but AI-assisted reporting could reduce this by 5–10 minutes per case. For a centre handling 2,000 scans per year, that saving translates into 300–400 hours—equivalent to approximately £30,000–£40,000 in radiologist time based on NHS salary benchmarks. With an annual subscription of £50,000, this delivers nearly a 1:1 return on investment purely from time savings, even before factoring in the added clinical benefits.

Example 3: Earlier Cancer Detection

Earlier detection of prostate cancer can cut late-stage treatment costs by £10,000–£20,000 per patient. If Quibim’s AI enables the identification of 5–10 additional early-stage cancers each year, this could generate savings of £100,000–£200,000. When combined with efficiencies from reduced reporting time and fewer unnecessary biopsies, the overall ROI has the potential to exceed 5–7x annually.

4.5.7.1. Pricing for Private Providers

While the NHS will anchor Quibim's UK pricing, the **private sector** operates differently:

- **Self-pay patients:** Typically pay £300–£600 for a prostate MRI scan. Adding AI at **£30–£50 per study** is reasonable if marketed as an upgrade that improves accuracy and patient confidence.
- **Private hospital groups:** More open to enterprise licenses, especially if AI accelerates patient throughput. Here, **£70,000–£100,000 annually** is realistic.
- **Diagnostic chains (Vista Health, LivingCare, Scan.com):** Particularly receptive to per-study models that can be easily passed on as itemised patient charges.

Thus, Quibim can afford to set **slightly higher per-study rates in the private sector (£25–£30 per scan)** without resistance, provided the clinical value is communicated.

4.5.7.2. Procurement Alignment

Quibim should structure its NHS pricing strategy in a way that directly aligns with procurement processes:

- **G-Cloud framework:** Submissions must include clear per-study and subscription options. Pricing transparency is essential.
- **NICE evaluations:** Submitting to NICE can legitimize Quibim's pricing, as NICE often publishes cost-effectiveness ranges.
- **AI in Health and Care Award:** Participating in this funding scheme can offset early adoption costs for trusts, making pricing more attractive.

Model	Proposed Price	Target Segment	Notes
Per-Study	£18–£22 per scan	NHS pilots, small trusts, diagnostic centres	Aligns with Lucida benchmark, signals value
Subscription	£50k–£70k annually	Large NHS trusts, private hospital groups	Predictable, scalable
Hybrid (Tiered)	£20k base + £10–£12/scan	Mid-sized trusts	Flexible, NHS-friendly
Outcome-Based (Pilot)	Negotiated	Research-focused NHS trusts	Differentiates Quibim as innovative

Table 15 Recommended Pricing Strategy

4.5.8. Market Entry Channels – NHS & Private Sector Pathways

The Importance of Entry Channels

Even the best AI imaging solution cannot succeed in the NHS if it lacks the right **distribution channels**. The NHS is a uniquely complex system, where adoption requires not only clinical validation but also procurement pathways that reduce IT burden, satisfy regulatory compliance, and minimise administrative friction.

Quibim must therefore consider both **indirect models (marketplaces, distributors)** and **direct engagements (NHS tenders, private contracts)**. The optimal strategy is a **hybrid approach**, leveraging marketplaces and distributors for NHS access while cultivating private provider partnerships to generate early traction.

4.5.8.1. AI Marketplaces – The Fastest Route to NHS Adoption

AI marketplaces have become central to NHS adoption strategies because they offer a “**one integration, many solutions**” model. For trusts, this simplifies procurement, billing, and IT integration. For vendors, marketplaces accelerate reach, reduce overhead, and lend credibility through trusted intermediaries.

Key Marketplaces:

- **Sectra Amplifier Marketplace**

The Sectra Amplifier Marketplace is widely adopted across NHS radiology departments and provides hospitals with plug-and-play access to AI solutions directly through the Sectra PACS. For Quibim, joining this marketplace would offer valuable visibility and streamlined access to NHS trusts that are already operating on Sectra infrastructure, making adoption easier and faster.

- **CARPL.ai**

CARPL.ai offers an evaluation and deployment platform that brings together multiple AI vendors, with a particular strength in enabling hospitals to “test before buying.” For Quibim, partnering with CARPL.ai could provide a useful entry route into NHS trusts that are cautious about committing to large-scale deployments upfront.

- **DeepcOS Marketplace (Deepc.ai)**

The DeepcOS Marketplace by Deepc.ai focuses on enterprise-wide infrastructure, offering embedded AI applications across hospital systems. For Quibim, this represents an opportunity to integrate its solutions into the broader digital ecosystems of major hospitals, facilitating seamless adoption and workflow integration.

- **Nuance AI Marketplace**

The Nuance AI Marketplace integrates directly into radiology workflows through Nuance PowerScribe and is already deployed in select NHS radiology environments. For Quibim, participation would offer both credibility and seamless workflow integration. This also reflects a broader trend: NHS buyers are increasingly favoring procurement of AI solutions via

established marketplaces rather than direct vendor contracts. As a result, Quibim should prioritize onboarding to platforms like Sectra and CARPL.ai as near-term entry points.

4.5.8.2. Distributors – Trusted NHS Intermediaries

Distributors act as **market navigators**, bringing established NHS relationships, contract expertise, and technical support. For international vendors, they can reduce the complexity of NHS sales cycles.

Key Distributors:

- **AMG Medtech Ltd** – Experienced in navigating NHS procurement.
- **Radiobotics Partner Network** – Offers distribution and technical support for AI vendors.
- **The ValueCare Group** – Focused on connecting medtech vendors to NHS buyers.

Implication: Partnering with one or more distributors would help Quibim scale NHS adoption without building a large UK sales team.

4.5.8.3. Direct NHS Engagement

Despite the growing role of marketplaces and distributors, direct NHS engagement remains critical. Quibim should focus on:

- **G-Cloud framework submissions** – A prerequisite for NHS digital procurement.
- **AI in Health and Care Award applications** – Provides funding and validation.
- **NICE evaluation submissions** – Essential for building credibility and publishing ROI evidence.

Direct engagement ensures Quibim remains visible as a standalone vendor while marketplaces and distributors act as amplifiers.

4.5.8.4. Private Sector Market Entry

The private sector provides a **faster, more agile entry route** than the NHS, with providers less constrained by procurement bureaucracy.

Key Targets:

- **Nuffield Health & Spire Healthcare** – National networks with MRI capacity.
- **Alliance Medical** – The largest independent provider of imaging services in Europe, with NHS contracts.
- **Circle Health & Vista Health** – Growing diagnostic chains.
- **LivingCare & Scan.com** – Patient-facing diagnostic providers, receptive to AI-enhanced imaging.

Implication: Early pilots with private providers could generate case studies and ROI evidence that Quibim can leverage in NHS tenders.

Recommended Entry Strategy for Quibim

1. **Short Term (0–12 months):** In the short term (0–12 months), Quibim should focus on joining the Sectra Amplifier and CARPL.ai marketplaces, establishing a partnership with a distributor such as AMG Medtech, and securing one to two pilot projects with private providers, for example Vista Health.
2. **Medium Term (12–24 months):** Quibim should submit its technology for NICE evaluation, scale NHS adoption through distributor-led contracts, and expand its presence to additional marketplaces such as Nuance and DeepcOS.

3. **Long Term(24+ months):** Quibim should aim to establish itself as a multi-organ platform across NHS trusts while negotiating enterprise-wide licenses with major NHS trusts and private hospital groups.

To maximise the likelihood of adoption, Quibim must also focus on key enablers. These include: proactive NHS engagement through the Imaging Services Transformation Programme and ICS-level pilots; ensuring regulatory readiness with UKCA marking and DTAC compliance; building partnerships with NHS-trusted vendors such as Brainomix; and prioritising high-impact actions such as pricing pilots and compliance in the first 12–18 months. Together, these enablers strengthen Quibim’s positioning and reduce barriers to adoption.

4.5.8.5. The Value-Based ROI Proposition

To succeed in the NHS, Quibim must present a clear value-based case that combines clinical, operational, and financial impact into a compelling return on investment (ROI).

Clinically, Quibim’s solutions improve prostate cancer detection rates, reduce false negatives, and lower the number of unnecessary biopsies. By incorporating quantitative biomarkers into reporting, the platform enhances radiologist confidence and supports earlier and more accurate diagnoses.

Operationally, Quibim saves 5–10 minutes per prostate MRI case, which translates to 300–400 hours saved annually across 2,000 scans. This equates to the workload of approximately 0.5–1 full-time radiologist, valued at £30,000–£40,000 per year. Beyond efficiency, Quibim also standardises reporting and facilitates smoother multidisciplinary team discussions.

Financially, the benefits are substantial. Each unnecessary biopsy costs the NHS £1,500–£2,000; reducing these by 10% across 1,000 patients saves £150,000–£200,000 annually. Earlier cancer detection further reduces treatment costs by £10,000–£20,000 per patient, while

operational efficiencies add further savings. Altogether, hospitals adopting Quibim can achieve a 5–7x ROI each year, even before considering long-term improvements in outcomes and resource allocation.

Conclusion

In summary, Quibim should adopt a value-based pricing strategy that balances affordability with proven clinical impact. By demonstrating cost savings to the NHS, aligning with tariff structures, benchmarking competitively, and ensuring compliance through UKCA and DTAC, Quibim can reduce adoption barriers and accelerate procurement across England.

5. Discussion (George Dreyer-Wilton – 240669537)

The findings of this study provide a multi-layered view of England's neuroimaging landscape and the adoption potential of Quibim's AI-enabled QP-Brain tool. Brain MRI demand continues to climb faster than workforce and infrastructure capacity, creating bottlenecks across the NHS (NHS England, 2023; Warner & Zaranko, 2024), while the private sector, though smaller, is growing quickly as patients seek alternatives to long waits (LaingBuisson, 2025; IHPN, 2025). At the same time, the introduction of AI-driven diagnostics promises to address precisely those inefficiencies that are generating systemic strain. Yet, the path to adoption is neither linear nor guaranteed. It depends on a combination of institutional readiness, financial incentives, regulatory approval, and clinical trust. To interpret the analyses presented earlier in the report, this discussion reflects on four key themes: the implications of demand trends in both NHS and private markets; the role of adoption scenarios in highlighting diffusion pathways; the strategic positioning of QP-Brain within a crowded competitive environment; and the structural limitations of the study that frame how confidently these findings can be applied.

One of the most striking elements of the NHS-focused analysis is the persistence of a supply–demand imbalance in neuroimaging. Annual brain MRI volumes have risen at a compound annual rate of nearly 5% over the past decade, outpacing both population growth and radiologist workforce expansion (Warner & Zaranko, 2024). The workforce shortfall is particularly notable. While radiologist numbers have grown, the increase has lagged behind rising demand, resulting in recurring shortages and backlogs (Royal College of Radiologists, 2023). Waiting lists remain persistently above NHS targets, with a significant proportion of patients waiting more than six weeks for imaging (NHS England, 2023). These trends have real-world consequences. For dementia patients, diagnostic delay can mean missing the critical

window for early intervention (Alzheimer’s Society, 2024). For brain tumor patients, slower detection leads to worse prognoses and more invasive treatment (Cancer Research UK, 2024). For stroke patients, delays can mean permanent disability (Stroke Association, 2023). Thus, the issue is not abstract but tied directly to health outcomes and financial costs. Against this background, the potential contribution of AI-enabled tools becomes clearer: by automating parts of the reporting process, standardizing interpretation, and accelerating turnaround times, QP-Brain could directly mitigate some of these pressures.

The private sector, though smaller in scale, plays a big role in shaping innovation trajectories. Estimates suggest private brain MRI volumes in England fall somewhere between 20,000 and 60,000 annually, with recent growth rates of 10–15% (LaingBuisson, 2025; IHPN, 2025). This expansion is driven largely by patients unwilling to face NHS delays and willing to self-pay for rapid access. In London and other metropolitan centres, this demand has fostered a highly competitive marketplace where providers differentiate themselves on convenience, speed, and quality of service. For Quibim, this landscape presents an opportunity. Early partnerships with large hospital groups and independent imaging providers could provide a proving ground for QP-Brain, demonstrating real-world impact and building an evidence base that can later be leveraged within the NHS. The private sector’s agility and willingness to trial new technologies can thus complement the slower processes of NHS procurement. However, this opportunity is tempered by affordability concerns. Private brain MRI already carries substantial costs, and adding AI surcharges may limit uptake unless insurers or providers absorb them (LaingBuisson, 2025). For this reason, value demonstration in the private sector is not only about clinical accuracy but about showing measurable efficiency and cost-effectiveness.

Forecasting demand and modelling adoption scenarios highlight both the inevitability of rising strain and the contingency of technological relief. The scenario-based analysis produced three

adoption curves (conservative, medium, and ambitious), each reflecting different degrees of policy support, investment, and organizational readiness. These trajectories are not merely statistical exercises but illustrations of how systemic levers shape diffusion. Under conservative adoption, uptake remains slow, leaving NHS backlogs largely unchanged and AI confined to niche pilots. Medium adoption represents a steady but incremental pathway, consistent with historical patterns of NHS innovation uptake (Charlesworth et al., 2024). Ambitious adoption, by contrast, sees rapid diffusion across trusts, leading to earlier systemic impact. What these scenarios underscore is that adoption is a choice as much as a forecast. The spread of innovation is heavily influenced by policy priorities, procurement frameworks, and cultural acceptance within the medical community (Galea et al., 2023). This in turn highlights the importance of aligning QP-Brain not only with technological needs but with institutional realities. Evidence alone will not guarantee adoption. It must be coupled with integration into procurement mechanisms, compliance with regulatory frameworks, and resonance with clinician priorities.

The adoption discussion also raises questions about equity and regional disparity. Analysis of provider-level activity showed significant variation in scan volumes, workforce availability, and waiting list lengths (NHS England, 2023). Major teaching hospitals and metropolitan trusts carry disproportionate workloads but often have greater resources to experiment with new technologies. Smaller or rural trusts face equally pressing demand pressures but often lack the capital or infrastructure to adopt advanced tools quickly. This suggests that adoption may initially cluster in resource rich areas, potentially exacerbating inequalities in diagnostic access (Galea et al., 2023). For Quibim, early adoption where the infrastructure is strongest is the most logical option, but for the healthcare system, it could end up widening existing gaps in access. Any future deployment strategy must therefore grapple not only with technical feasibility, but also with the fairness and ethical issues around how new technology spreads.

Competitor dynamics further complicate the landscape. While Quibim distinguishes itself with a multi-organ platform, several competitors are already well established in NHS pathways, particularly in prostate and stroke imaging (Brainomix, 2023; Lucida Medical, 2023). Brainomix, for example, has deep penetration in NHS stroke networks, while Lucida Medical has positioned itself strongly within prostate MRI. Quibim therefore needs to focus on what makes it different. It has a wider platform, strong biomarker analysis, and ability to work efficiently within hospital systems, rather than just competing on single-organ tools. In practice, this means emphasizing how a multi-organ tool reduces IT fragmentation, eases procurement by consolidating vendor relationships, and provides broader value across neurology and oncology. Competitor analysis also points to pricing benchmarks: NHS tenders suggest per-scan AI costs in the £15–£20 range (NHS Supply Chain, 2024), with potential for higher value-based models once strong clinical validation is established. For Quibim, this indicates that initial pricing must remain competitive, but longer-term strategy should focus on building evidence that justifies premium positioning. The competitive environment thus reinforces the importance of coupling technical innovation with strategic commercial positioning.

While the report provides valuable insights, it is important to acknowledge its limitations. First, data limitations are significant, particularly in the private sector. Lack of transparency forces reliance on secondary estimates (LaingBuisson, 2025; IHPN, 2025), leading to wide ranges in market size. Second, forecasting methods, though rigorous, are sensitive to assumptions around seasonality, demand drivers, and decay factors. The projections should therefore be viewed as directional indicators rather than precise predictions (Charlesworth et al., 2024). Third, adoption modelling relies on simulations that cannot capture the full complexity of real-world decision-making, which is influenced by factors such as political priorities, procurement negotiations, and individual clinical champions (Galea et al., 2023). Fourth, the study does not

incorporate a full health-economic evaluation, which NHS procurement bodies generally require (NICE, 2023). While adoption scenarios reveal diffusion patterns, they do not quantify cost savings or patient outcome improvements in monetary terms, which are critical in procurement decisions. Fifth, while competitor insights were considered, the analysis did not undertake a detailed financial benchmarking or model different pricing strategies for Quibim beyond high-level references. Finally, although ethical and data governance issues were noted in earlier sections, the study did not fully integrate these into adoption modelling. Concerns around algorithmic bias, data security, and patient consent remain unresolved in the broader literature (Topol, 2019; Galea et al., 2023).

Limitations also extend to scope. While the study focused on brain MRI as the central application, in practice, procurement decisions may be influenced by multi-organ considerations, integrated care pathways, or broader hospital IT strategies (NHS England, 2023). Furthermore, the analysis centered on England, with only brief references to UK-wide data. This may limit applicability, as Scotland, Wales, and Northern Ireland operate distinct healthcare systems with their own procurement frameworks. Another scope limitation is the absence of patient and clinician perspectives. Adoption scenarios model demand and capacity pressures, but they do not capture the lived experience of radiologists navigating workload strain or patients deciding whether to self-pay for private scans (Alzheimer's Society, 2024). Incorporating qualitative data from these stakeholders could enrich future analyses by grounding modelling assumptions in frontline realities.

Despite these limitations, the analysis carries clear implications. For policymakers, it demonstrates the scale of the diagnostic challenge and the potential role of AI in alleviating it (Warner & Zaranko, 2024). For providers, it highlights how adoption could be targeted to trusts under the greatest strain. For Quibim, it provides strategic insight into where to focus market

entry efforts, how to position against competitors, and what barriers must be overcome. The broader message is that while the opportunity for AI in brain imaging is substantial, success depends on careful alignment of technology, evidence, and strategy with the realities of the healthcare ecosystem. Innovation must be embedded within procurement frameworks, validated through clinical outcomes, and supported by a sustainable pricing model (NICE, 2023). Without this alignment, even the most sophisticated AI tools may remain underutilized, confined to pilots rather than achieving systemic impact.

In sum, the discussion highlights both promise and caution. Rising demand and systemic bottlenecks create fertile ground for innovation, and AI tools such as QP-Brain are well positioned to deliver measurable value. Yet adoption will not occur automatically. It requires deliberate strategy, robust evidence, and institutional alignment. The limitations of the study highlight the importance of further research. What's clear, however, is that without change, waiting lists will remain long and patient care will suffer. The question is not whether AI can play a role, but how quickly and effectively it can be integrated into the fabric of the healthcare system.

6. Conclusion (George Dreyer-Wilton – 240669537)

This report has explored the adoption potential of Quibim's QP-Brain AI platform within England's neuroimaging market, situating it against the broader realities of demand, capacity, and technological change in diagnostic imaging. The earlier sections set out the pressures facing providers, the opportunities for innovation, and the practical barriers to adoption. The purpose of the conclusion is not to repeat these findings, but to draw out their broader meaning, to highlight the study's contributions, and to identify clear directions for both future research and practical action.

This work provides several important contributions that advance understanding of AI adoption in medical imaging. First, it offers a consolidated view of brain MRI activity across the NHS and private providers, addressing a common gap in which datasets are considered separately (NHS England, 2023; LaingBuisson, 2025). Bringing these sources together allows a more realistic appreciation of the total scale of neuroimaging demand and the distribution of capacity across providers.

Second, the study applies scenario modelling to AI adoption. Rather than treating uptake as a fixed trajectory, the analysis illustrates how adoption might unfold under different assumptions about investment, clinical culture, and policy frameworks (Charlesworth et al., 2024; Galea et al., 2023). This approach acknowledges uncertainty while still offering structured insights for decision-makers.

Third, the report situates the technical promise of QP-Brain within a strategic and commercial context. It shows that clinical accuracy alone will not determine success. Procurement requirements, affordability, clinician trust, and integration into existing workflows are equally decisive (NICE, 2023; Royal College of Radiologists, 2023). By doing so, the report moves the discussion from abstract enthusiasm for AI toward a more grounded understanding of how adoption might actually occur in practice.

The study is also clear about its own boundaries. Data limitations meant that estimates of private-sector activity remain approximate, and forecasting results should be read as indicative rather than predictive (IHPN, 2025). The competitor analysis provided useful context but was not exhaustive. These limitations matter not only because they restrict the certainty of the findings, but because they highlight precisely where future work is most needed.

Several priorities emerge from this analysis. The first priority is clinical validation. QP-Brain must demonstrate its real-world impact on reporting times, diagnostic accuracy, and patient outcomes (Topol, 2019; Alzheimer’s Society, 2024). Without embedded pilot studies in live clinical settings, the evidence will remain abstract and procurement decisions are likely to remain cautious. Alongside this, economic evaluation is critical. For the NHS, cost-effectiveness needs to be shown using health-economic frameworks familiar to policymakers, while for private providers, evidence of return on investment in terms of increased throughput and improved patient satisfaction will be decisive (NICE, 2023).

Another important area is equity and access. Adoption is likely to begin in larger, better-resourced centers, but without deliberate planning, smaller trusts and rural providers could be left further behind (Galea et al., 2023). Future research should therefore focus on how AI can be scaled equitably so that benefits reach patients regardless of geography. Closely related is the question of workforce integration. Radiologists and other clinicians must view AI as a supportive tool rather than a threat. Research into training, co-design, and change management will be essential to build acceptance and ensure sustainable implementation (Royal College of Radiologists, 2023).

Finally, two broader priorities emerge. The first is policy and procurement engagement. Understanding and aligning with the evolving frameworks through which the NHS adopts digital health technologies will be decisive, and companies such as Quibim must work proactively within these structures while contributing to the shaping of future guidelines (NHS Supply Chain, 2024). The second is global comparative analysis. Examining adoption dynamics across different health systems internationally would not only help Quibim position QP-Brain as a globally credible solution but also generate insights that are transferable across markets.

From a strategic perspective, the findings carry several implications for Quibim. The company must position itself not only as a provider of accurate AI tools but as a partner that understands the institutional and systemic realities of adoption. This means developing evidence tailored to procurement frameworks, offering pricing structures aligned with NHS sensitivities, and investing in relationships with clinicians and policymakers (NICE, 2023).

Quibim should also leverage its distinctive advantage in multi-organ MRI quantification. While competitors may offer strong organ-specific solutions, the breadth of Quibim's platform allows it to offer efficiency across a range of imaging needs. Communicating this breadth, along with the quality of biomarker extraction and workflow integration, can provide a basis for differentiation that does not depend on competing head-to-head in crowded niches (Brainomix, 2023; Lucida Medical, 2023).

Finally, Quibim should view the UK not only as a market in its own right but as a proving ground. The NHS is both a large healthcare system and a global reference point (Warner & Zaranko, 2024). Demonstrating successful adoption in the UK could therefore create reputational benefits that extend well beyond England, strengthening the company's positioning in other health systems.

More broadly, the findings of this report underscore the dual challenge of urgency and caution in healthcare innovation. On the one hand, neuroimaging demand is rising sharply, backlogs are entrenched, and patients are already experiencing delays with real consequences for outcomes (Cancer Research UK, 2024; Stroke Association, 2023). On the other hand, adoption of AI in healthcare carries risks: premature implementation could undermine trust, exacerbate inequalities, or generate new errors (Topol, 2019). The challenge is to move quickly enough to address immediate pressures while embedding evaluation mechanisms that protect safety and fairness.

The study also highlights the importance of seeing AI adoption as a systemic rather than purely technical process. Introducing new tools is not only about demonstrating accuracy but about understanding incentives, trust, culture, and equity (Galea et al., 2023). In this sense, adoption is as much a social and organisational challenge as it is a technological one.

In conclusion, this study provides one of the most comprehensive views to date of the demand landscape for brain MRI in England and the adoption potential of AI solutions such as QP-Brain. It demonstrates both the urgency of the problem and the possibility of meaningful solutions. At the same time, it makes clear that success will require more than algorithms: it will require evidence, strategy, and partnership.

For Quibim, the path forward is to combine technological ambition with institutional sensitivity, positioning QP-Brain as not only accurate but affordable, trusted, and scalable. For the health system, the challenge is to embrace innovation without losing sight of safety, equity, and long-term value. The stakes are high, but so too are the potential benefits. If approached with rigour, caution, and commitment, AI-enabled neuroimaging has the potential to improve diagnostic pathways, strengthen capacity, and deliver better outcomes for patients.

AI Declaration

We admit to using AI for this report. AI was used for research purposes, structuring and streamlining the content.

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